SAEngine User Manual

Structural analysis and design to BNBC 2020

Version 0.0.42 (Pre Alpha 5)

SAEngine is free software under the GNU Affero General Public License v3.0. This manual describes the program as it is in this version; check every result independently before using it for construction. All projects, boreholes and models in it are synthetic.

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SAEngine User Manual

SAEngine version 0.0.42 (Pre Alpha 5). This manual describes what the program does in this version, part by part, with four step-by-step tutorials. It is written for structural engineers who know BNBC 2020 and programs such as ETABS and AutoCAD; the glossary maps SAEngine's terms to theirs.

The same content is compiled into one PDF, **SAEngine User Manual** (SAEngine-User-Manual.pdf), by build.py.

> **Pre-alpha.** SAEngine is under active development. Check its results independently before using them for construction. Every chapter ends with a list of what is limited or could not be confirmed in this version.

Conventions

- **Bold** words are buttons, menu entries, commands and field names as they appear on screen. *Italic* text in quotes is a message SAEngine shows.
- Units are SAEngine's fixed SI units: mm for sections and plan positions, m for levels and site positions, kN, kN/m^2, kN·m, MPa.
- Screenshots were taken from the development build of 0.0.42 with the sample project and synthetic data. Your window may differ slightly.
- All projects, boreholes and models in this manual are synthetic. None of them is a real project or a real soil investigation.

Getting started

This chapter explains the parts of the SAEngine window that every project shares: the start page, the Project sidebar, the Overview, **Run All**, Versions and project archives. The eleven parts each have their own chapter.

SAEngine is a desktop program for the structural analysis and design of buildings and special structures to **BNBC 2020** (ACI 318-19 and AISC 360-22 can be chosen instead). It works in fixed SI units: millimetres for geometry, metres for site positions and levels, kN and kN/m² for loads, MPa for strengths. There is no unit setting.

> Installing SAEngine is described in docs/install (`docs/install/README.md`). The Python backend (analysis, design, reports, drawings) comes with the installer. If a command says "*The SAEngine Python backend is not available*", run **Check Python Runtime** from the Command Palette; it reports the Python, OpenSeesPy and opensees-studio versions it found.

The start page

When SAEngine starts without a project, the start page fills the window.

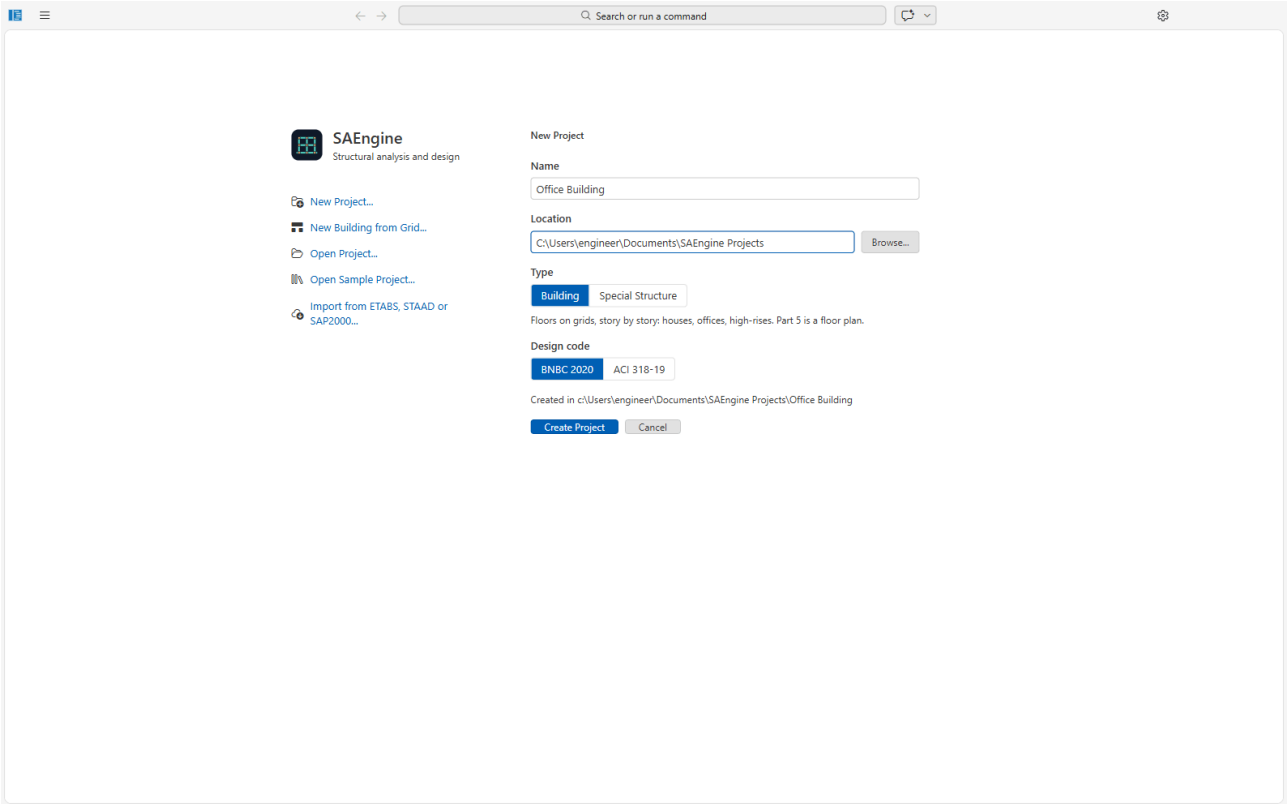


Figure 1: The start page

Button	What it does
New Project...	Shows the New Project form on the right (below).
New Building from Grid...	Opens the New Building wizard: a whole RC building from a grid in one step (Part 5).
Open Project...	Picks a project folder (default `Documents\SAEngine Projects`). If the folder is not a project, SAEngine offers **Create Project Structure** existing files are kept.
Open Sample Project...	Copies the *Sample Frame* project to a folder you choose and opens the copy. If a copy is already there, it offers to open that copy.
Import from ETABS, STAAD or SAP2000...	Creates a project from a text export of another program (Importing models).

Recent Projects on the right lists up to eight recently opened projects.

New Project

Field	Meaning
Name	The project name; it becomes the folder name and cannot contain `\ / : * ? " < > |`.
Location	The parent folder (default `Documents\SAEngine Projects`), with **Browse...**.
Type	**Building**: floors on grids, storey by storey; Part 5 is a floor plan. **Special Structure**: drawn in 3D (roofs, stadiums, towers, bridges); Part 5 is a 3D structural model.
Design code	**BNBC 2020** (default) or **ACI 318-19**. BNBC 2020 sets concrete to *BNBC 2020 Part 6 Chapter 6* and steel to *Part 6 Chapter 10* ACI 318-19 sets ACI 318-19 and AISC 360-22. Both can be changed later in Project Settings.

Create Project creates the folder and opens it in the same window.

What a project is

A project is a folder with a `<name>.saengine` manifest and one folder per part:

```
Office Building/
├─ Office Building.saengine
├─ 01 Soil Report/soil-report.json
├─ 02 Project Settings/project-settings.json
├─ 03 Materials/materials.json
├─ 04 Objects/objects.json
├─ 05 Floor Plan/floor-plan.json        (05 Structural Model/structure.json for a special
structure)
├─ 06 BIM/bim.json
├─ 07 MEP/mep.json
├─ 08 FEM/model.osmodel                  (derived, with its results in model.results/)
├─ 09 Substructure/substructure.json
├─ 10 BOQ/boq.json
└─ 11 Exports/                           (report, drawings, IFC, DXF, CSV)
```

The manifest, the part folders and each part's main file are protected: SAEngine refuses to delete, rename or move them and recreates them if they are removed outside the program. You can add other files (for example architect's DXF underlays) freely.

Every editor saves to its file like a document: **Ctrl+S** saves, **Ctrl+Z / Ctrl+Y** undo and redo, and each change in a form or on a plan is one undoable step.

The Project sidebar

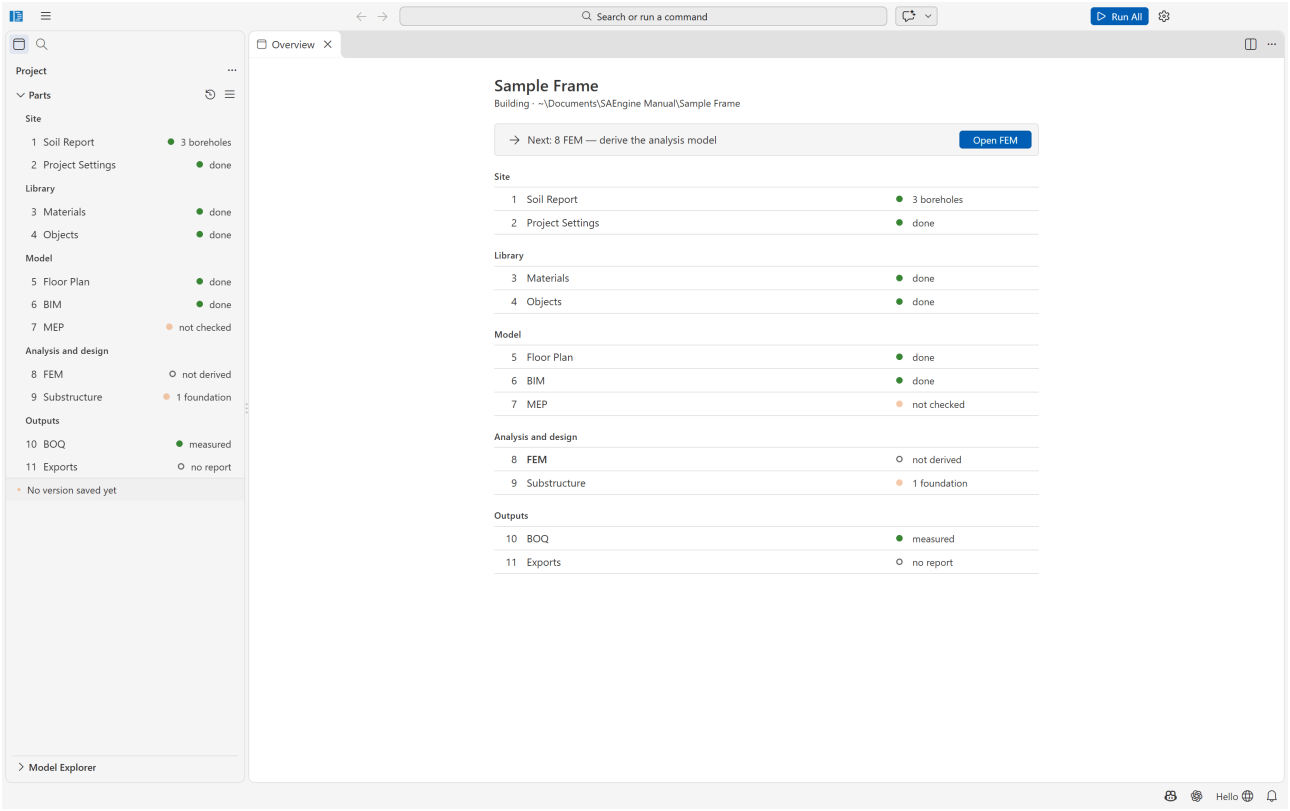


Figure 2: The Overview and the Project sidebar of the sample project

The **Project** sidebar lists the eleven parts in five groups. Click a part to open its editor.

Group	Parts
Site	1 Soil Report, 2 Project Settings
Library	3 Materials, 4 Objects
Model	5 Floor Plan (or Structural Model), 6 BIM, 7 MEP
Analysis and design	8 FEM, 9 Substructure
Outputs	10 BOQ, 11 Exports

Each part shows a status dot and a short text, for example *done*, *in progress*, *not started*, *out of date* ("stale"), *checks fail* or *invalid file*, or a count such as *3 boreholes* or *1 foundation*.

Below the parts, a line summarises the versions (for example *No version saved yet* or *2 parts changed since v3*); click it to open Versions.

The sidebar's ≡ menu holds **Project Overview**, **Project Settings**, **Versions**, **Save Version…**, **New Building from Grid…**, **Open Folder in File Explorer**, **Export Project…** and **Open Project Archive…**.

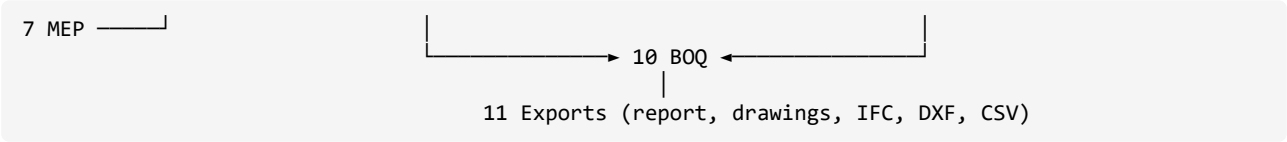
The second sidebar, **Model**, holds the **Model Explorer**: the members of the model in a tree. Selecting a member there selects it in the floor plan and the 3D view too.

The Overview

Project Overview opens when a project opens. It shows the project type and path, a **Next:** line with the part that needs attention next and a button to open it (for example *Next: 8 FEM — derive the analysis model* / **Open FEM**), and the same part list as the sidebar. When nothing is left to do it says *Every part is up to date.*

How the parts depend on each other

```
1 Soil Report ─┐
2 Project Settings ─┤
3 Materials ─┤
4 Objects ─┼─► 5 Floor Plan ─► 6 BIM ─► 8 FEM (derive ► analyse ► design) ─► 9 Substructure
```



- The **building model** (6 BIM) is built from the floor plan when you click **Build from Floor Plan** / **Rebuild from Floor Plan** in the BIM editor. Run All does **not** rebuild it.
- The **analysis model** (8 FEM) is derived from the building model, Project Settings, Object Types and Materials (and the soil report, MEP and substructure where they matter). It is never edited by hand.
- Every derived file remembers what it was made from. When a source changes, the part shows **Out of date** with a link to derive, measure or build it again.

Run All

Run All (the blue button in the title bar, or *SAEngine: Run All* in the Command Palette) runs the analysis and design of the whole project. It first saves any unsaved edits of the part files, then runs these steps in order, skipping those that are not needed:

Step	What it does	Skipped when
1 analysis model	Derives `08 FEM/model.osmodel` from the building model	it is up to date
2 analysis	Runs every analysis case with OpenSeesPy	—
3 fixed-base run for the soil springs	A second run on fixed supports, used by the foundation design	supports are not soil springs
4 design checks	Checks every member against the load combinations, then the steel each needs	—
5 foundation design	Checks the foundations of 9 Substructure	there are no foundations
6 MEP coordination, plumbing, HVAC and electrical checks	Openings, clashes, plumbing, HVAC and electrical results	MEP is empty
7 bill of quantities	Measures 10 BOQ	it is up to date
8 report	Prints `11 Exports/report.pdf`	—
9 drawings	Regenerates the structural drawings	none were generated yet, or they are current

While it runs, the left of the status bar shows *Running 2 of 9: analysis*. **Click that entry to cancel.** When it finishes, a notification sums up the result, for example *Done: 15 pass, 2 fail; steel short in 2 element(s) · foundations 1 pass, 0 fail · BOQ updated · report written*, with **Open Report**.

If a step cannot be done, Run All stops there: *Run All stopped at step 2 of 9 (analysis): …* (with **Show Log** for analysis errors). **Failing design checks do not stop Run All**: they are results, shown in 8 FEM and the report.

Run All does not:

- rebuild 6 BIM from the floor plan (do that in the BIM editor after editing the plan);
- generate the first drawing set (start it once from 11 Exports);
- change any section size (see 8 FEM, design proposal).

Run All took four to ten minutes for the sample on the PC used for this manual, and 20 to 35 minutes for the six-storey building of Tutorial 2; the design checks take most of the time.

Versions

A *version* is a snapshot of all the part files that you save at a milestone. SAEngine keeps versions inside the project folder (in a hidden `.git` folder, written by SAEngine itself; Git does not need to be installed). Computed results are not stored in versions.

Open **Versions** from the sidebar line, the History button in the sidebar title or the ≡ menu.

Action	What it does
Save Version…	Asks for a message (e.g. *Column sizes after the wind check*) and optionally marks it **Issued** (a submission or a drawing issue). Unsaved editors are saved first.

Action	What it does
Compare with Current	Lists the parts and values changed since the selected version. Ctrl+click a second version to compare two versions.
Restore…	Pick the parts to restore. Before restoring, unsaved changes are saved as a version *Before restoring …* the restore itself becomes a new version. History is never rewritten.
Mark as Issued…	Labels a version, e.g. *Rev A, Issued for Approval*.

The author name of versions is the setting `saengine.versions.author` (empty: *SAEngine user*). A project that lies inside another Git repository cannot keep versions; the sidebar then says *Versions are not kept for this project*.

Project archives (.civil)

Export Project… writes the whole project into one `.civil` file (a zip with a manifest and the part files), for sending or archiving. Options: **Include Versions** and **Include Results** by default computed results are left out (they are recomputed) and `11 Exports` is included. **Open Project Archive…** (or double-clicking a `.civil` file) unpacks it into a folder you choose and opens it.

Useful settings

Open them with the gear in the title bar (*Settings*), and search for `saengine`.

Setting	Meaning
`saengine.python.path`	The Python interpreter of the backend (empty: the one bundled by the installer).
`saengine.cad.program`	The CAD program DXF drawings open in (empty: the bundled LibreCAD, then an installed LibreCAD, then the Windows default for DXF).
`saengine.map.tileUrl`, `saengine.map.tileAttribution`	The map tiles of the soil-report site map (default OpenStreetMap).
`saengine.qgis.program`	QGIS, for **Open in QGIS** on the site map.
`saengine.versions.author`	The name saved with versions, as `Name <email>`.
`saengine.developerMode`	Shows the code-editor features of the underlying Code - OSS (Explorer, terminal, …). Off by default. **Toggle Developer Mode** is in the Help menu.

There are **no SAEngine keyboard shortcuts** beyond the ones listed in each part (editor keys such as Ctrl+S, Ctrl+Z, Delete, Esc). Every command can be found in the Command Palette (**Ctrl+Shift+P**, or the search box in the title bar) under the category *SAEngine*.

Language: English or Bangla

SAEngine's own screens are available in Bangla (বাংলা) through the built-in *Bangla Language Pack for SAEngine*. Run **Configure Display Language** from the Command Palette, choose Bangla (বাংলা) and restart SAEngine. The rest of the underlying Code - OSS workbench stays in English, and the `saengine` command line always writes English. This manual uses the English names.

Light and dark

SAEngine follows the colour theme of the window (gear → *Themes* → *Color Theme*). All editors, plans and the 3D views have light and dark variants.

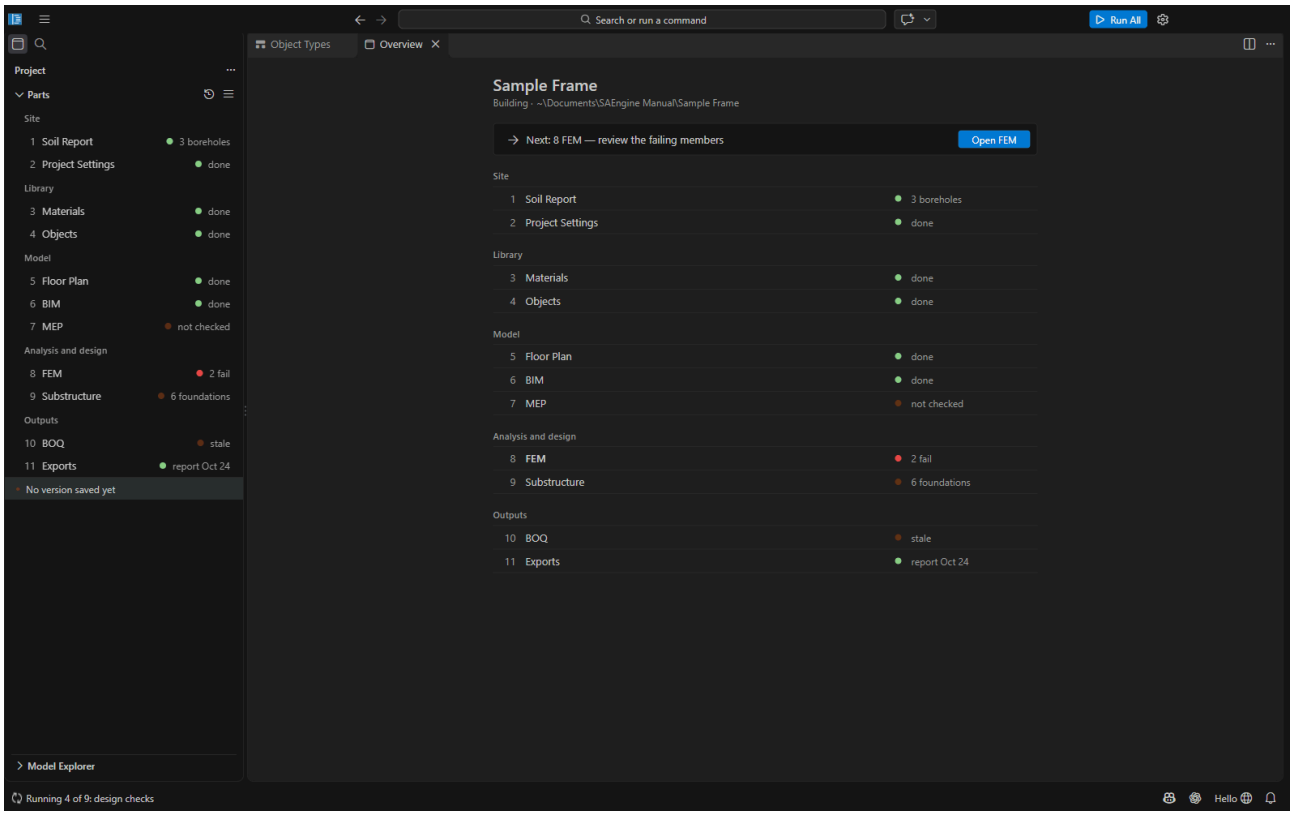


Figure 3: The Overview in a dark theme

Part 1: Soil Report

The soil report holds the boreholes of the site investigation: strata, SPT blow counts and lab test results. From them SAEngine works out the BNBC site class, allowable bearing pressures, settlements, liquefaction, the modulus of subgrade reaction and pile design parameters. These results feed Project Settings (site class), 8 FEM (soil springs) and 9 Substructure (bearing capacity, pile capacity).

The soil report is optional: without one, the site class and allowable bearing pressure entered in Project Settings apply.

File: `01 Soil Report/soil-report.json`. Open it with **1 Soil Report** in the sidebar.

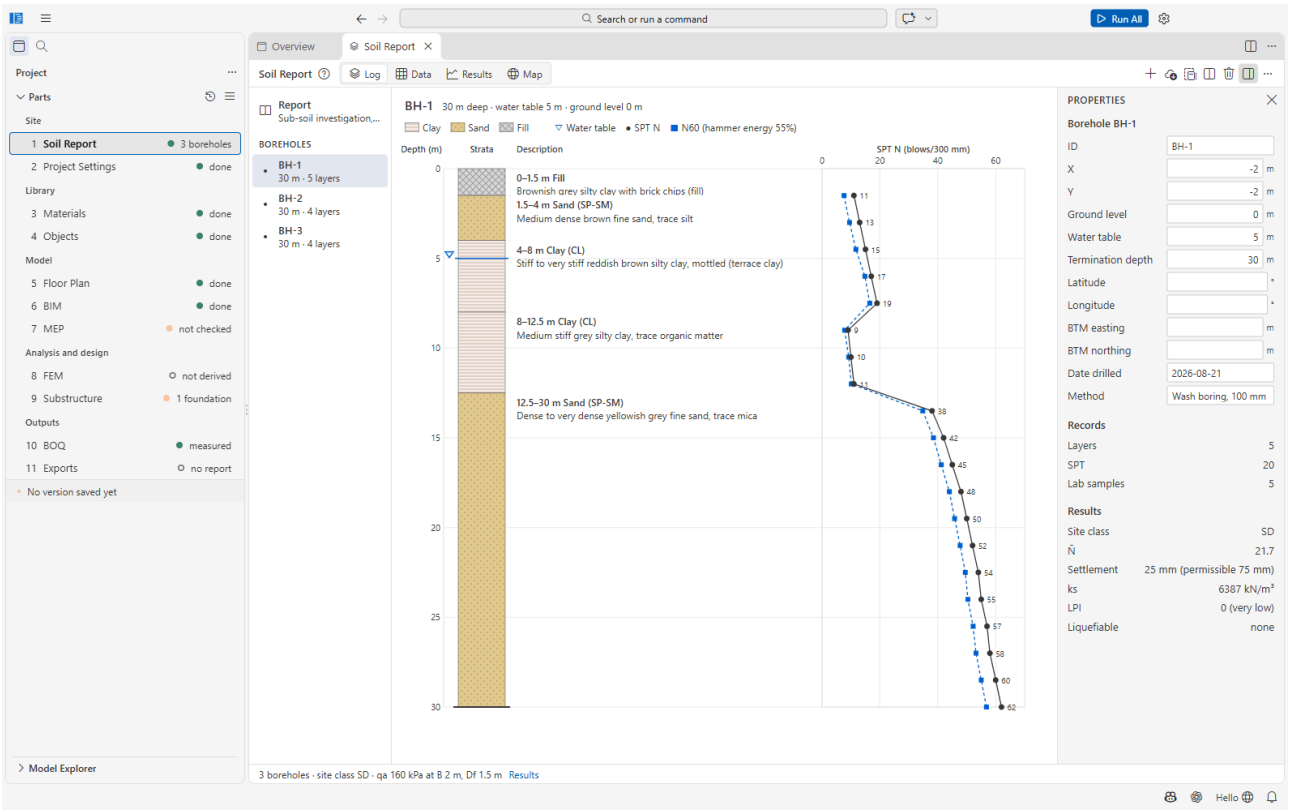


Figure 4: The borehole log of BH-1 in the sample project

Layout

- **Left:** the borehole list. *Report* at the top (the whole report), then each borehole with its depth and number of layers (or its site class once the results are computed). A warning icon marks a borehole with errors. Up/Down arrows move through the list.
- **Middle:** four views, **Log**, **Data**, **Results** and **Map**.
- **Right:** the Properties panel.
- **Toolbar: Add Borehole**, **Import CSV/Excel...**, **Save Template...**, **Show All Boreholes Side by Side** (Log view, two or more boreholes), **Delete** under **...**: **Duplicate Borehole** and **Compute Results Again**.
- **Status line:** e.g. *3 boreholes · site class SD · qa 160 kPa at B 2 m, Df 1.5 m*.

An empty report shows *No boreholes yet* with **Add Borehole**, **Import CSV/Excel...** and **Save a template to fill in**.

Report properties

Select **Report** in the list. The Properties panel shows:

- **Report:** Title, Consultant, Report number, Date, Ground level (m).
- **Design** (the basis of the computed results):

Field	Choices / default
Practice	**Textbook** (default) or **Bangladesh consultant practice** (see SPT corrections)
Liquefaction method	**Idriss & Boulanger 2014** (default) or **BNBC 2020 Sec 3.9.5**

Field	Choices / default
Hammer energy	% of theoretical, default 60 (30 to 100)
Factor of safety	on bearing, default 3
Allowable settlement	mm, default 25
Earthquake magnitude Mw	default 7.5
PGA	g; empty = taken from Project Settings (Z × S)

Empty values use the defaults.

Entering a borehole log sheet (Data view)

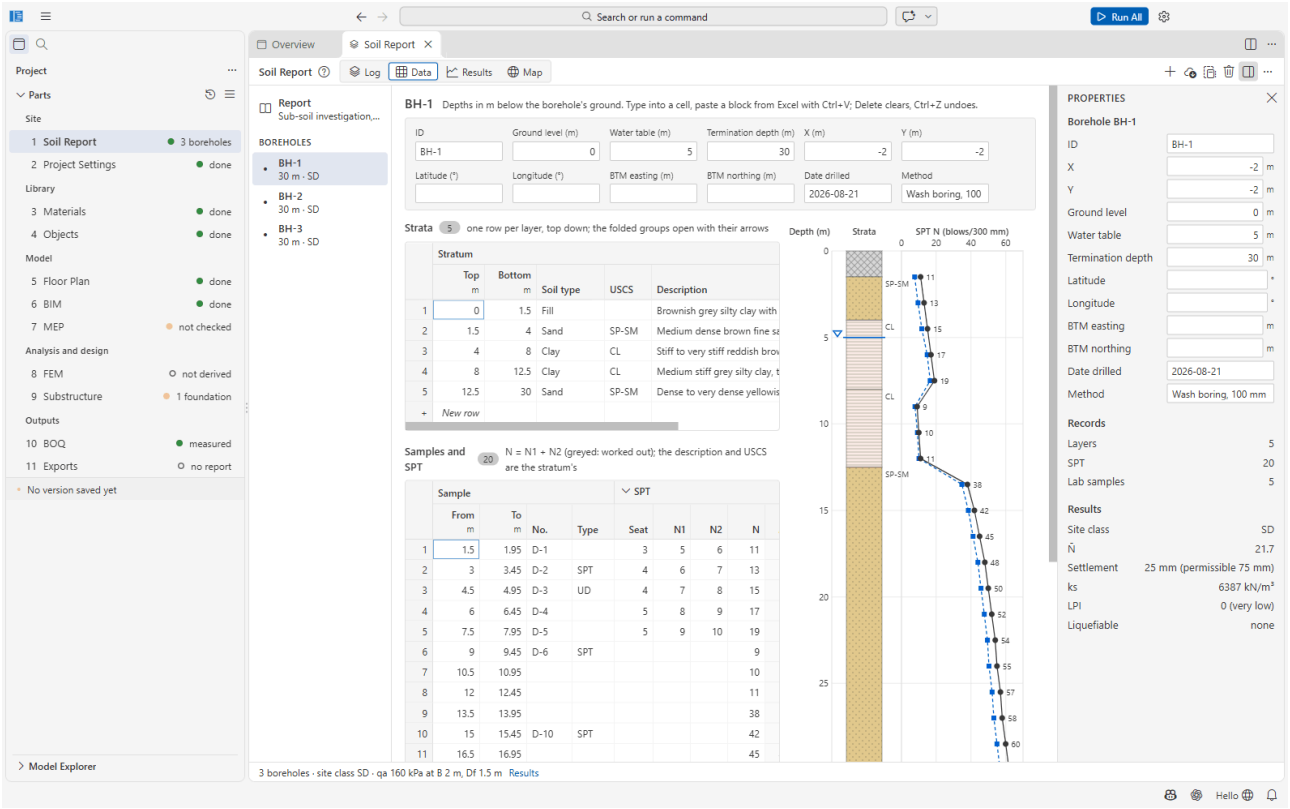


Figure 5: The Data view: the log sheet of a borehole

1. Click **Add Borehole**. The new borehole gets the next ID, 10 m to the right of the last one, copies its water table, is 10 m deep and starts with one clay layer from 0 to 10 m.
2. Fill in the **header strip**: ID, Ground level (m), Water table (m, depth below ground), Termination depth (m), X and Y (m, in the project's coordinates), or Latitude/Longitude (°), or BTM easting/northing (m); Date drilled, Method. When latitude/longitude or BTM are given, X and Y are worked out from them and become read-only.
3. Fill in the **Strata** grid, one row per layer from the top: *Top* and *Bottom* (m), *Soil type* (Clay, Silt, Sand, Gravel, Organic, Fill, Rock), *USCS*, *Description*, and as far as known: unit weights γ, γsat (kN/m³), strength cu, c′ (kPa), φ′ (°), E (MPa), index values FC, PI, LL (%), and consolidation Cc, Cr, e0, OCR. The column groups *Unit weight*, *Strength*, *Index* and *Consolidation* start folded; click a group heading to unfold it.
4. Fill in the **Samples and SPT** grid, one row per test depth: *From* / *To* (m; To defaults to From + 0.45 m), sample *No.* and *Type*, the SPT blow counts *Seat*, *N1*, *N2* (N is worked out as N1 + N2 and shown greyed unless you type N), *Ref.* for refusal (R, or e.g. `50/100`), recovery and the rod energy ratio *ER* if measured, then the index and strength results of the sample (w, γ, LL, PL, FC, cu, c′, φ′, Cc, e0; PI is computed). SAEngine shows the corrected **N60** and **(N1)60** in the last columns.
5. For a row with a sample, the **Lab tests** panel below takes the raw test data:
   - **Grain size:** sieve size (mm) and % passing; draws the curve, and **Use** buttons take the fines content, D10/D30/D60, Cu and Cc from it.
   - **Consolidation:** load steps σ′v (kPa) and void ratio e; gives Cc, Cr, e0 and the preconsolidation pressure σ′p (Casagrande).
   - **Triaxial:** UU, CU or CD specimens (σ3, σ1−σ3, u); gives cu, or c′ and φ′.

- **Direct shear:** σn and τ; gives c′ and φ′ by least squares.

The grids work like a spreadsheet: **Ctrl+V pastes a block copied from Excel**, Delete clears, Ctrl+Z undoes. A compact log next to the sheet updates as you type, and results recompute about a second after each change.

SAEngine checks the sheet as you type and lists problems, for example *BH-1: a gap between 4 m and 4.5 m.*, *layers overlap*, *an SPT at 31.5 m is below the termination depth*, *two SPT records at 6 m*, or *N 20 is entered, but N1 + N2 = 18; N 20 is used. Delete it to use N1 + N2.* A silt without PI or cu gets the hint *is it plastic? Give PI or cu*.

Importing boreholes from CSV or Excel

Save Template... writes `boreholes-template.xlsx` (or a .csv) next to the soil report: a sheet *Soil data* with an example borehole and a *Read me* sheet. Fill it in and import it with **Import CSV/Excel...**.

- Accepted files: **.csv / .txt** and **.xlsx / .xlsm**. (The file dialog also lists .xls, but old .xls files are refused: save them as .xlsx first.)
- One flat table, one record per row. Column `Borehole` is required; `Record` says whether a row is a `layer`, `spt` or `sample` then Top, Bottom, Depth, N (or Seat/N1/N2) and the property columns. An Excel workbook is read from the sheet *Soil data*, else the first sheet.
- Lengths are in metres unless the header names mm, cm or ft; densities and pressures are converted from g/cc, t/m³, pcf, kg/cm², tsf, psi and other common units.
- **A borehole whose ID already exists is replaced** new IDs are added.

The Log view

Draws the borehole log: depth, strata hatching by soil type, descriptions, the water table ▽, and SPT N against depth with the corrected N60 points. **Show All Boreholes Side by Side** draws all logs on the project datum (elevations); click one to open it.

Results

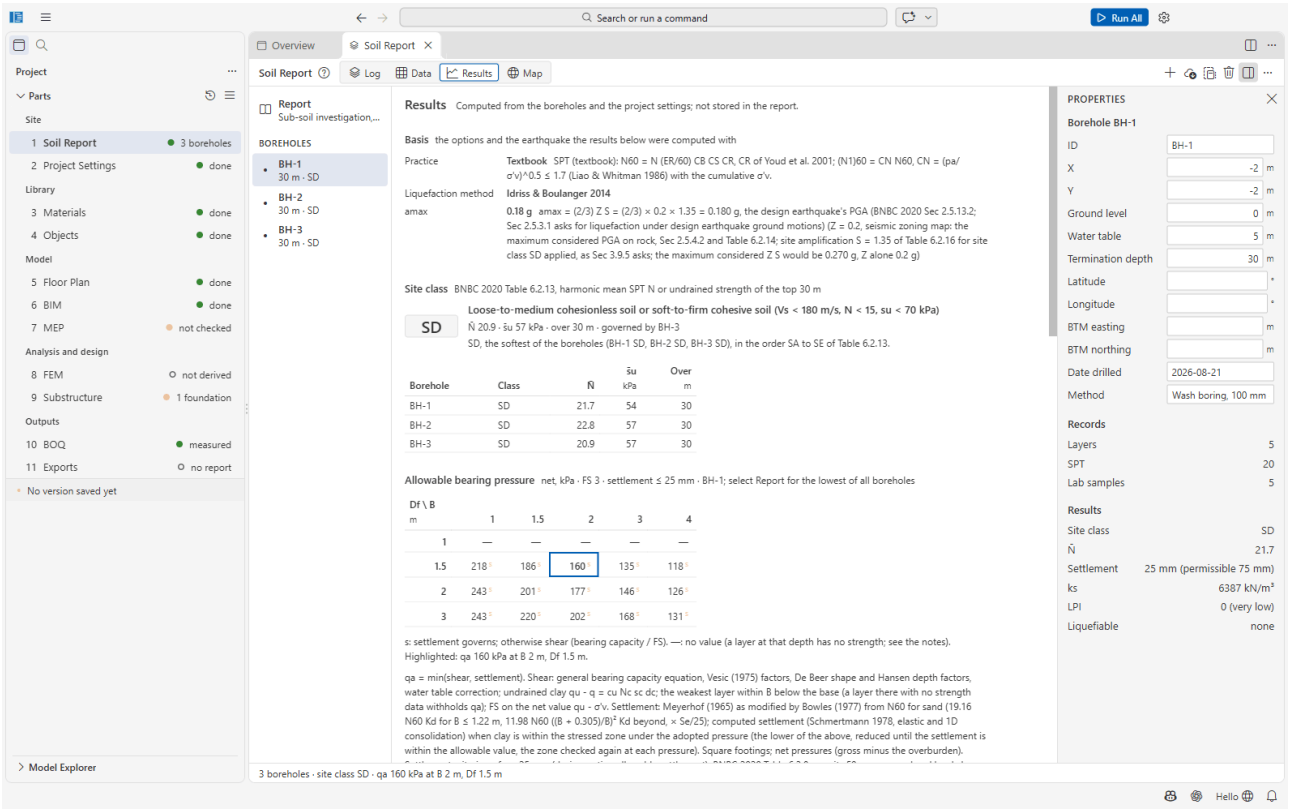


Figure 6: Results: site class and allowable bearing pressure

The Results view is computed from the boreholes and the project settings; **results are not stored in the report file**. Sections:

- **Basis:** the practice, the liquefaction method and the peak ground acceleration used, with its derivation.
- **Site class:** BNBC 2020 Table 6.2.13 from the harmonic mean SPT N (N̄) or undrained strength (s̄u) of the top 30 m, per borehole and for the site (the softest governs). Flags for S1/S2 soils are shown when they apply.

- **Allowable bearing pressure:** a table of net qa (kPa) against footing width B (1, 1.5, 2, 3, 4 m) and depth Df (1, 1.5, 2, 3 m), with the factor of safety and allowable settlement. A superscript *s* marks values governed by settlement; — means a layer without strength lies under the base. With *Report* selected the table shows the lowest of all boreholes, otherwise the selected borehole.
- **Settlement:** immediate, consolidation and total settlement of a reference footing, with the permissible value of BNBC Table 6.3.8.
- **Liquefaction screening:** factor of safety against depth, minimum FS, the liquefaction potential index LPI (Iwasaki) and the risk class, per borehole.
- **Modulus of subgrade reaction** ks (kN/m^3) per borehole.
- **Pile design parameters:** skin friction fs and end bearing qb per layer, with the method.
- **Notes**.

How the values are worked out

Quantity	Method
Allowable bearing qa	The lower of shear capacity (general bearing capacity equation with Vesic factors, De Beer shape and Hansen depth factors; undrained cu·Nc in clay; FS on the net value) and the pressure giving the allowable settlement (Meyerhof as modified by Bowles from N60 in sand; computed settlement where clay is in the stressed zone).
Settlement	Schmertmann (1978) in granular soil; elastic (Steinbrenner) plus 1D consolidation in clay.
Site class	BNBC 2020 Table 6.2.13, harmonic $\bar{N}$ or $\bar{s}u$ over 30 m.
Liquefaction	Idriss & Boulanger (2014), or BNBC 2020 Sec 3.9.5; LPI after Iwasaki et al. Saturated, non-clay-like soil above 20 m is screened.
Piles	Bored piles: O'Neill & Reese (1999). Driven piles: API RP 2A (clay) and Meyerhof (1976) (sand). Rock is not covered.
ks	qa divided by the settlement under qa, for a 2 m footing at 1.5 m.

SPT corrections

- **Textbook:** N60 = N · ER/60 · CB · CS · CR (Youd et al. 2001); (N1)60 with CN of Liao & Whitman (≤ 1.7) on the cumulative effective stress.
- **Bangladesh consultant practice:** effective stress at mid-depth of each SPT; CN = 0.77 log10(2000/σ′v) (Peck, Hanson & Thornburn); Skempton's rod-length factors; CB = CS = 1; the dilatancy correction N′ = 15 + ((N1)60 − 15)/2 below the water table when (N1)60 > 15 (Terzaghi & Peck). Bearing from settlement then uses N′.

The site map

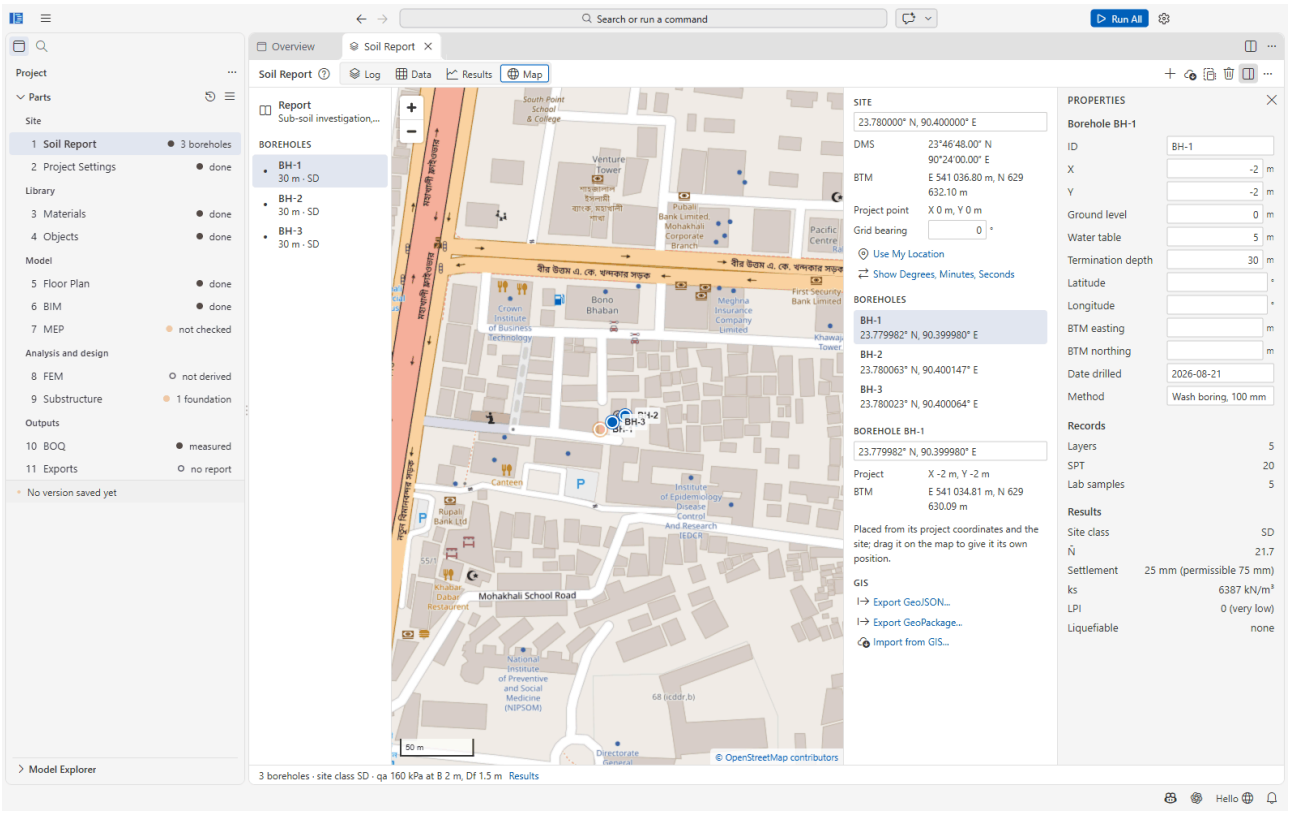


Figure 7: The Map view with the boreholes of the sample

The **Map** view shows the site on a street map (OpenStreetMap tiles by default; see the `saengine.map.*` settings). Offline, the boreholes are drawn on the project grid instead.

- **Site:** type the latitude/longitude of the project point (decimal, e.g. `23.8103, 90.4125`, or degrees-minutes-seconds), and the **Grid bearing** (°) between the project's X axis and east. The DMS and BTM (Bangladesh Transverse Mercator) coordinates are shown. **Use My Location** fills in the computer's location.
- **Boreholes:** a borehole placed by X/Y is drawn from the project point and bearing. **Drag a marker** to give it its own latitude/longitude, or type them under *Borehole BH-n*.
- **BNBC location:** when the nearest BNBC town's seismic zone or wind speed differ from those Project Settings use, a banner names them with **Apply to Project Settings**, which writes them into the site of Project Settings (save the settings to keep it).
- **GIS: Export GeoJSON...** and **Export GeoPackage...** write layers *site*, *site_boundary*, *boreholes* (with depth, water table, N̄ and site class) and *building* (the outline from 6 BIM) in WGS 84, by default into `11 Exports`. **Import from GIS...** reads points (boreholes) and a boundary polygon from .gpkg or .geojson. **Open in QGIS** appears when QGIS is installed (setting `saengine.qgis.program`). Export reads the saved file: save the soil report first.

Commands

Command (category SAEngine)	Does
Add Borehole to the Soil Report	Adds a borehole
Import Boreholes from CSV/Excel...	Imports a file
Save Borehole Template...	Writes the template

Not confirmed / limitations

- Old Excel **.xls** files are offered by the file dialog but refused by the importer.
- Some options exist only in the JSON file (open it with *Reopen Editor With... → Text Editor*): the SPT correction factors CB, CS and rod length, the pile type used for Part 1 pile parameters, the widths and depths of the bearing table, and sample permeability and chemistry.
- GeoPackage export has not been confirmed in QGIS by the developers; BNBC town coordinates are approximate.

Part 2: Project Settings

Project Settings hold the codes, occupancy, site, loads, load combinations and analysis options of the project. BNBC 2020 governs; every code value can be overridden. The floor plan, the building model and the analysis read these settings.

File: `02 Project Settings/project-settings.json`. Open it with **2 Project Settings** in the sidebar or *SAEngine: Project Settings*.

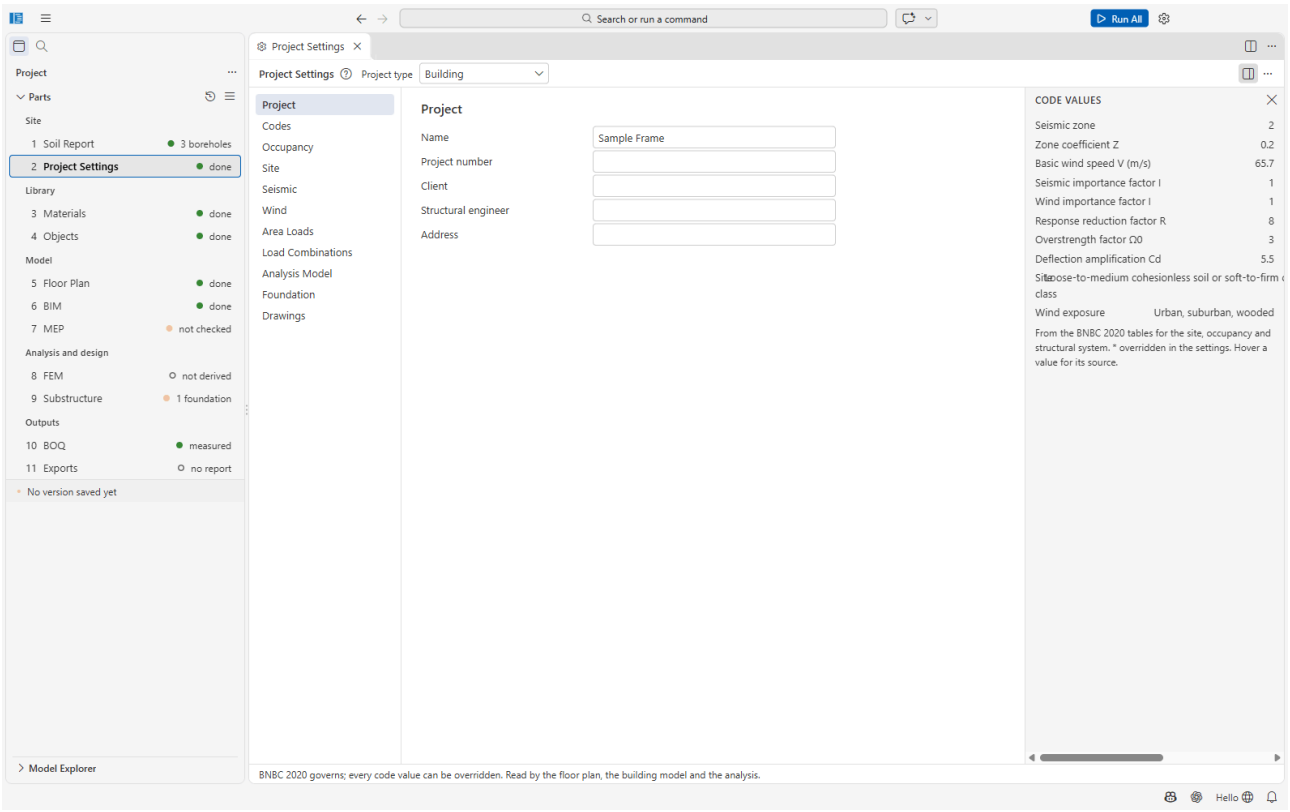


Figure 8: Project Settings with the Code Values panel

Layout

- **Toolbar:** the **Project type** (Building / Special Structure). Changing it asks *Make this project a …?* the old Part 5 folder is kept but no longer used. **Reset Combinations to BNBC 2020…** replaces the combinations (with confirmation; undoable).
- **Left:** the sections. **Middle:** the form of the section. **Right:** the **Code Values** panel: the values SAEngine takes from the BNBC tables for your site, occupancy and system. Values you override are marked *; hover a value for its source.

Sections and fields

Project

Name, Project number, Client, Structural engineer, Address. They appear in the report and in the drawing title block.

Codes

Field	Choices
Loads and general	BNBC 2020 (fixed)
Concrete	**BNBC 2020 Part 6 Chapter 6** (default) or ACI 318-19
Steel	**BNBC 2020 Part 6 Chapter 10** (default) or AISC 360-22
Design method	Strength design (LRFD), fixed

The design checks use the ACI 318-19 clauses and factors that BNBC 2020 Part 6 Chapter 6 follows.

Occupancy

- **Building use:** Residential (default), Office, Educational, Health care, Assembly, Retail, Industrial, Storage, Parking, Mixed use.
- **Occupancy category:** I to IV (BNBC Table 6.1.1), default II. It sets the importance factors and the allowable storey drift.

Site

- **Location:** one of ten BNBC towns, which give the seismic zone and the basic wind speed, or *Custom (enter Z and V)*:

Town	Zone	Z	V (m/s)
Dhaka	2	0.20	65.7
Chattogram	3	0.28	80.0
Cox's Bazar	3	0.28	80.0
Khulna	1	0.12	73.3
Rajshahi	1	0.12	49.2
Sylhet	4	0.36	61.1
Barishal	1	0.12	78.7
Rangpur	3	0.28	65.3
Mymensingh	4	0.36	67.4
Cumilla	2	0.20	61.4

- **Site class source:** *From the soil report* (default; the softest class of the boreholes, see Part 1) or *Entered*.
- **Site class:** SA to SE (default SD), used when entered or when there is no soil report.
- **Seismic zone coefficient Z** and **Basic wind speed**: optional overrides.

The site can also be filled from the soil report's map (**Apply to Project Settings**).

Seismic

Structural system (BNBC Table 6.2.19), which sets R, Ω0 and Cd:

System	R	Ω0	Cd
RC special moment frame	8	3	5.5
RC intermediate moment frame (default)	5	3	4.5
RC ordinary moment frame	3	3	2.5
Building frame with RC special shear walls	6	2.5	5
Building frame with RC ordinary shear walls	5	2.5	4.25
Dual: RC special moment frame and special shear walls	7	2.5	5.5
Steel special moment frame	8	3	5.5
Steel ordinary moment frame	3.5	3	3
Steel special concentrically braced frame	6	2	5

R, Ω0 and Cd can each be overridden.

Wind

Exposure A (urban, suburban, wooded; default), B or C (BNBC 6.2.9.3).

Area Loads

The floor loads of the project, in kN/m². Each area load has an ID, a name, a **use** (BNBC Table 6.2.3, which sets the minimum live load), **live**, **Floor finish** and **partitions**. **Floors** and **Roof** choose which area load applies to every floor and to the roof.

Default	Use	Live	Finish	Partitions
FL Floors	Residential	2.0	1.2	1.0
ST Stairs	Stairs	4.8	1.2	—
RF Roof	Roof	1.0	1.5	—

A live load below the BNBC minimum for its use is flagged (*FL: live load 1.5 kN/m² is below the BNBC 2020 minimum of 2.0 kN/m² for residential*).

> **Where loads come from.** Self-weight is computed from the object types and material densities. Floor finish and partitions become the superimposed dead case SD, the live load the case L (Lr on the roof). Non-structural walls become line loads on the beams under them. MEP equipment and tanks add their weights (Part 7). The floor plan editor has **no per-slab load entry**: one area load applies to all floors and one to the roof. A slab can carry its own area load (`areaLoad` in `floor-plan.json`, written by the importers when a model's slabs have different loads); there is no field for it in the editor in this version.

Load Combinations

Each combination has a name and the factors of the cases **D, L, Lr, WX, WY, EX, EY, T, F** (D also applies to SD; F follows D when left empty). New projects get the BNBC 2020 strength combinations (Part 6, 2.7.3), expanded to ±X and ±Y for wind and earthquake:

- 1.4D
- 1.2D + 1.6L + 0.5Lr
- 1.2D + 1.6Lr + 1.0L
- 1.2D + 1.6Lr + 0.8W
- 1.2D + 1.6W + 1.0L + 0.5Lr
- 1.2D + 1.0E + 1.0L
- 0.9D + 1.6W
- 0.9D + 1.0E

Combinations are formed from the case results (exact for the linear analysis). The service combinations used for foundation bearing are built in by 9 Substructure.

Analysis Model

Field	Default	Meaning
Slabs	Loads to beams	Slab loads go to the beams by the 45° rule; *Shells* models the slabs as shell elements
Floor diaphragm	Rigid	or flexible
Mesh size	1000 mm	for shells
Response spectrum	On	adds the response-spectrum cases RX, RY (BNBC design spectrum, CQC, 5 % damping)
Section stiffness	Cracked (ACI) for lateral cases	
Second-order analysis	None (moment magnification)	or P-Delta analysis
Earthquake design forces	Response spectrum, scaled (BNBC 2.5.9.4)	or Equivalent static (EX, EY)
Orthogonal effects (100 % + 30 %)	Auto	by Seismic Design Category
Accidental torsion	On	±5 %
Vertical earthquake component	On	
Redundancy factor ρ (drift)	1.0	
Construction sequence, time histories, pushover, temperature	Off	advanced analyses
Column supports	Fixed	or **Soil springs** from the soil report and 9 Substructure

Foundation

Allowable bearing override qa (kPa; empty: from the soil report), **Soil unit weight γ** (18 kN/m³) and **Ground level** (0 mm).

Drawings

Firm, Engineer, Drawn by, Checked by, Approved by, Project number, status (default *FOR REVIEW*), revisions, **Sheet size** (A1, A2, A3; default A1), scales (plans 1:100, slab plans 1:50, elevations 1:50, details 1:25), development length (*simplified*), **Cut off beam top bars** (on) and **BOQ reinforcement** (*Measured by the BOQ's rules* or *From the bar bending schedule*).

Code Values panel

Seismic zone, Z (Table 6.2.14), basic wind speed V (Table 6.2.8), seismic and wind importance factors (Tables 6.2.17 and 6.2.9), R, Ω0, Cd (Table 6.2.19), the site class (Table 6.2.13, *from 1 Soil Report* when used) and the wind exposure.

Occupancy category	Seismic I	Wind I
I	1.00	0.87
II	1.00	1.00
III	1.25	1.15
IV	1.50	1.15

Not confirmed

- The section names are shown as rendered in the app (*Project, Codes, Occupancy, Site, Seismic, Wind, Area Loads, Load Combinations, Analysis Model, Foundation, Drawings*); some field labels inside the advanced analysis options were read from the schema and may be worded slightly differently on screen.
- Per-slab area loads (*Floors* applies "unless the floor plan says otherwise") can only be set by the importers or in the JSON file; the floor plan editor shows no field for them.

Part 3: Materials

The materials library holds the physical materials of the project: concrete, reinforcement, structural steel, masonry, mortar and others. Object types (Part 4) refer to them; the analysis takes E, ν and the density from them, the design f′c and fy, and the bill of quantities the concrete and mortar mixes.

File: `03 Materials/materials.json`. Open it with **3 Materials**.

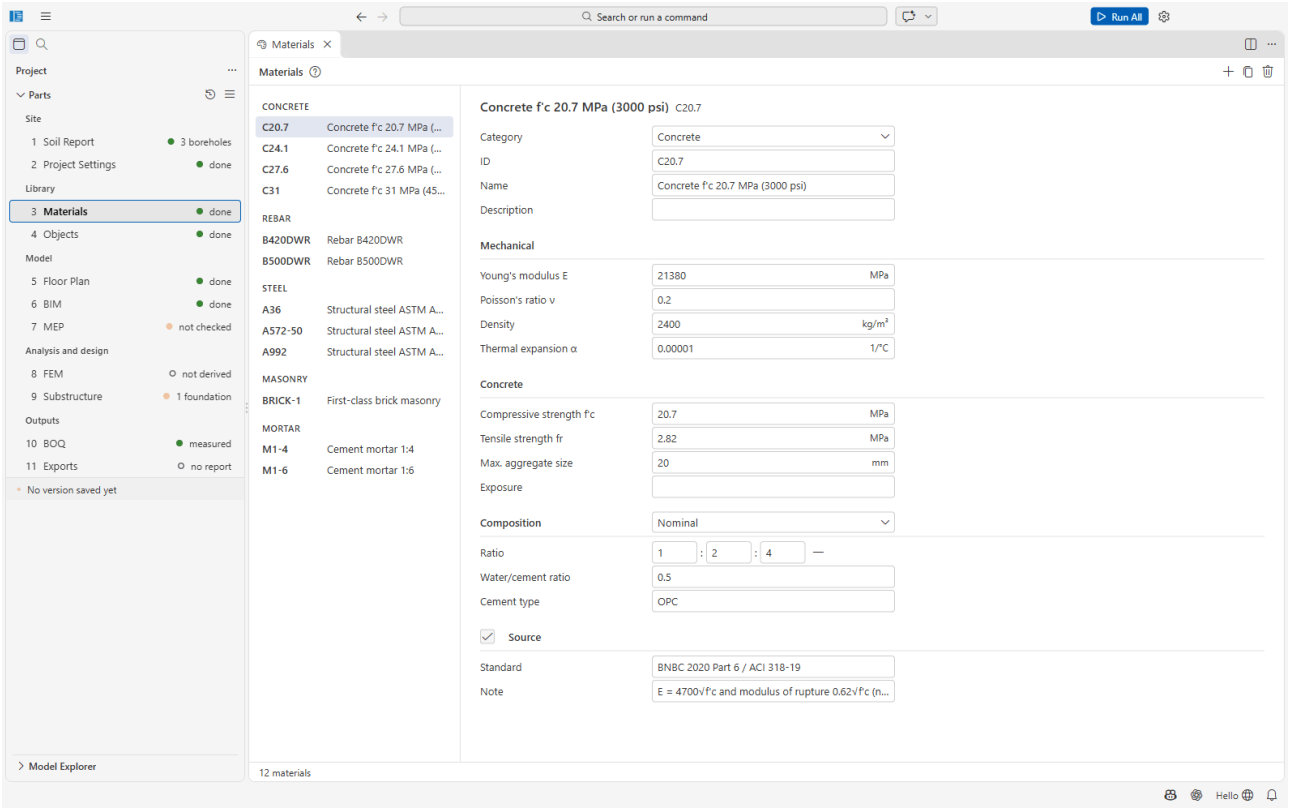


Figure 9: The Materials editor

Using the editor

- **Left:** the materials grouped by category. Up/Down arrows move through them; right-click for **Duplicate** and **Delete**.
- **Right:** the form of the selected material, with units, dropdowns and per-field validation. Only valid values are written to the file; each change can be undone with Ctrl+Z.
- **Toolbar:** **+** (**Add Material**, by category), **Duplicate** (creates *name (copy)* with a new ID), **Delete** (asks *Delete 'name'?* Ctrl+Z restores it).
- The status line counts the materials and the problems to fix, with a **Show** link to the first problem.

Fields

Group	Fields
General	Category (Concrete, Rebar, Steel, Masonry, Mortar, Other), ID (letters, digits, `.` `_` `-`), Name, Description
Mechanical	Young's modulus E (MPa), Poisson's ratio ν, Density (kg/m³), Thermal expansion α (1/°C)
Concrete	Compressive strength f′c, tensile strength fr (MPa), max. aggregate size (mm), exposure
Rebar / Steel	Grade, fy, fu (MPa), elongation, actual fy and fu from mill tests
Masonry	Unit, unit strength, f′m, mortar (a mortar material)
Composition (concrete and mortar)	*None*, *Nominal* (volume ratio such as 1:1.5:3, water/cement ratio, cement type) or *Design* (cement, water, fine and coarse aggregate in kg/m³, admixtures)
Source	Standard, note

Rules: concrete, rebar and steel need E; fu must not be below fy; a nominal mix has three parts for concrete and two for mortar; IDs are unique; a masonry mortar must exist.

> The **composition** is what the bill of quantities uses to work out cement (kg and 50 kg bags), sand and stone chips (cft). A concrete without a mix gets no cement/sand/stone take-off.

Default materials of a new project

ID	Material	Main values
C20.7	Concrete f'c 20.7 MPa (3000 psi)	E 21 380 MPa, nominal mix 1:2:4, w/c 0.5
C24.1	Concrete f'c 24.1 MPa (3500 psi)	E 23 070 MPa, nominal mix 1:1.5:3, w/c 0.45
C27.6	Concrete f'c 27.6 MPa (4000 psi)	E 24 690 MPa
C31	Concrete f'c 31 MPa (4500 psi)	E 26 170 MPa
B420DWR	Rebar B420DWR	fy 420 MPa, BDS ISO 6935-2:2016
B500DWR	Rebar B500DWR	fy 500 MPa
A36	Structural steel ASTM A36	fy 250, fu 400 MPa
A572-50	Structural steel ASTM A572 Grade 50	fy 345, fu 450 MPa
A992	Structural steel ASTM A992	fy 345, fu 450 MPa
BRICK-1	First-class brick masonry	density 1920 kg/m³, mortar M1-4
M1-4, M1-6	Cement mortar 1:4 and 1:6	

For all concrete: E = 4700√f'c, fr = 0.62√f'c, density 2400 kg/m³, ν = 0.2. Each value records its standard or code formula in *Source*.

A new material copies the first default of its category.

Part 4: Objects (Object Types)

Object types are the sections of the project, what ETABS calls *frame sections* and *shell properties*: a column type, a beam type, a slab type, with their material, profile and reinforcement. Every member on the floor plan refers to one object type. **To change the size of a member, change its object type here** (or assign another type to the member on the floor plan).

File: `04 Objects/objects.json`. Open it with **4 Objects** (also **Open Object Types** in the floor plan and the design proposal).

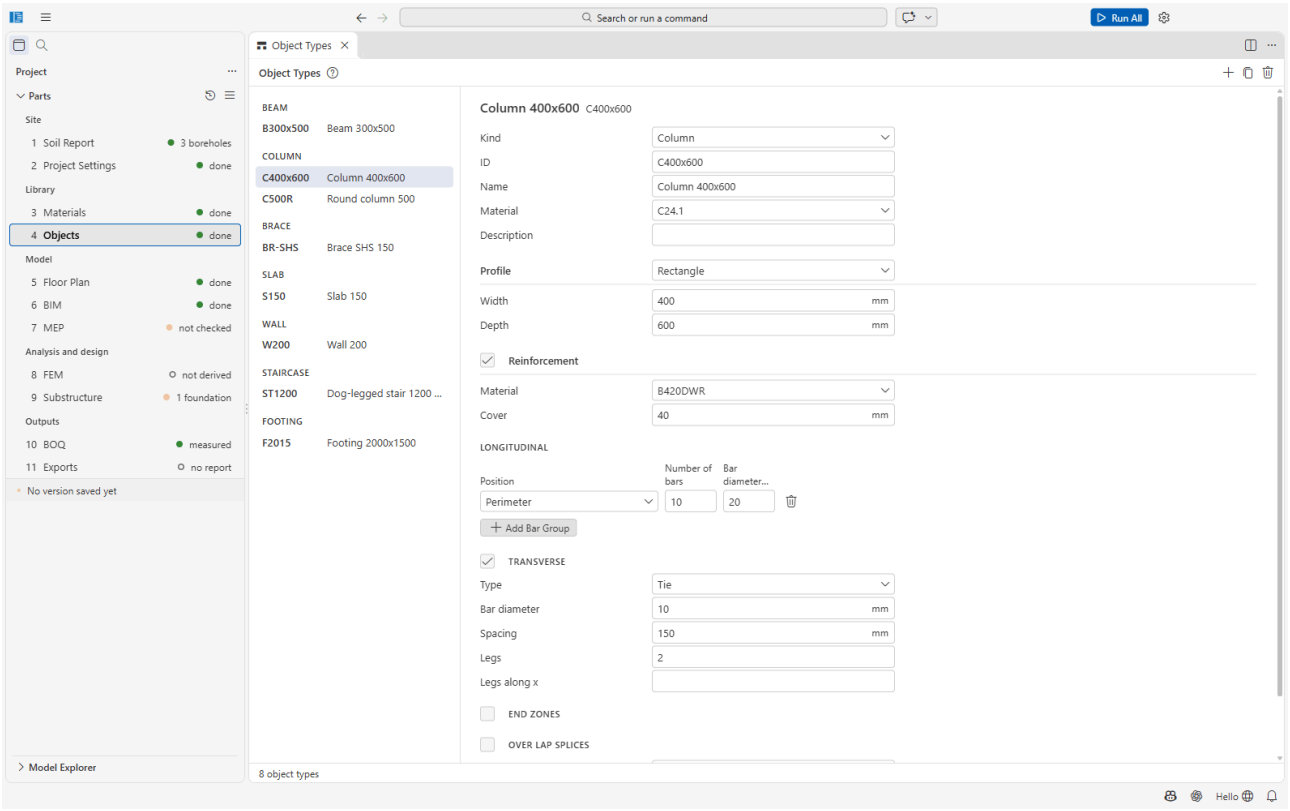


Figure 10: The Object Types editor with a column type

The editor works like the Materials editor: a list grouped by kind on the left, a form on the right, **+** (**Add Object Type**, by kind), **Duplicate** and **Delete**.

Kinds

Kind	Main fields	New-type default
Beam	Material, profile, reinforcement, seismic role, Ω0 combinations; steel beams: fin-plate or end-plate connection	B, 250×450, C24.1
Column	Material, profile, reinforcement, seismic role	C, 400×400
Brace	Material, profile, gusset connection	BR, L75×75×6, A36
Slab	Thickness, system (one-way, two-way, flat), planar reinforcement, punching reinforcement	S, 150 mm two-way
Wall	Thickness, structural or not, reinforcement (incl. boundary elements), braced, restrained ends, basement	W, 200 mm
Staircase	Waist, riser, tread, flight width, arrangement (straight or dog-legged), landings, supports, headroom	ST, 150/150/250/1200
Footing	Isolated, combined or strip; length, width, thickness	F, 2000×2000×450
Pile cap	Size, the pile type it uses	PC, 2400×2400×900
Pile	Bored, driven or precast; diameter, length	P, bored, 600 mm, 18 m
Solid	Material only	MASS

Profiles follow IFC: Rectangle, Circle, Rectangular hollow (RHS), Circular hollow (CHS), I-section, T-section, Angle (L), Channel (U). Section properties are computed from the profile.

Reinforcement

For beams and columns:

- **Material** (a rebar) and **Cover** (mm).
- **Longitudinal** bar groups: position (Top, Bottom, Sides, Perimeter), number of bars, bar diameter. **Add Bar Group** adds a row.
- **Transverse**: Stirrup, Tie or Spiral; diameter, spacing, legs.
- **End zones** and **Lap splices**.

For slabs and walls: planar meshes (bar and spacing per face and direction).

You do not have to enter bars before the analysis: a type without bars is checked for the **steel it requires**, and **Design Reinforcement** in 8 FEM proposes bars and writes them here (Part 8).

Seismic role

Auto, *Lateral* (part of the seismic force-resisting system) or *Gravity* and **Ω0 combinations** (*Auto*, *Transfer*, *Collector*, *None*) for members that need the overstrength combinations.

Rules

IDs are unique; the material must exist; reinforcement must be a rebar material; a pile cap's pile type must exist. A new project starts with no object types (New Building from Grid and the importers create them).

Not confirmed

- The pile template radius: the template stores r = 300 mm (600 mm diameter); the Substructure editor's default pile type is 500 mm (see Part 9).

Part 5: Floor Plan

The floor plan is where a building is laid out: grids, storeys, and the columns, beams, walls and slabs on each storey. It is drawn on the grid, storey by storey, like a structural framing plan. The 3D building model (Part 6) and everything after it are generated from it.

File: `05 Floor Plan/floor-plan.json`. Open it with **5 Floor Plan**. (A *special structure* project has a 3D **Structural Model** in Part 5 instead; it is not covered by this manual.)

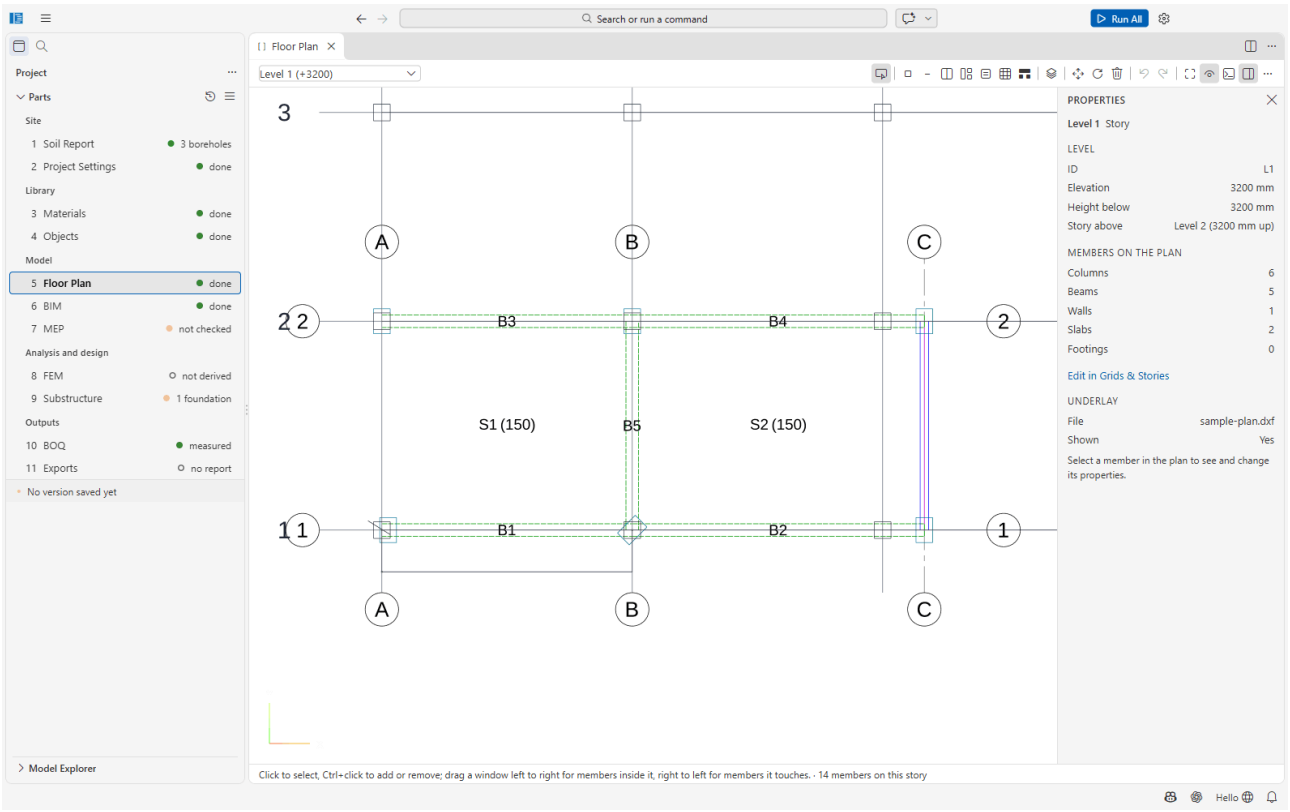


Figure 11: The floor plan of Level 1 of the sample, with the architect's underlay

The screen

- **Storey picker** (top left): one storey at a time, shown as *Level 1 (+3200)*. A storey that repeats a typical floor shows = *Level 2* the typical floor itself shows *(typical)*.
- **Toolbar** (top right), in order: **Select Column**, **Beam**, **Wall**, **Slab**, **Polygon Slab**, **Fill Place Footings in 9 Substructure Typical Floors… Move**, **Rotate**, **Delete Undo**, **Redo Zoom Extents**, **Show/Hide Underlay**, **Show/Hide Command Line (:)** and the Properties panel. On a narrow window some of these move into the **…** menu.
- **… menu:** Select All on Story (Ctrl+A), Select by Type…, Copy Selection to Stories…, Mirror Selection…, Delete Selection, Members Table, Edit Grids and Stories, Open Object Types, Attach/Replace/Remove Underlay, Export DXF.
- **Status line:** a hint for the current tool and the member count of the storey. The cursor position is shown as *x* …, *y* … *mm*.
- **Properties panel:** the storey (nothing selected), one member, or several members.

All positions are in **mm**. Every edit is one undoable change of `floor-plan.json` (Ctrl+Z / Ctrl+Y, or the toolbar). Save with Ctrl+S.

Grids and storeys

Grids and storeys are not drawn on the plan. Choose **Edit Grids and Stories** (or *Edit in Grids & Stories* in the Properties panel): a form opens beside the plan and the plan follows your edits as you type.

- **X lines** (vertical on the plan) are labelled A, B, C, …; **Y lines** 1, 2, 3, …; labels must be unique.
- A **storey** has an ID, a name and an **elevation** (top of floor, mm). New storeys are named *Level n*. The lowest storey is the base: columns and walls rise *to* a storey *from the one below*, so nothing is placed on the base storey except footings (which belong to Part 9).

- Members are stored relative to the nearest grid intersection, so they follow the grid when you move a grid line.

Placing members

Pick the object type in the **Type** picker that appears next to the tools (types come from Part 4 if there is none of that kind: *Define a column type in Object Types first.*).

Tool	How
Column	Click a grid intersection. The column rises from the storey below to this storey. A second column at the same place is refused.
Beam / **Wall**	Click two intersections. Placement chains from the last point until Esc or a right-click. Walls also rise from the storey below.
Slab	Click inside a bay (the rectangle between neighbouring grid lines).
Polygon Slab	Click the corners (they snap to intersections and grid lines); finish on the first corner, with Enter or a right-click (at least three corners).
Move	Click a member, then click the new place, or type the displacement *Δx, Δy* in mm (e.g. `150, 0`) and press **Move**.
Rotate	Columns and footings only: click one; type an angle and press **Rotate**, or **Turn 90°**. Beams, walls and slabs follow their points.
Delete	Click a member on this storey.

Members get automatic IDs: C1, C2… for columns, B for beams, W for walls, S for slabs, F for footings.

Selecting members

Gesture	Selects
Click	one member
Ctrl+click	adds or removes one member
Shift+click	adds one member
Drag left to right	a **window** (solid outline): the members **fully inside** it
Drag right to left	a **crossing window** (dashed outline): every member it **touches**
Ctrl or Shift while dragging	adds the window's members to the selection
Ctrl+A	all members of the storey
Esc	cancels the current gesture, then clears the selection, then returns to Select

The status line confirms *Window: 6 member(s) fully inside.* or *Crossing window: 9 member(s) touched.* The selection is shared with the Model Explorer and the 3D view; selecting a member there switches the plan to its storey.

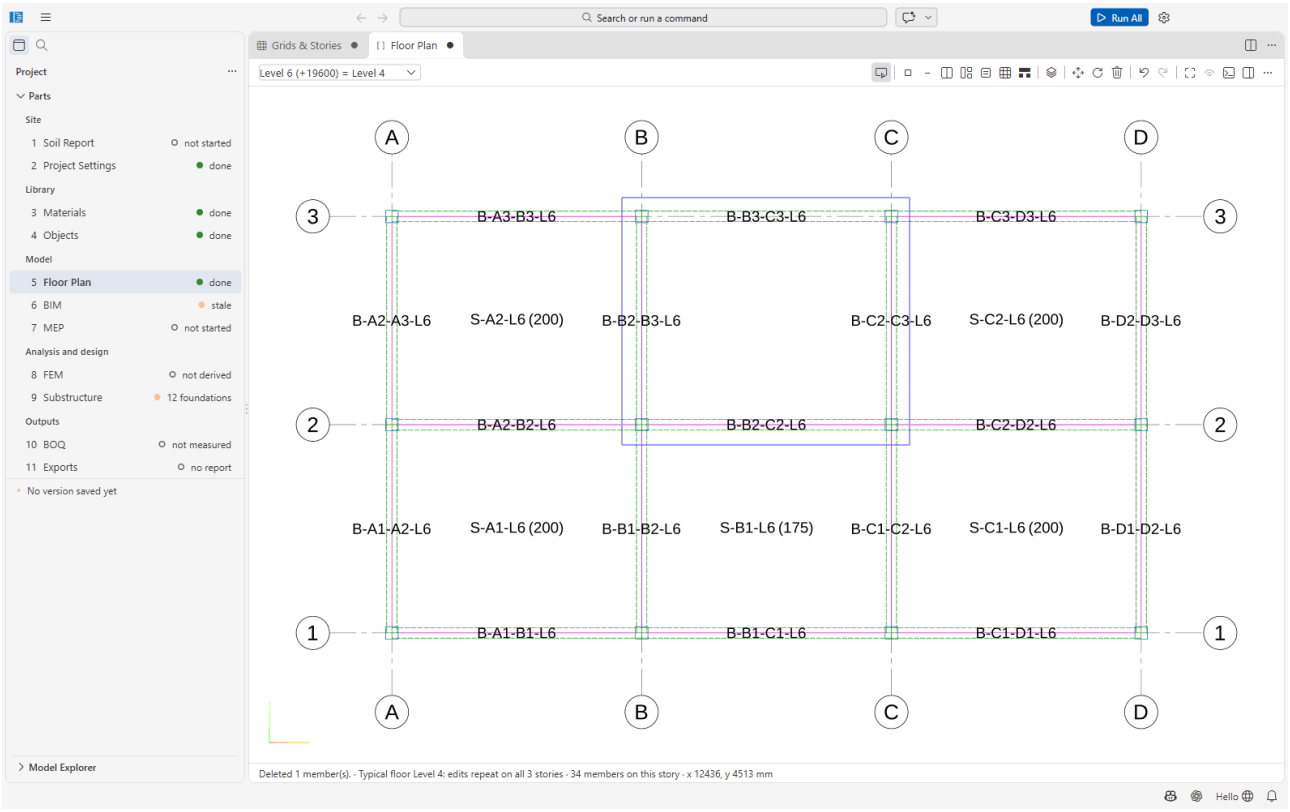


Figure 12: A window selection, left to right

With members selected, the Properties panel lets you change, for all of them at once: the **Type** (per kind: *Type of columns*, *Type of beams*…), the **Rotation (°)**, **Move by (mm)** and **Copy by (mm), times** (e.g. `6000, 0, 3` makes three copies 6 m apart). Press Enter to apply. Links: **Mirror…**, **Copy to Stories…**, **Delete n Members**.

Fill: many members at once

Fill places members on the grid inside a window you drag:

Fill choice	Places
Columns at Intersections	a column at every grid intersection
Beams on Grid Segments / Beams along X / Beams along Y	a beam on every grid segment
Slabs in Bays	a slab in every whole bay
Walls on Grid Segments / Walls along X / Walls along Y	a wall on every segment

Choose the type in **Type**, the kind in **Fill**, then drag a window over the grid. Places that are taken are skipped: *Placed 12, skipped 3 existing.* The new members become the selection. Fill works on the grid only; it does not look for beams around a bay.

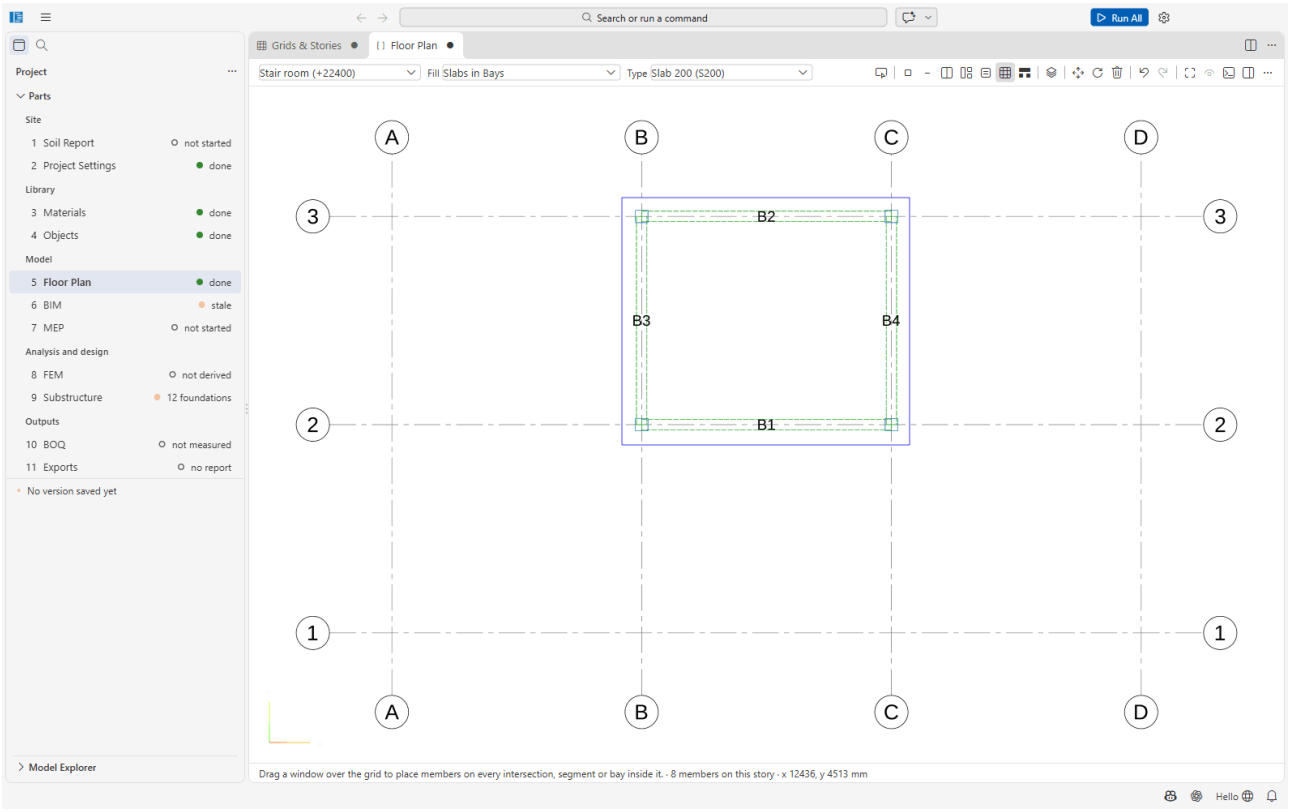


Figure 13: Fill: slabs in every bay of the window

Copying to other storeys and typical floors

- **Copy Selection to Stories…** copies the selection to the storeys you tick (*Copied 14 member(s) to Level 2, Level 3; skipped 0 already there.*). Columns and walls rise from the storey below each target. Footings are not copied.
- **Ctrl+C / Ctrl+V:** copy the selection and paste it at a base point (the lower-left corner of the copied members), on this storey or another.
- **Typical Floors…** (three or more storeys): first pick the typical floor, then the storeys that repeat it. Their plans are replaced by the typical floor's plan, and from then on **every edit on any storey of the family is made on all of them** (*Typical floor Level 2: edits repeat on all 8 stories*). If the storeys of a family differ (for example after editing the file), the status line offers **Copy the typical floor to them** edits are refused until then.
- **Mirror Selection…** mirrors about a grid line; the originals are kept.

The command line

Press **:** (or **Show Command Line**) for a command line. Grid points are written `A-1` (or `A/1`).

Command	Does
`select all`, `select none`, `select type C400x600`, `select C1 C2`	selects
`fill columns`, `fill beams x`, `fill slabs A-1 C-3`	fills the whole storey, or a window between two grid points
`copy 6000,0 3`	copies the selection 3 times, 6 m apart
`array 0,5000 4`	the same, 1 to 100 copies
`move 150,0`	moves the selection
`mirror x B`, `mirror B move`	mirrors about grid line B (`move` drops the originals)
`rotate 90`, `rotate to 45`	rotates selected columns and footings
`tostories all`, `tostories L2 L3`	copies the selection to storeys
`delete`	deletes the selection

Moved or copied members that would land on another member or off the grid stay where they were (*Moved 4, kept 1 in place*).

Underlay (architect's drawing)

Attach Underlay... shows a DXF drawing greyed under the plan of the current storey. A DXF from outside the project is copied into `05 Floor Plan/Underlays/`. **Hide/Show Underlay** toggles it. The underlay's offset, scale and rotation can only be changed in the JSON file.

Members Table

Members Table (in **...**) lists every member: ID, Kind, Object Type, Story, Location, with filters and a search box (*Find ID, type, grid*). Select rows (Ctrl/Shift/Ctrl+A) to change the type or storey of all of them; **Export CSV...** writes the table.

Export DXF

Export DXF writes the plan of every storey to `11 Exports/Floor Plan/<id> <name>.dxf`, with layers for the grid, columns, beams (dashed), walls, slab edges, footings and labels.

Footings

There is no footing tool on the plan: **Place Footings in 9 Substructure** opens Part 9, where foundations are placed and designed. Footings already in the plan are shown and can be selected, moved and rotated.

Loads

The floor plan editor has **no load entries**. Floor and roof loads are the area loads of Project Settings self-weight comes from the object types. A slab may name its own area load (`areaLoad` in the file, written by the importers); the editor does not show or edit it in this version.

New Building from Grid

New Building from Grid... (start page, ≡ menu, or File menu) creates a complete new RC building project in one step, with preliminary member sizes you edit afterwards.

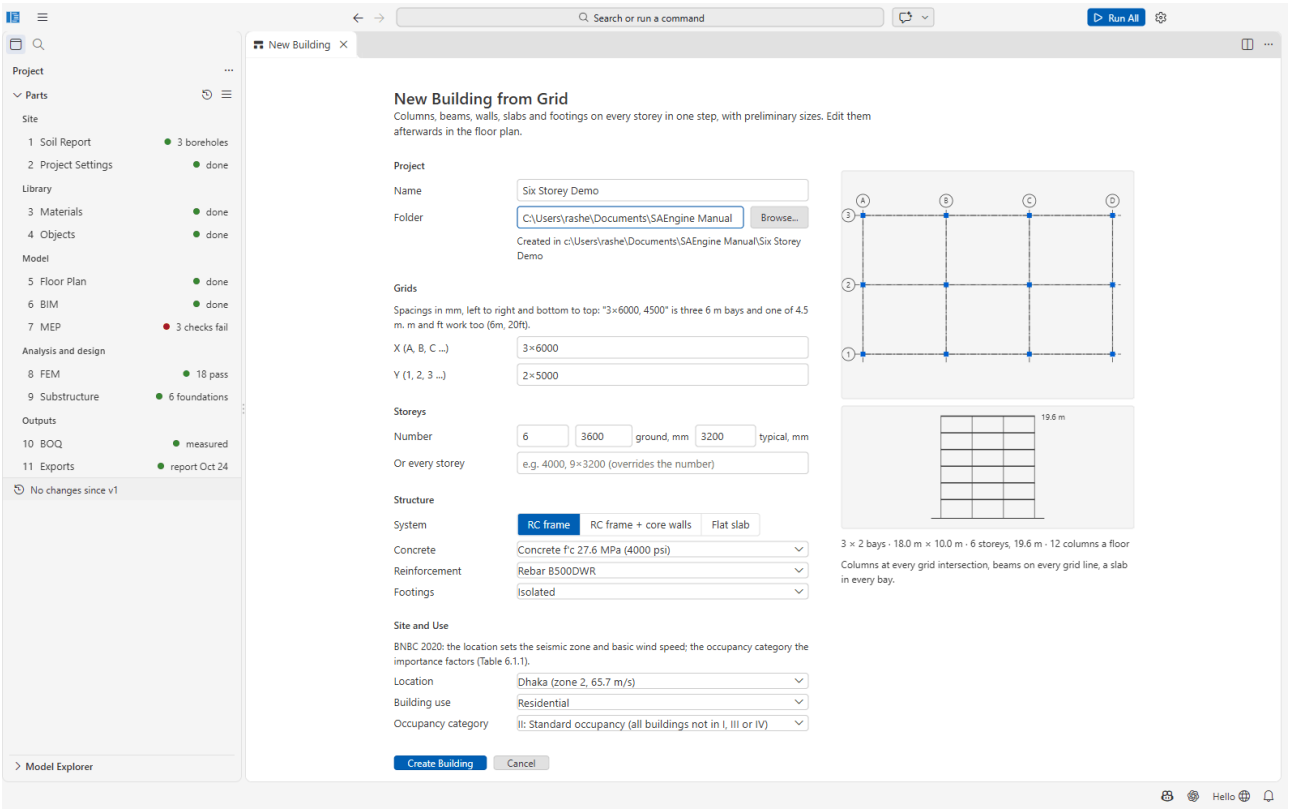


Figure 14: The New Building wizard

Input	Default	Notes
Name, Folder		the new project folder

Input	Default	Notes
X (A, B, C ...)	`3×6000`	bay widths; shorthand like `3×6000, 4500` units mm, cm, m, ft; 1 to 30 m per bay, at most 60 bays
Y (1, 2, 3 ...)	`2×5000`	
Number of storeys	6	at most 80
ground, mm / typical, mm	3600 / 3200	storey heights (2 to 12 m)
Or every storey		e.g. `4000, 9×3200` overrides the number
System	RC frame	*RC frame*, *RC frame + core walls*, *Flat slab*
Concrete	C27.6	
Reinforcement	B500DWR	
Footings	Isolated	*Isolated*, *Mat (raft)*, *None (add in 09 Substructure)*
Location	Dhaka	with zone and wind speed
Building use, Occupancy category	Residential, II	

A live plan and elevation preview shows the bays, storeys and the number of columns per floor. **Create Building** writes the project: default materials; column, beam, slab, wall and footing types **without bars** the floor plan with typical-floor families; the building model; footings (or a mat) in 9 Substructure; and Project Settings with the occupancy, live load, site and the structural system the seismic design category requires. A page *<name> Is Ready* shows the **Generator Notes** (the sizing rules used) with **Open Building**. The sizes are preliminary and the allowable bearing pressure defaults to 100 kPa until you enter a soil report. See also docs/new-building.md.

Not confirmed

- The underlay's offset, scale and rotation have no controls in the editor.
- `docs/new-building.md` calls the table command *SAEngine: Members Table* in the app it is in the category *Floor Plan*.

Part 6: BIM (Building Model)

The building model is the 3D model of the building: every member of the floor plan as an element with its storey, section and material, plus what the floor plan cannot show (openings, stairs, inclined members). It is **generated from the floor plan** you do not draw members here. The analysis model, the bill of quantities, the MEP coordination and the IFC export are made from it.

File: `06 BIM/bim.json`. Open it with **6 BIM**. The editor is titled **Building Model**.

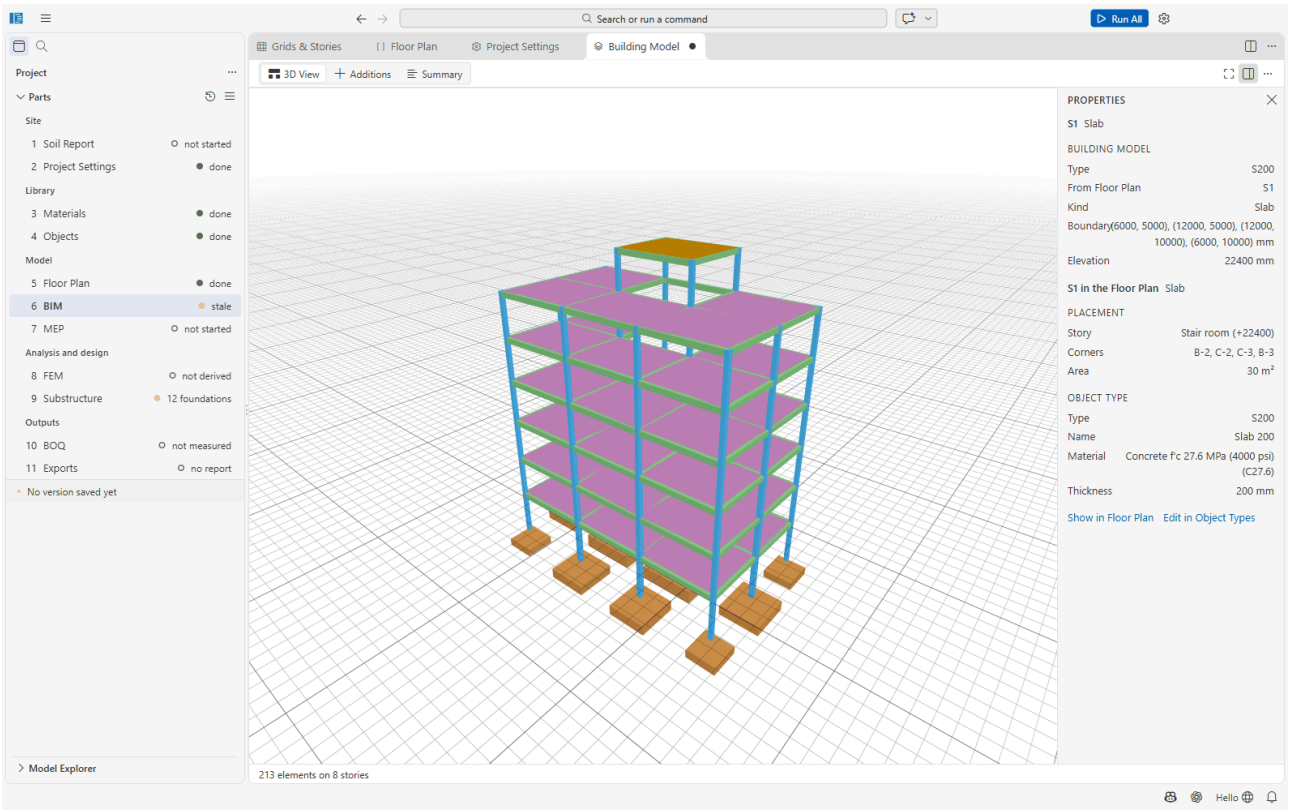


Figure 15: The building model of the six-storey tutorial building in 3D

Building and rebuilding

- **Build from Floor Plan** builds the model the first time; afterwards the button reads **Rebuild from Floor Plan**. It is highlighted when the model is out of date: the status line then says *the Floor Plan changed since it was built* with a **Rebuild** link.
- **Rebuild after every floor-plan edit**, before Run All: Run All derives the analysis model from the building model, not from the floor plan.
- Rebuilding keeps the *Additions* (below).
- If the floor plan is empty: *The Floor Plan has no layout yet. Open it and create it first.*

Views

View	Shows
3D View	The model in 3D, coloured by element class like the floor plan. Drag to orbit, and pan and zoom with the mouse (the viewer's standard camera); a click without dragging selects an element. **Zoom to Fit** frames the model.
Additions	What the building model adds to the floor plan: slab openings (O..), wall openings and doors (D..), stairs between storeys (ST..), and inclined members such as braces, sloped beams or raking columns (M..). They are edited in a form and kept when you rebuild; openings follow their slab or wall.
Summary	The storeys (ID, elevation, height) and the number of elements by kind, from the floor plan and own.

Clicking an element selects the floor-plan member it came from (also in the Model Explorer and the floor plan) and shows its properties.

The 3D view is drawn from the same IFC file that **Export IFC** writes, so what you see is what is exported.

Export IFC

Export IFC (in **…**, or on the Exports page) writes `11 Exports/IFC/<project>.ifc` with the elements, storeys and materials, for other BIM software.

Not confirmed

- The exact mouse buttons for orbit and pan are the defaults of the 3D library (That Open Components camera controls) and are not set by SAEngine.
- The 3D view has no section, measure or hide tools in this version.

Part 7: MEP

Part 7 covers the building services **as they affect the structure**: equipment that loads the floors (water tanks, chillers, generators…), sleeves and openings through beams, slabs and walls, vertical shafts, and the plumbing, HVAC and electrical design a structural engineer needs (water demand and tank sizes, RC tank design, plant weights, duct and tray openings).

File: `07 MEP/mep.json`. Open it with **7 MEP**.

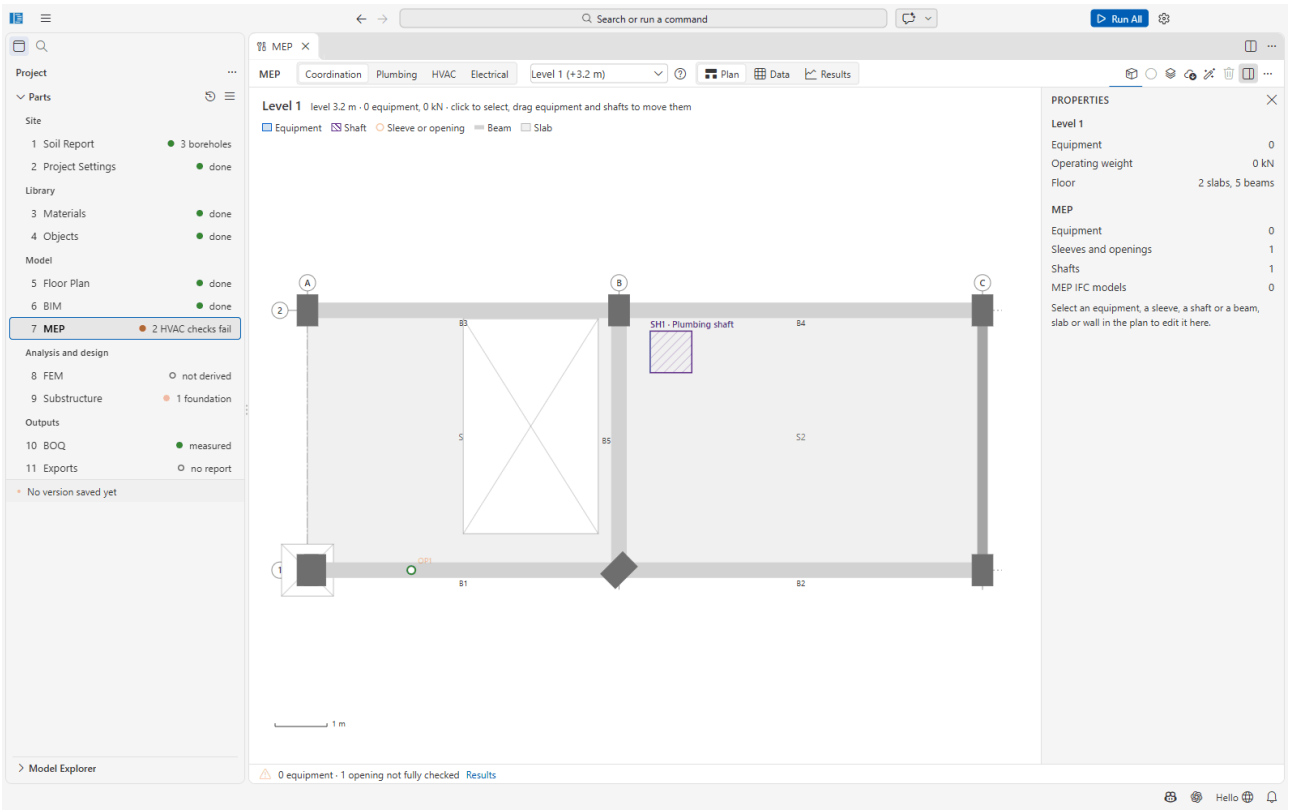


Figure 16: The MEP coordination plan of the sample

Layout

- **Discipline tabs: Coordination**, **Plumbing**, **HVAC**, **Electrical**.
- **Storey picker** and **views: Plan**, **Data**, **Results**.
- The **Plan** draws the storey from the building model (grid, slabs with openings, beams, columns, walls) with the MEP items. Click to select; drag equipment, risers, tanks, plant and shafts to move them. **Delete** removes the selected item, **Esc** deselects.
- Plan positions are in **m**, sizes in **mm**, weights in **kN**.
- The checks are computed by the backend about a second after each edit. Results are not stored in the file.

Coordination

Toolbar: **Add Equipment** (presets: *Overhead water tank, Chiller, Air handling unit, Generator, Transformer, Lift machine room, Pump*), **Add Sleeve through the Selected Beam, Slab or Wall**, **Add Shaft**, **Import MEP IFC Model**, **Suggest Sleeves from Clashes**, **Delete**.

- **Equipment:** operating weight and empty weight (kN), plinth height, length/width/height (mm), position X/Y (m), rotation. Replace the preset values with the supplier's data.
- **Sleeves and openings:** select a beam, slab or wall in the plan, then add a sleeve (circle) or opening (rectangle). In a beam the position is *From start* and *Above bottom* in a wall *Along* and *Up*. Columns cannot take openings.
- **Import MEP IFC Model:** reads the IFC of a services model (copied to `07 MEP/IFC/`), finds where pipes and ducts cross the structure and lists the **clashes**. A clash with a large enough sleeve or a shaft counts as coordinated; otherwise SAEngine proposes a sleeve, which **Suggest Sleeves from Clashes** adds. Clashes with columns or foundations get no sleeve: *reroute the service*.

Results (*Loads, openings and clashes*): equipment loads by storey, operating weights, the **opening checks** (beam sleeves against the h/3 and 3-diameter limits of practice; small openings by the net-section method, ACI 318-19 shear; large openings flagged as needing a detailed design, which SAEngine does not do), and the clashes.

How MEP loads the structure: equipment weight goes to the superimposed dead case SD; the contents (operating minus empty weight) go to the fluid case F (or to L until the analysis model has an F case).

Plumbing (BNBC 2020 Part 8)

Toolbar: **Add Riser or Stack** (cold water riser, hot water riser, soil, waste, soil and waste, vent stack, rainwater leader, pumping main), **Add Tank** (overhead tank, underground reservoir, septic tank, soak well, rainwater tank), **Add Roof Drain**, **Size Tanks**, **Add Sleeves for Risers**.

Data:

- **Supply and roof:** source (*City main* / *Deep tubewell*), mains residual head, pumping hours, minimum residual head, rainfall intensity (mm/h).
- **Fixtures by storey:** WC, urinal, basin, shower, bathtub, kitchen sink, floor drain, washing machine, bidet, others; population and occupancy (BNBC Table 8.5.1, e.g. *Moderate apartment*, 180 lpcd).
- **Tanks:** length, width, water depth, freeboard, compartments, wall/base/roof thickness, top level; construction RC, brick or prefabricated.

Size Tanks writes the sizes the demand requires into the tanks (undoable). **Add Sleeves for Risers** adds slab sleeves where a riser is not in a shaft.

Results (*Plumbing coordination*): daily demand and storage, the **RC tank design** (walls, base and roof analysed as shells, with reinforcement and quantities), riser and rainwater leader sizes (Table 8.7.2), septic tank and soak well (Appendix S), and the **hydraulics**: residual head at every floor against the required head, solved with the EPANET 2.2 engine (WNTR).

HVAC

Toolbar: **Add Plant** (chiller, cooling tower, AHU, FCU, VRF and split outdoor units, pump, exhaust fan, fresh air unit), **Draw Duct** (supply, return, exhaust, fresh air, kitchen exhaust, smoke extract; riser ducts in shafts), **Propose Plant**, **Add Duct Openings**, **Calculate Loads**.

Data: the system (VRF, chilled water, split, packaged, ventilation only) and design conditions, the envelope, the storeys (use from Table 8.2.3), plant and ducts. While drawing a duct: click the points, double-click or Enter to end, Backspace removes the last point, Esc cancels.

Calculate Loads runs EnergyPlus for the design day when it is installed (about half a minute); without it, a hand estimate is used and labelled as such. **Propose Plant** sizes plant entries; **Add Duct Openings** proposes and checks beam web openings for the ducts.

Electrical

Toolbar: **Add Equipment** (transformer, HT switchgear, LT panel, generator, fuel tank, UPS, battery bank, PFI capacitor bank, distribution board, solar PV array), **Add Riser** (busbar trunking, cable tray, cable ladder, conduit bank), **Draw Tray**, **Propose Equipment**, **Add Openings**.

Data: supply, loads by storey (Table 8.1.17), lightning protection, equipment, risers and trays. Results: maximum demand (kVA), transformer, generator and UPS sizes, rooms, risers, trays and the lightning protection risk with the down-conductor columns needed. Equipment weights are typical catalogue values: replace them with the supplier's data.

Files

`07 MEP/mep.json` imported IFC models in `07 MEP/IFC/` computed summaries and tank designs in `07 MEP/mep.results/` the EnergyPlus cache in `07 MEP/hvac.results/`.

Not confirmed / limitations

- Large beam openings (Vierendeel action) and Imhoff tanks (above 300 residential users) are reported, not designed.
- Some values are marked in the code as typical or to be checked: catalogue equipment weights, the 40 % cable tray fill, the HVAC fallback irradiance.
- Clashes, tank sizing, hydraulics and proposals all need the Python backend.

Part 8: FEM (analysis and design)

Part 8 is the analysis model and its results, and the design of the members. It is what ETABS calls the *analysis* and *concrete frame design*: SAEngine derives a finite-element model from the building model, analyses it with **OpenSeesPy**, combines the results with the load combinations of Project Settings and checks every member to BNBC 2020 (ACI 318-19 clauses and factors) or AISC 360-22.

File: `08 FEM/model.osmodel`, with the results in `08 FEM/model.results/`. Open it with **8 FEM**. The editor is the **FEM Model Viewport**.

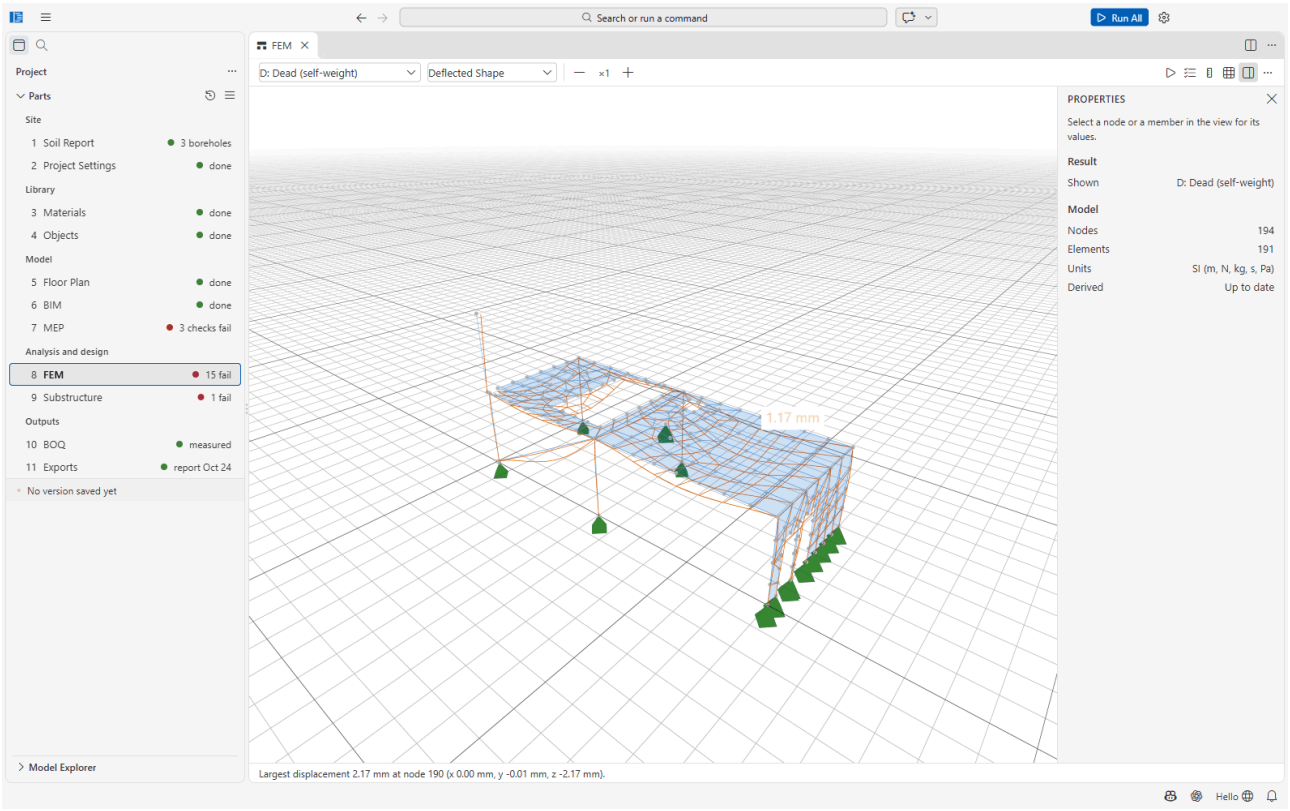


Figure 17: The analysis model of the sample: deflected shape under dead load

The analysis model is derived, never drawn

The analysis model is made from the **building model** (6 BIM), **Project Settings**, **Object Types** and **Materials**, plus the soil report (when the site class comes from it), MEP (when it has equipment, openings or shafts) and 9 Substructure (with soil-spring supports). You do not edit it by hand: change the sources and derive again.

- **Derive from Building Model** (empty model) or **Derive Again from Building Model** (shown when a source changed). Run All derives automatically when needed.
- When a source changes, the toolbar shows an **Out of date** pill: *Changed since the analysis model was derived: Object Types. Click to derive it again.*
- After deriving: *Derived 36 nodes and 63 elements; base shear 276.5 kN, seismic weight 3071.7 kN.*

What the derivation does:

- Columns, beams and inclined members become elastic beam-column elements; walls (and slabs when *Slabs* is set to *Shells*) become shell elements meshed with Gmsh.
- Column and wall bases on the lowest storey are fixed (or on soil springs).
- With *Slabs: Loads to beams* (default), slab loads go to the beams by the 45° rule; a rigid diaphragm ties each floor.
- Steel members with simple connections are pinned at their ends.
- Wind and earthquake are applied as BNBC equivalent static storey forces; the response spectrum cases are added when *Response spectrum* is on.
- Cracked section stiffness (BNBC 6.3.10.4.1: columns 0.70 Ig, beams 0.35 Ig, walls 0.70 or 0.35 Ig, slabs 0.25 Ig) is used for the lateral cases, gross sections for gravity.

Load cases

Load cases are created by SAEngine; you do not define them.

Case	Meaning
D	Dead (self-weight)
SD	Superimposed dead (finishes, partitions)
L	Live
Lr	Roof live
WX, WY	Wind in X, Y (BNBC equivalent static)
EX, EY	Earthquake in X, Y (equivalent static: T = Ct·hn^m, base shear from Z, I, R, site class)
F	Fluids (MEP equipment contents), when there are any
T	Temperature, when asked for in Project Settings
Modal	up to 12 modes
RX, RY	Response spectrum X, Y (BNBC design spectrum, CQC, 5 % damping)

The **load combinations** are those of Project Settings, expanded to ±X and ±Y (*1.2D + 1.0(-EX) + 1.0L*). They are formed as factored sums of the case results.

With *Earthquake design forces: Response spectrum* (the default), the design scales the response-spectrum forces to 85 % of the static base shear (BNBC 2.5.9.4), adds the 100 % + 30 % orthogonal effects, accidental torsion (amplified by Ax where the plan is torsionally irregular) and the vertical component, and checks the storey drift. The Properties panel shows this under **Seismic Design (Response Spectrum)**, e.g. *EX scaled × 1.99 to 85 % of the static base shear*, the Seismic Design Category, and *Largest drift 0.20 % (allowed 2.00 %)*.

Running the analysis

- **Run Analysis** (toolbar ▷) runs every case; **Run Selected Cases…** (in **…**) only some. While it runs the button becomes **Stop Analysis**.
- The log is in the output channel *SAEngine Analysis* (**Show Log** in **…**). It ends with *Analysis complete: 11 case(s). Results are in …*.
- Only one run per model at a time: *The analysis of this model is already running.*
- If the model changed after the run, the status line says *The model changed after this analysis ran.* with a **Run Analysis** link.

Viewing results

Two pickers in the toolbar choose **the result** (a case, a combination, *Modal*, *RX*/*RY*) and **what to show of it**:

For	Choices
Cases and combinations	Model Only, Deflected Shape, Support Reactions, Axial Force N, Shear Force Vy, Shear Force Vz, Torsion T, Moment My, Moment Mz; with shells: Shell Moment M11, M22, M12, membrane forces N11, N22, N12, transverse shear Q13, Q23
Design	**Design Utilisation**, **Joint Checks**
Modal	Mode Shape, with a Mode picker (*Mode 1 (T = 0.72 s)*) and **Animate the Mode Shape**
Response spectrum	Peak Displacements (response spectrum results have no sign)

Draw Smaller / **Draw Larger** (− and +) scale the diagrams. The status line gives the extreme value (*Largest displacement 2.17 mm at node 190*, *Total reaction: Fx…, Fy…, Fz…*); select a node or member for its values in the Properties panel. Units on screen: mm, kN, kN·m, kN/m, MPa.

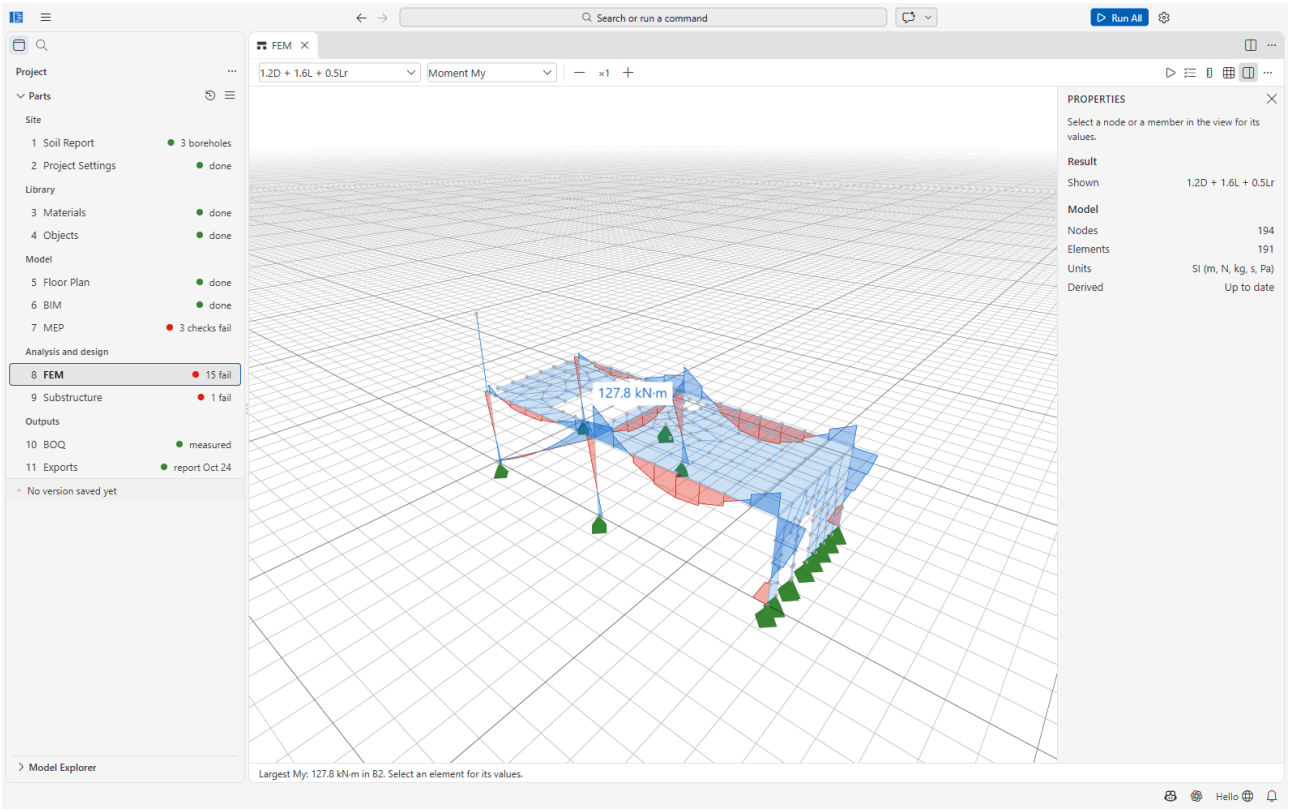


Figure 18: Moment My under 1.2D + 1.6L + 0.5Lr

Result Tables (toolbar) opens tables of the results by case (first 500 rows on screen); **Export Result Tables…** (in **…**) writes all rows to CSV.

Design checks (Design Utilisation)

Choose **Design Utilisation** after a run. SAEngine checks every member against every combination (a few minutes on the sample, much longer on a large building: the checks run again each time you choose this view after a change) and colours the members: **below 0.9** green, **0.9 to 1.0** amber, **above 1.0** red.

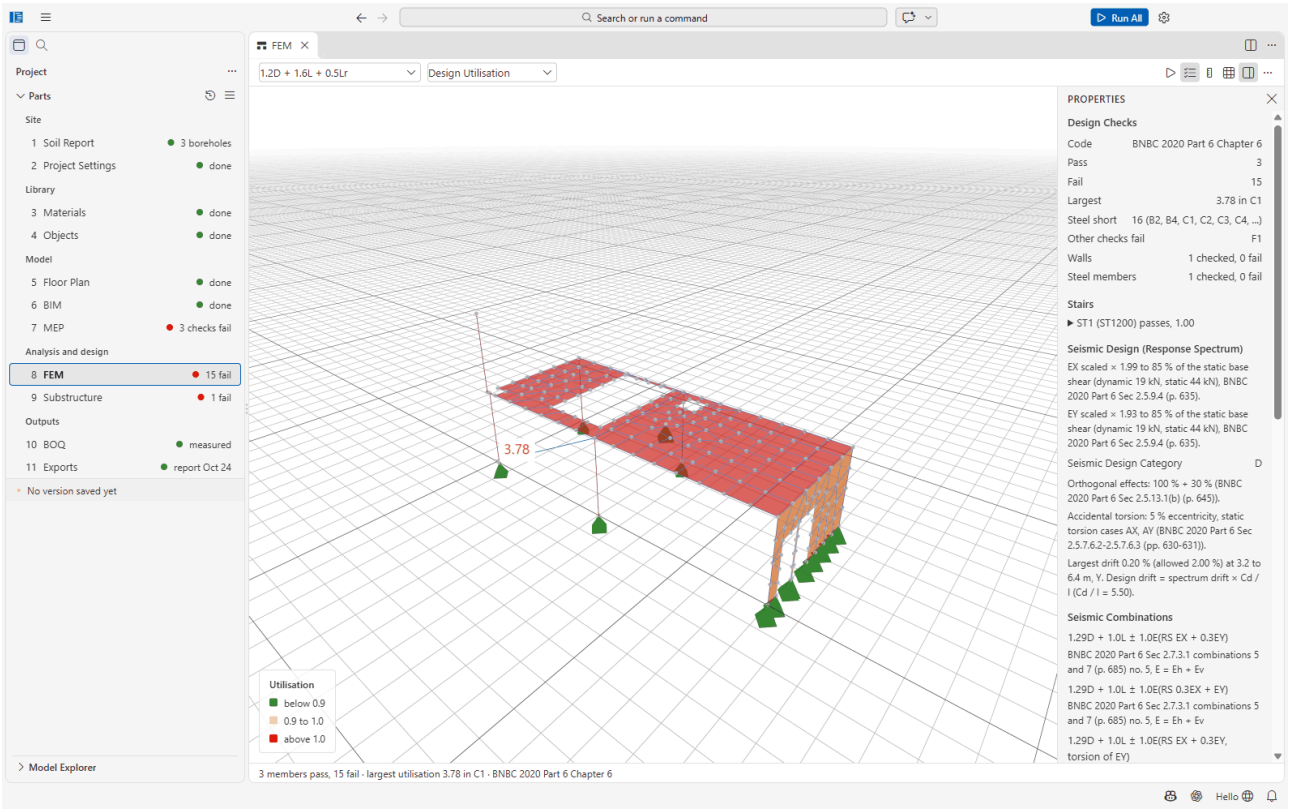


Figure 19: Design utilisation of the sample before the bars were designed

The Properties panel (nothing selected) shows **Design Checks**: the code, the number that pass and fail, the largest utilisation, and lists:

- **Steel short**: members whose bars give too little steel (fixed by Design Reinforcement);
- **Too small**: sections that no practical bars can make work (fixed by a bigger section);
- **Other checks fail**: checks bars cannot fix;
- **No bars yet**, **Second-order analysis required**, **Walls**, **Steel members**, **Stairs**.

Select a member for its checks: *Passes* or *Fails*, each check with its utilisation, values and governing combination, and a **Required Steel** section (*required 1,257 mm², provided 1,473 mm²* for columns *ρ required 0.0123, provided 0.0201*). Each row ends in *enough*, *short*, *required* (no bars yet) or **section too small**.

Element	Checked
RC beams (rectangular)	Sagging and hogging flexure, shear φ(Vc + Vs). Not checked: weak-axis bending, torsion, axial force in beams.
RC columns (rectangular, circular)	Axial force with biaxial bending (interaction surface); slenderness by moment magnification or P-Delta (*k lu / r*, *Moment magnifier*).
Slabs	Per strip and layer (Wood-Armer), one-way shear.
Structural walls	Piers, boundary elements, coupling beams.
Stairs	By hand methods (span/thickness).
Steel beams, columns, braces	BNBC Part 6 Chapter 10 / AISC 360-22, with simple connections.
Seismic detailing (BNBC 8.3)	Beam-column joints, joint shear, strong column / weak beam, capacity shear (*Joint Checks* view).
Foundations	**Not here**: in 9 Substructure.

Elements of other kinds show *This element is not designed*.

Design Reinforcement: the design proposal

Design Reinforcement (FEM toolbar, Project menu, or the Command Palette) proposes bars for every object type from the steel its members require under the combinations of the last run, and applies them to 04 Objects in one undoable edit.

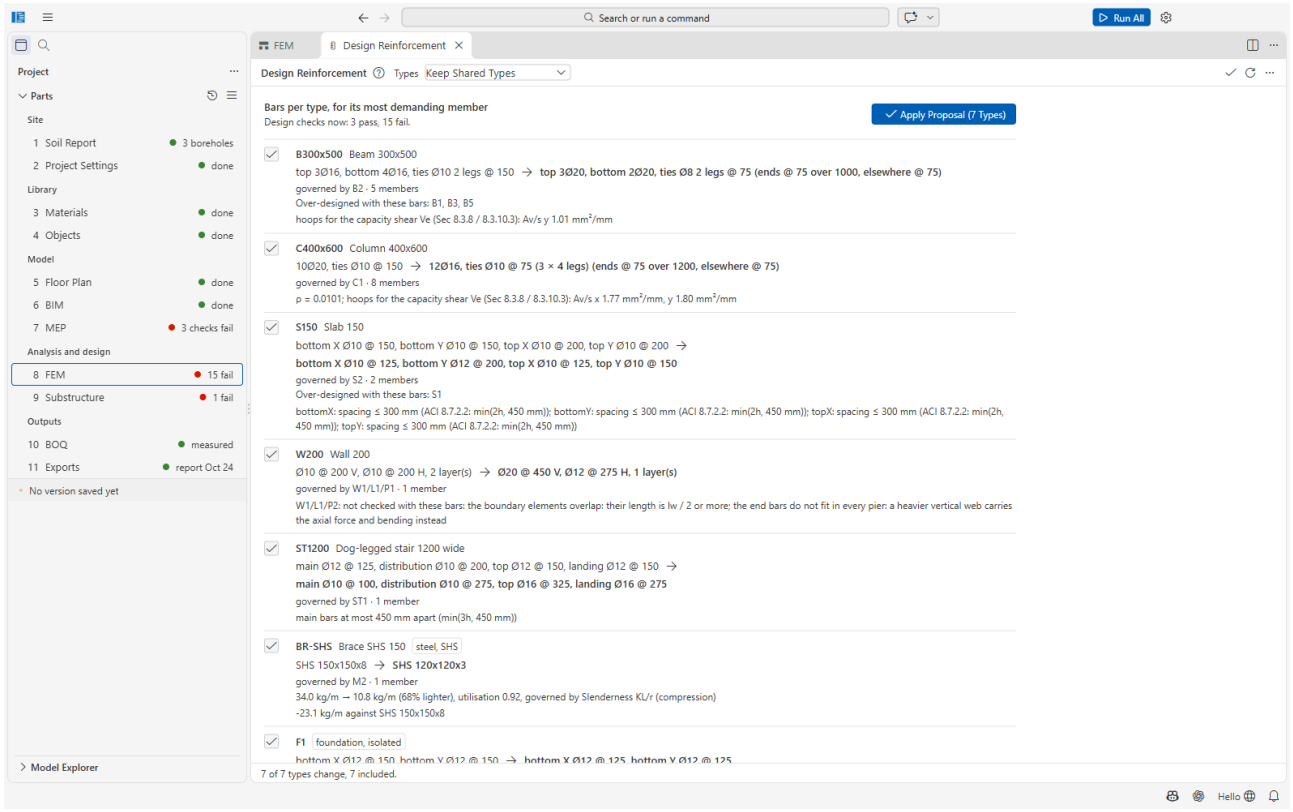


Figure 20: The design proposal of the sample

- **Types: Keep Shared Types** designs each type for its most demanding member; **Split into New Types** moves members that need much less into new, lighter types (this also edits 06 BIM and 05 Floor Plan).
- Each row shows the type, its current bars → the proposed bars (*10Ø20, ties Ø10 @ 150 → 12Ø16, ties Ø10 @ 75 (3 × 4 legs)*), *governed by C1 · 8 members*, members that are *Over-designed with these bars*, and notes. Steel types get the lightest section of the same family that passes. Foundations get their bars too.
- Tick or untick rows (**Include All Changes**, **Include None**), then **Apply Proposal (n Types)**. SAEngine saves the files and checks the members again: *Proposal applied — 3 pass, 15 fail → 18 pass, 0 fail*. **Undo Apply** reverts it.

Detailing rules: beams use one diameter per face, at most two sizes and two rows, stirrups Ø8/10/12 with 2 or 4 legs; columns equal bars round the section with ties per ACI 25.7.2; slabs Ø10/12/16 meshes.

The proposal never changes a concrete section. After applying it, the model shows *Out of date* (the object types changed): derive and run again, or simply Run All, before issuing results.

When a section is too small

When no practical bars can carry the demand, the type gets the badge **section too small**, it cannot be included, and the row says why and what to try, for example:

- column: *Section too small: no layout with ρ up to 0.08 (ACI 10.6.1.1) carries the demand; try 500 x 700 mm (+100 mm) with about ρ = 0.025; the analysis must be run again*
- beam: … *try 300 x 600 mm (depth +100 mm); the analysis must be run again, the stiffness changes*
- slab: *try 175 mm (+25 mm)* foundation: *a thicker foundation*.

To fix it:

1. Open **04 Objects** (**Open Object Types** in the proposal) and enlarge the section of that type, or create a bigger type and assign it to the members concerned on the floor plan.
2. If you changed the floor plan, **Rebuild from Floor Plan** in 6 BIM.
3. **Run All** (it derives the model again and reruns the analysis, because the stiffness changed).
4. **Design Reinforcement** again, and apply.

Run All's summary counts these as , *2 section(s) too small*.

Commands

Command (category SAEngine)	Does
Run All	Analysis, design and outputs (Getting started)
Run Analysis	Runs every case of the open model
Run Selected Analysis Cases…	Runs the cases you pick
Show Analysis Log	The *SAEngine Analysis* output
Design Reinforcement	The design proposal
Check Python Runtime	Checks the backend

Files

`08 FEM/model.osmodel` (derived), `08 FEM/model.results/` (results, deleted and rewritten by each run; `design-summary.json`, `design-required.json`, `design-report.json`), and for soil springs `model-fixed.osmodel` and `ssi-compare.json`.

Not confirmed / limitations

- No beam or slab deflection check (L/240 and similar) was found in the design; static wind and earthquake drifts are tabulated in the report, the response-spectrum drift is checked.
- Column shear is covered by the seismic capacity-shear checks, not by the general strength checks.
- *Section too small* suggestions are information only; there is no button that enlarges a section.
- Advanced analyses (pushover, time history, construction sequence, temperature, soil-structure interaction) exist in Project Settings but are not described in detail in this manual.

Part 9: Substructure

Part 9 is where foundations are placed and designed: isolated, combined and strip footings, pile caps with piles, rafts, and tie beams. The loads are the support reactions of the analysis (8 FEM); the soil capacity comes from the soil report (Part 1) or from Project Settings.

File: `09 Substructure/substructure.json`. Open it with **9 Substructure** (or **Place Footings in 9 Substructure** on the floor plan).

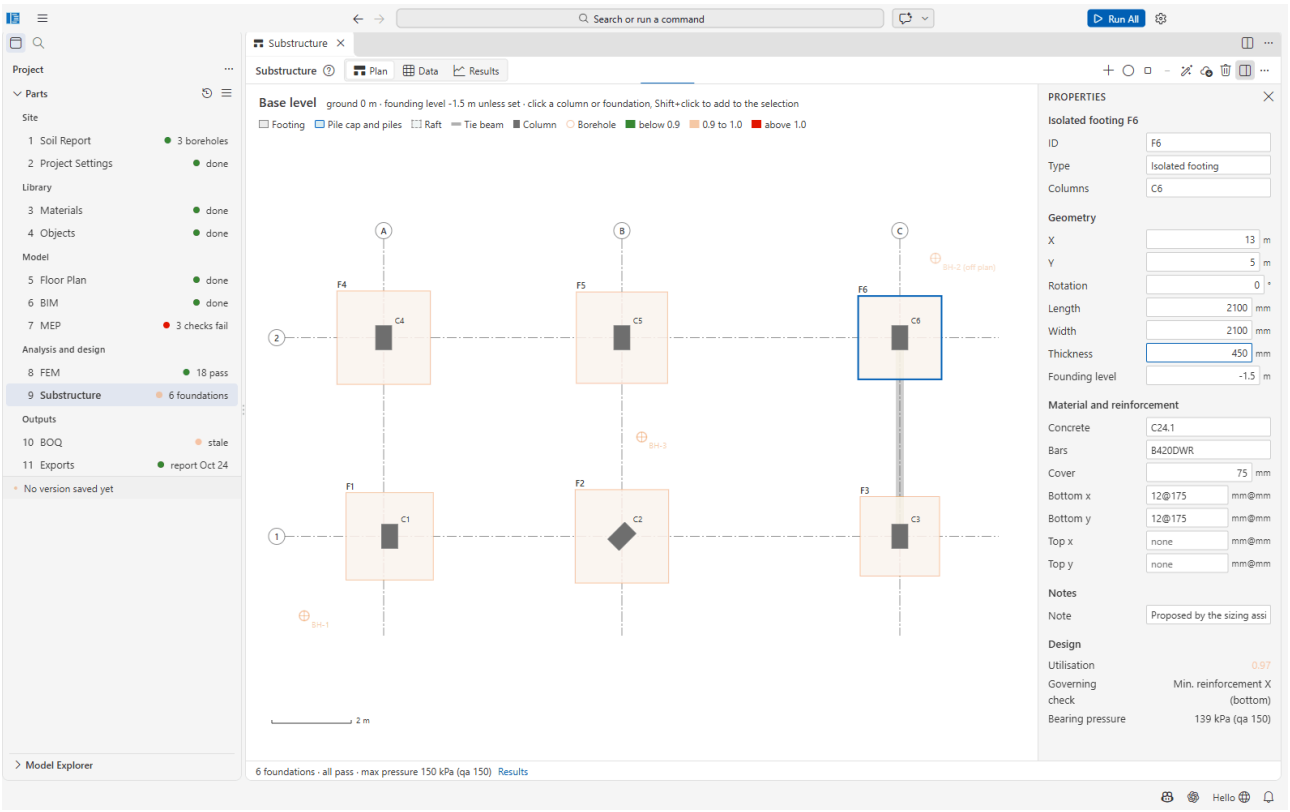


Figure 21: The foundation plan of the sample after sizing

Layout

- **Views: Plan**, **Data**, **Results**.
- **Plan** (*Base level*): the grid, the columns, the boreholes and the foundations, coloured by utilisation (**below 0.9**, **0.9 to 1.0**, **above 1.0**). Click a column or a foundation to select it, **Shift+click** to add to the selection; **Delete** removes the selection; **Esc** discards a proposal or clears the selection.
- **Data:** tables of foundations (ID, type, columns, sizes, founding level, piles, utilisation), pile types and tie beams. Sizes in mm, levels and positions in m.
- **Properties panel:** the selected foundation, or (nothing selected) the counts, *Columns without one*, and the **Design** options.
- **Status line:** e.g. *6 foundations · 0 fail · max pressure 150 kPa (qa 150)*.

Toolbar (the labels follow the selection):

Button	Does
Add Footing under the Selected Columns	an isolated footing; with several columns selected **Add Combined Footing under the n Selected Columns** with nothing selected **Add Footings under the Columns without a Foundation**
Add Pile Cap under the Selected Columns	a pile cap (needs a pile type)
Add Raft under the Selected Columns	a raft round the columns
Add Tie Beam between the Two Selected Foundations	a tie beam
Size Footings	proposes sizes (below)

Button	Does
Import Footings from the Building Model	takes over footings drawn in an older building model
Delete	
…	**Add Strip Footing along the Selected Columns**, **Add Pile Type**, **Select Columns without a Foundation**, **Check the Foundations Again**

Foundation properties

Group	Fields (defaults)
General	Type (*Isolated footing, Combined footing, Strip footing, Pile cap, Raft*), Columns
Geometry	X, Y (m; default the columns' centroid), Rotation (°), Length, Width, Thickness (mm), Founding level (m; default ground − 1.5 m)
Piles (pile caps)	Pile type; layout *n piles at 3 diameters* with n = 2, 3, 4, 5, 6 or 9
Material and reinforcement	Concrete, Bars, Cover (75 mm), Bottom x/y and Top x/y as `12@150`
Design (read only)	Utilisation, governing check, bearing pressure *p kPa (qa q)*

New foundations: isolated 1.5 × 1.5 × 0.45 m; combined: the columns' extent + 750 mm each side, 600 mm thick; raft: the columns' extent + 1 m, 600 mm thick; bottom mesh 12@150 both ways. New pile type: bored, 500 mm, 18 m.

Design options (nothing selected): pile capacity from the *Soil report (per layer)* or the *BNBC SPT equations* factors of safety for pile compression and uplift (default 2.5, BNBC Table 6.3.10a; compression raised to 3.0 without a pile load test at the site) and for bearing (default: the soil report's); *Pile load test at the site* raft mesh size (500 mm); subgrade modulus ks (default: the soil report's).

Sizing footings

Size Footings proposes isolated footings for the columns without a foundation (or for the foundations and columns you selected): the width grows in 50 mm steps until bearing passes, then the thickness from 300 mm for punching and one-way shear, then the bars. Footings that would overlap are merged into combined footings. The proposal is drawn dashed (*Proposed sizes for 5 foundations*) with **Accept** and **Discard** nothing is written until you accept.

It needs analysis results: *The sizing assistant proposed no foundations. Run the analysis first: it sizes from the support reactions.*

Where the loads and soil capacity come from

- **Loads:** the support reactions of `08 FEM/model.results`, reversed. With soil-spring supports, the fixed-base run is used. Without results: *there are no analysis results; run the analysis in 8 FEM first*.
- **Bearing combinations:** the service combinations of BNBC 2020 Part 6 2.7.2 — D, D + L + Lr, D + W, D + 0.75L + 0.75Lr + 0.75W, 0.6D + W, D + 0.7E, D + 0.75L + 0.75Lr + 0.525E, 0.6D + 0.7E, with W and E in ±X and ±Y. **Strength** checks use the factored combinations of Project Settings.
- **Allowable bearing qa** of a footing: the *Allowable bearing override qa* in Project Settings → Foundation if set; else the soil report: the lowest of the boreholes within 10 m (else the nearest), at the footing's width and depth. The check's note says which applied (*qa = 150 kPa, the override in Project Settings → Foundation; 1 Soil Report gives 153 kPa here (BH-1, 2.9 m away …)*).

Checks

Foundation	Checks
Footings (isolated, combined, strip)	Bearing (service), Uplift (service), Resultant within the base (service), One-way shear X/Y, Punching at each column, Flexure X/Y bottom and top, Flexure between columns (top), Min. reinforcement. ACI 318-19 Ch. 13, 22 and 7; plain concrete per 14.5.
Pile caps	Pile load in compression and uplift, group capacity, block failure (clay), group settlement, pile spacing, cap edge beyond the piles, cover, flexure, one-way shear, punching at columns and round a pile; strut-and-tie for deep caps.

Foundation	Checks
Pile capacity	Soil-report method or BNBC SPT equations (6.3.21 to 6.3.34): Qult = Qs + Qb − W, uplift ≤ 0.7 Qs + W.
Rafts	OpenSees plate on compression-only springs: net contact pressure, maximum and differential settlement, angular distortion (BNBC Table 6.3.8), punching, slab flexure and shear (Wood-Armer). A contact pressure / settlement map can be shown on the plan.
Tie beams	Tie force (10 %, BNBC Part 6 1.8.5.2), size (BNBC 8.3.11.3), flexure and shear from settlement.

The **Results** view (*Design checks — computed from the analysis reactions and the soil report; not stored in the file*) shows tiles for the number checked and failing, the largest utilisation and the largest bearing pressure; a table **By foundation** with the governing check and combination; and **all checks** of the selected foundation with demand, capacity, utilisation, combination and a note explaining the numbers. Then pile loads, contact pressure, pile capacity, notes and the method.

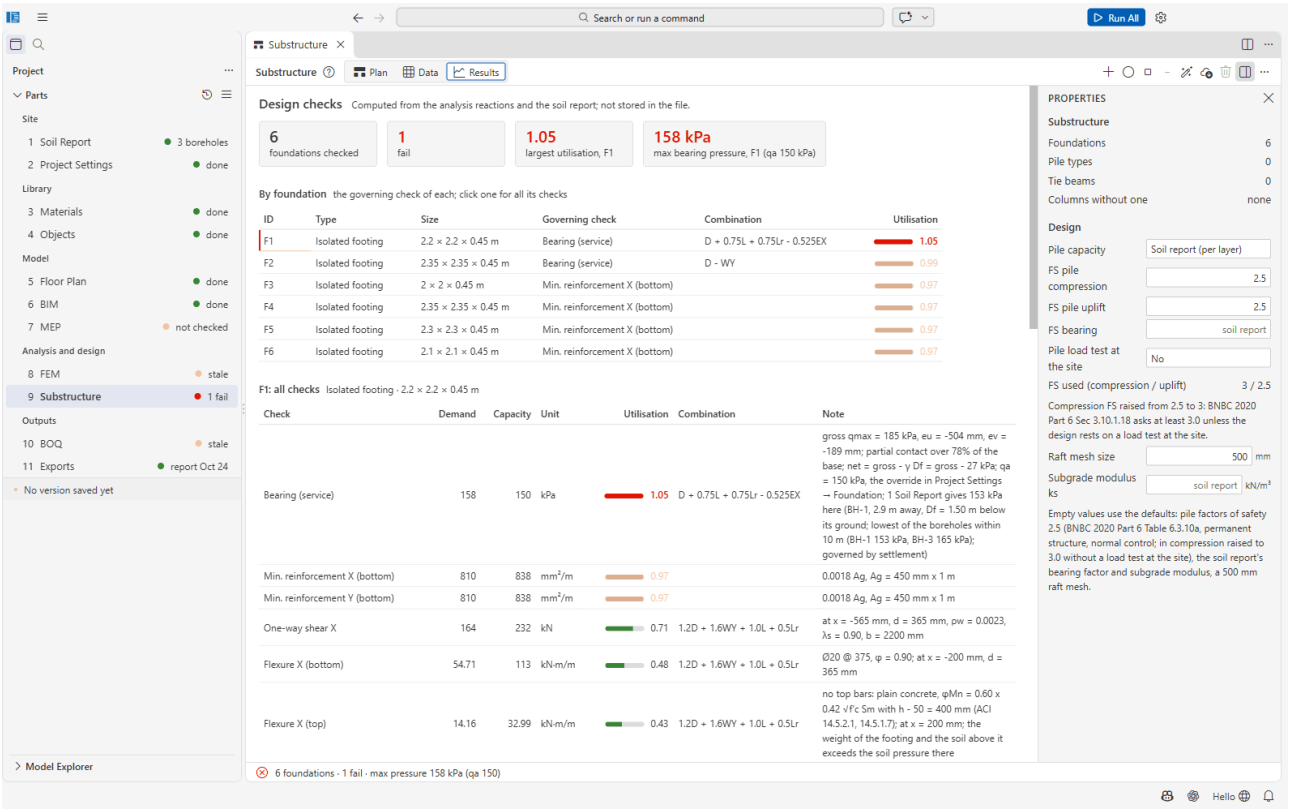


Figure 22: Results of the foundation checks

The checks run on the saved file: save (Ctrl+S) to check your latest edits; the view warns *These are the checks of the saved file. Save to check the latest edits.*

Outputs

The foundations are measured in the BOQ, drawn on the *Foundation plan* and *Foundation details* sheets of the drawings and reported in the design report. `09 Substructure/substructure.results/` holds the summary.

Not confirmed / limitations

- Not checked: sliding and overturning of footings, development lengths, rock sockets of piles.
- Walls on rafts are not supported yet (*walls on rafts: not yet*).
- Size Footings proposes isolated and combined footings only (not piles or rafts).
- Raft outlines and strip footing paths cannot be edited graphically.
- A per-foundation qa override, punching reinforcement and separate skin/end-bearing factors exist only in the JSON file.

Part 10: BOQ (Bill of Quantities)

The bill of quantities is measured from the building model, the object types and the materials (and the substructure and MEP): concrete by grade and element, reinforcement by bar diameter, formwork, structural steel, earthwork, masonry, MEP items, and the cement, sand, stone chips and water of the concrete and mortar mixes. **It contains quantities only, no rates or costs.**

File: `10 BOQ/boq.json`. Open it with **10 BOQ**. The editor is **Bill of Quantities**.

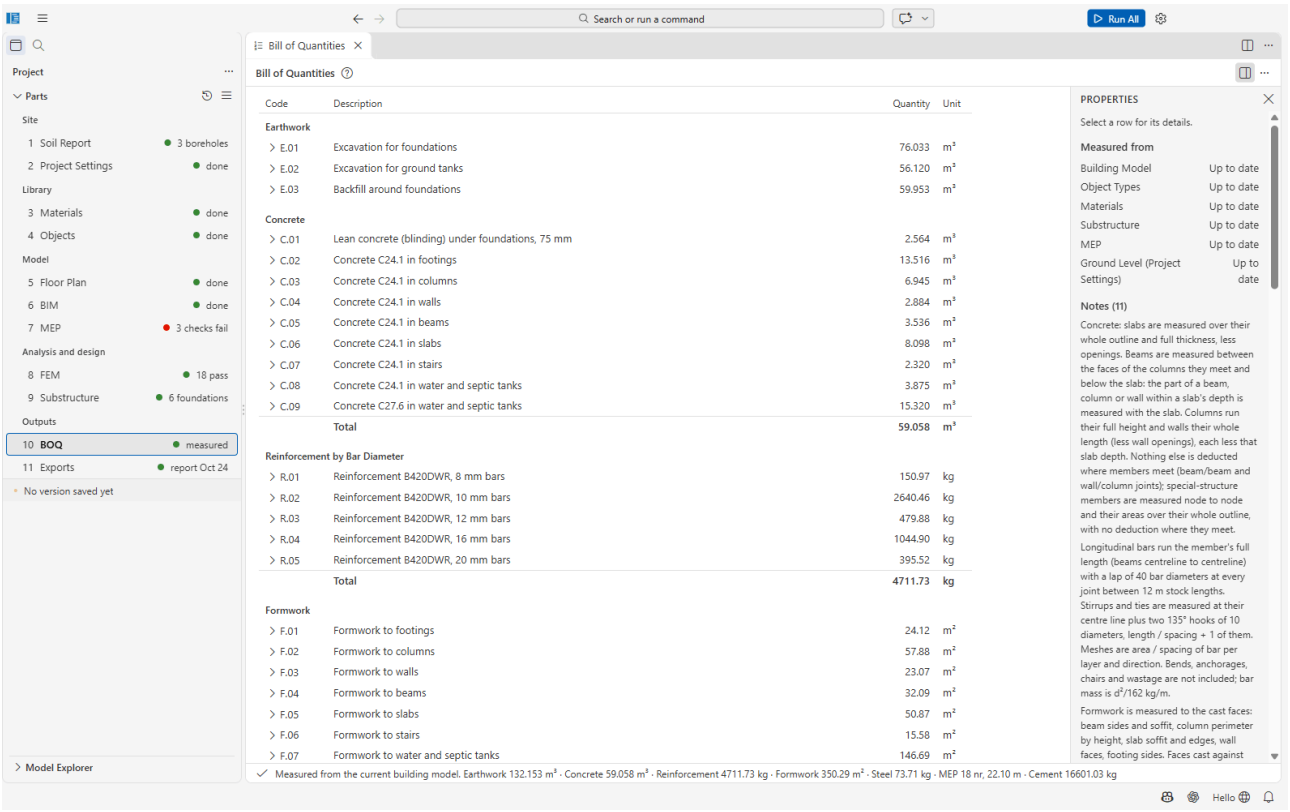


Figure 23: The bill of quantities of the sample

Measuring

- An empty bill shows **Measure from Building Model**. Afterwards the bill is kept; when a source changes it shows an **Out of date** pill (*Changed since the bill was measured: Building Model. Click to measure it again.*) and **Update from Building Model** becomes the main button.
- Run All measures the bill when it is out of date.
- The bill is never edited by hand: it is measured again on request and saved automatically.

The table

Columns **Code**, **Description**, **Quantity**, **Unit**, in groups with totals:

Group (code)	Items
Earthwork (E)	Excavation for foundations and ground tanks, backfill (300 mm working space each side)
Concrete (C)	*Lean concrete (blinding) under foundations, 75 mm* *Concrete C24.1 in columns*, … per grade and element kind, m³
Masonry (M)	Masonry per material, m³
Reinforcement by Bar Diameter (R)	*Reinforcement B420DWR, 16 mm bars*, kg
Formwork (F)	*Formwork to columns*, …, m²
Structural Steel (S)	per section, with its kg/m, kg
MEP Coordination (P)	tanks, pipes, fixtures, ducts, cable trays, sleeves and openings

Group (code)	Items
Materials: Cement, Sand, Stone Chips and Water (MT)	cement (kg) and **50 kg bags**, sand (fine aggregate, **cft**), stone chips (coarse aggregate, **cft**), water (l)

Click a row (or Enter / Space) to expand its **breakdown by element** (Element, Story, Detail, Quantity); the Properties panel shows the group, material, bar diameter, element count, the quantity by storey and which sources are up to date. Right-click: **Show Breakdown**, **Copy Quantity**, **Copy Row**, **Export CSV...**.

Measurement rules

The bill's notes state the rules it uses:

- laps of 40 d per joint between 12 m stock bars; stirrups at their centre line plus two 135° hooks of 10 d; bar mass $d^2/162$ kg/m; *bends, anchorages, chairs and wastage are not included*
- materials from the mixes with dry-volume factors 1.54 (concrete) and 1.30 (mortar), 0.23 m^3 of wet mortar per m^3 of brickwork, cement 1440 kg/m^3, sand 1600 and stone 1450 kg/m^3 loose, 50 kg bags, 35.3147 cft per m^3. These are estimating-practice values, not BNBC values; **no wastage is added and bags are not rounded up**
- a material without a mix gets no cement/sand/stone take-off (add the mix in 03 Materials).

Project Settings → Drawings → **BOQ reinforcement** chooses between *Measured by the BOQ's rules* (default) and *From the bar bending schedule* of the drawings; by default the notes reconcile the two per bar size.

Export

Export CSV... (or *Export Bill of Quantities as CSV...*) writes `11 Exports/boq.csv` (you can choose another file): one row per item followed by one row per element of its breakdown, with the columns `Row, Code, Group, Description, Element, Story, Detail, Item Quantity, Element Quantity, Unit`. Open it in Excel to add rates.

Not confirmed / limitations

- No rates, costs or xlsx export.
- There is no item named *brickwork*: brick walls are in the Masonry group, their mortar in the Materials group.

Part 11: Exports

Part 11 is what the project hands over: the design report, the structural drawings, the building model as IFC, the floor plans as DXF, the bill of quantities as CSV, and the project archive. Everything is written to the `11 Exports` folder.

Open it with **11 Exports** in the sidebar (or *SAEngine: Exports*).

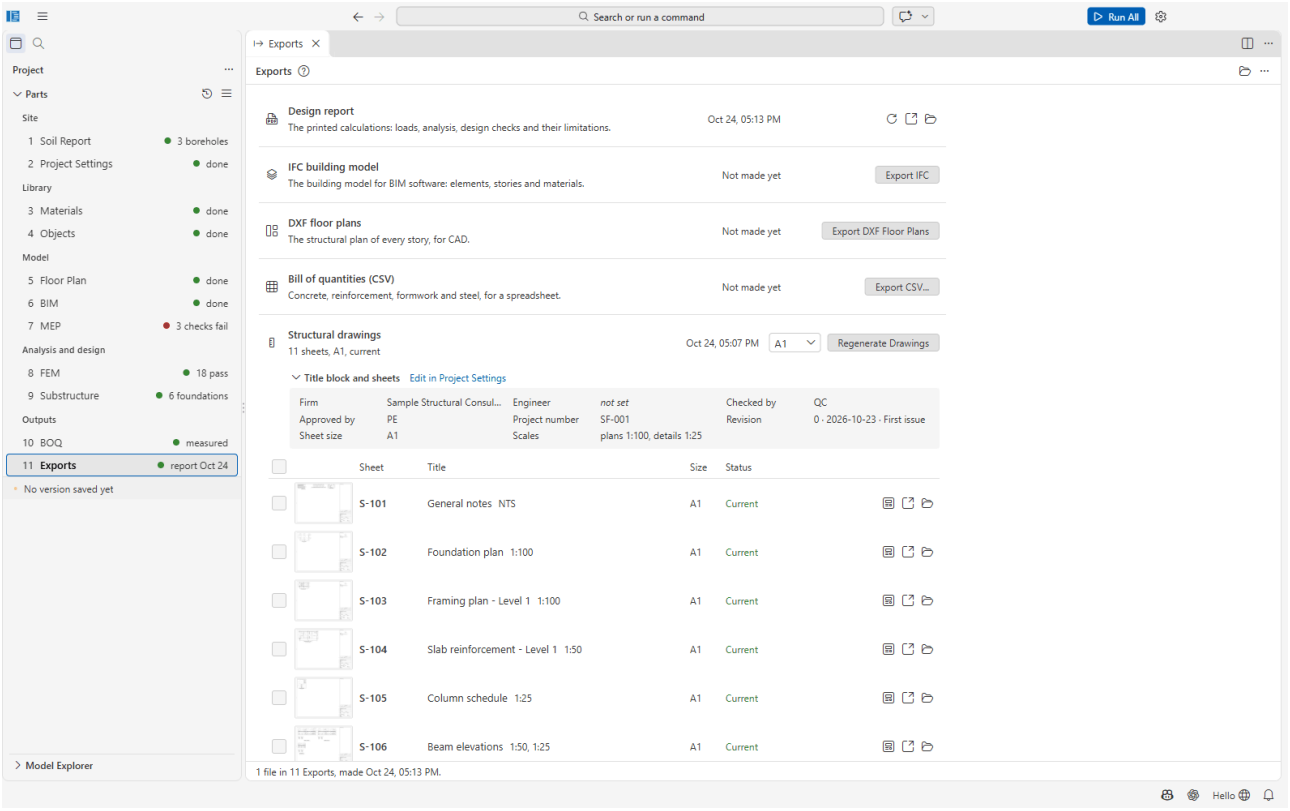


Figure 24: The Exports page

The Exports page

Row	Make it with	Writes
Design report — the printed calculations: loads, analysis, design checks and their limitations	**Print Design Report** / **Update Report**	`11 Exports/report.pdf`
IFC building model — for BIM software	**Export IFC**	`11 Exports/IFC/<project>.ifc`
DXF floor plans — the structural plan of every storey, for CAD	**Export DXF Floor Plans**	`11 Exports/Floor Plan/`
Bill of quantities (CSV)	**Export CSV...**	`11 Exports/boq.csv`
Structural drawings	**Generate Drawings**	`11 Exports/Drawings/`
Project archive (.civil)	**Export Project (.civil)...**	where you save it

A row that cannot be made yet says what is missing (*Build the building model (6 BIM) first.*, *Measure the bill of quantities (10 BOQ) first.*). Each file has **Open**, **Reveal in File Explorer** and, for DXF, **Open in LibreCAD**. The toolbar has **Open Exports Folder**.

The design report

Print Design Report (Exports page, File menu, or Command Palette) typesets the report as a PDF with Typst and opens it: *Design report written (48 pages)*. Warnings are listed in the report's *Limitations* section. It needs the analysis results (*There are no analysis results: run the analysis first.*). Run All rewrites it every time.

Contents:

1. **Design basis:** codes and method, building and site, seismic force-resisting system, materials, area loads, load cases, load combinations, analysis model and analyses.
2. **Soil and foundations** (with boreholes) and **Foundations**.
3. **Analysis model:** model summary, analysis run.
4. **Analysis results:** modal periods and mass participation, base reactions per case, displacements and storey drifts, earthquake design from the response spectrum (scaling, torsion, design drift), section stiffness and second-order effects.
5. **Member design.**
6. **Bill of quantities.**
7. **Limitations and notes.**
8. **Appendix:** the design checks of every element.

Structural drawings

The **Structural drawings** section of the Exports page makes the drawing set: *framing plans, column and beam schedules, slab, footing, wall and stair details with a title block, as DXF for LibreCAD and one PDF set, with the bar bending schedule.*

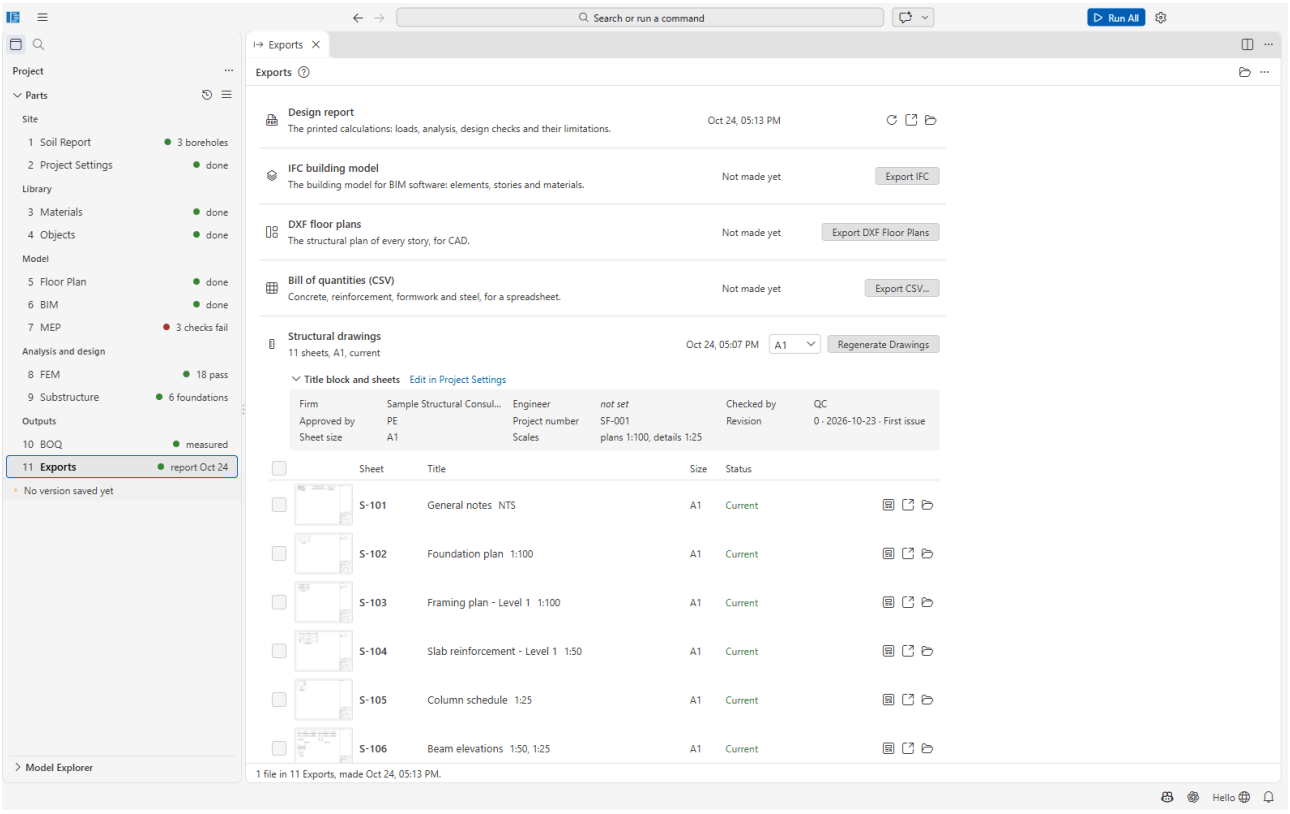


Figure 25: The drawings on the Exports page

1. Fill in the **Title block and sheets** (Firm, Engineer, Checked by, Approved by, Project number, Revision, Sheet size, Scales) with **Edit in Project Settings** (Project Settings → Drawings).
2. Choose the **sheet size** (A1, A2 or A3; default A1).
3. Click **Generate Drawings**. SAEngine runs the design checks, details the reinforcement, makes the bar bending schedule, draws the sheets and writes DXF, PDF and previews (about 15 s for the sample, minutes for a large building).
4. The sheet list shows each sheet (number, title, size, *Current* or *Out of date — regenerate*) with a thumbnail, **Preview**, **Open in LibreCAD** and **Reveal in File Explorer**. **Open PDF Set** and **Bar Bending Schedule (CSV)** open the whole set.

Sheets are numbered from **S-101** in this order: general notes, foundation plan, framing plans (one per level), slab reinforcement plans, column schedule, beam elevations, foundation details, wall details, stair details, bar bending schedule. A plan that does not fit its scale is drawn at the next scale (*drawn at 1:150, 1:100 does not fit*). Every plan notes *Levels are top of slab.*

If design checks fail or bars cannot be detailed, the general notes sheet says **NOT FOR CONSTRUCTION until these are resolved: ...**. Fix the design first.

Run All regenerates the drawings when a set exists and is out of date; **the first set is always started here**.

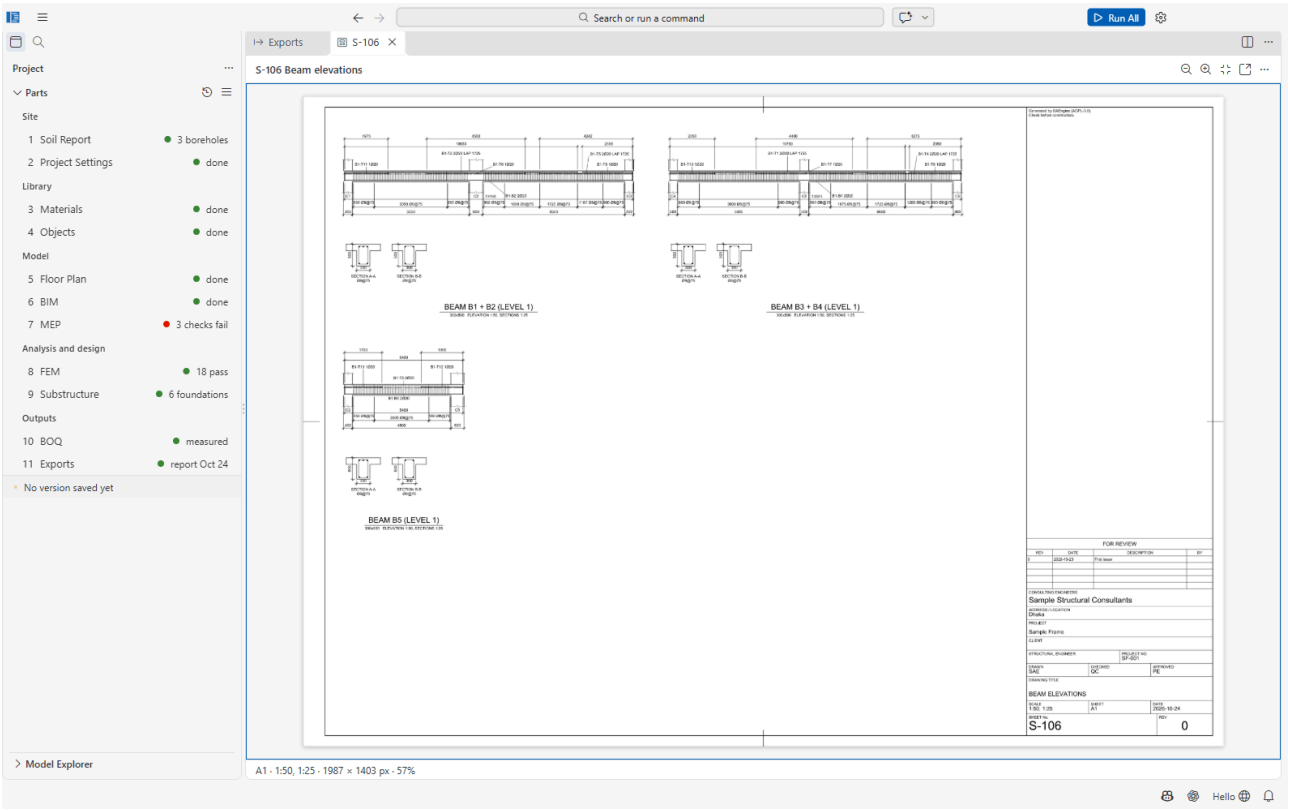


Figure 26: A sheet in the drawing preview

Preview opens the sheet in SAEngine (**Zoom In**, **Zoom Out**, **Fit Sheet**, **Actual Pixels**, **Open in LibreCAD**, **Open PDF Set**).

Files in `11 Exports/Drawings/`: one DXF per sheet (`S-101 General notes.dxf`, …), `drawings.pdf`, `previews/*.png`, `bbs.csv` (UTF-8 with BOM, for Excel) and `drawings.json`.

Opening the DXF sheets in CAD

The sheet DXFs are written for AutoCAD, BricsCAD and LibreCAD: layers follow the US National CAD Standard pattern (`S-GRID`, `S-COLS`, `S-BEAM`, …), text uses Arial, and each sheet plots from model space and from its layout (see docs/drawings-layers.md and, for checking them in AutoCAD, docs/drawings-qa.md). **Open in LibreCAD** opens a sheet in the LibreCAD bundled with SAEngine (or the program in the setting `saengine.cad.program`). **Open DXF in CAD…** (File menu) opens any DXF. DXF files also open inside SAEngine in a read-only CAD Drawing Editor.

Bar bending schedule

`bbs.csv` lists member, bar mark, type and size, BS 8666 shape code, dimensions, number, cutting length (rounded up to 25 mm), total length and mass, with totals per bar size.

DXF floor plans and IFC

- **Export DXF Floor Plans** writes one DXF per storey with layers S-GRID, S-COLS, S-BEAM and others (also **Export DXF** in the floor plan).
- **Export IFC** writes the building model (elements, storeys, materials) as IFC.

Not confirmed / limitations

- The BS 8666 shape-code values in the bar bending schedule are marked in the code as *to be checked against BS 8666:2020 before the schedules are issued for fabrication*.
- Whether rafts and tie beams are drawn on the foundation plan was not confirmed.

Importing ETABS, STAAD.Pro and SAP2000 models

SAEngine creates a **new project** from the text export of another analysis program. The model (stories, grids, members, sections, materials, supports, area loads and linear combinations) is carried over; SAEngine's own BNBC wind and earthquake loads replace those of the file.

Program	File	Export it there with
ETABS	`.e2k` (also `.$et`)	File → Export → ETABS .e2k Text File
STAAD.Pro	`.std`	the input file itself
SAP2000	`.s2k` (also `.$2k`)	File → Export → SAP2000 .s2k Text File

See also docs/import.md for the full list of what each importer reads.

In the app

1. **Import from ETABS, STAAD or SAP2000...** (start page, File menu, or ≡ menu).
2. Pick the file, then **Choose Where to Put the Imported Project** and **Name the Imported Project** (default *Imported Model*).
3. SAEngine writes the project and opens the **Import Report**.

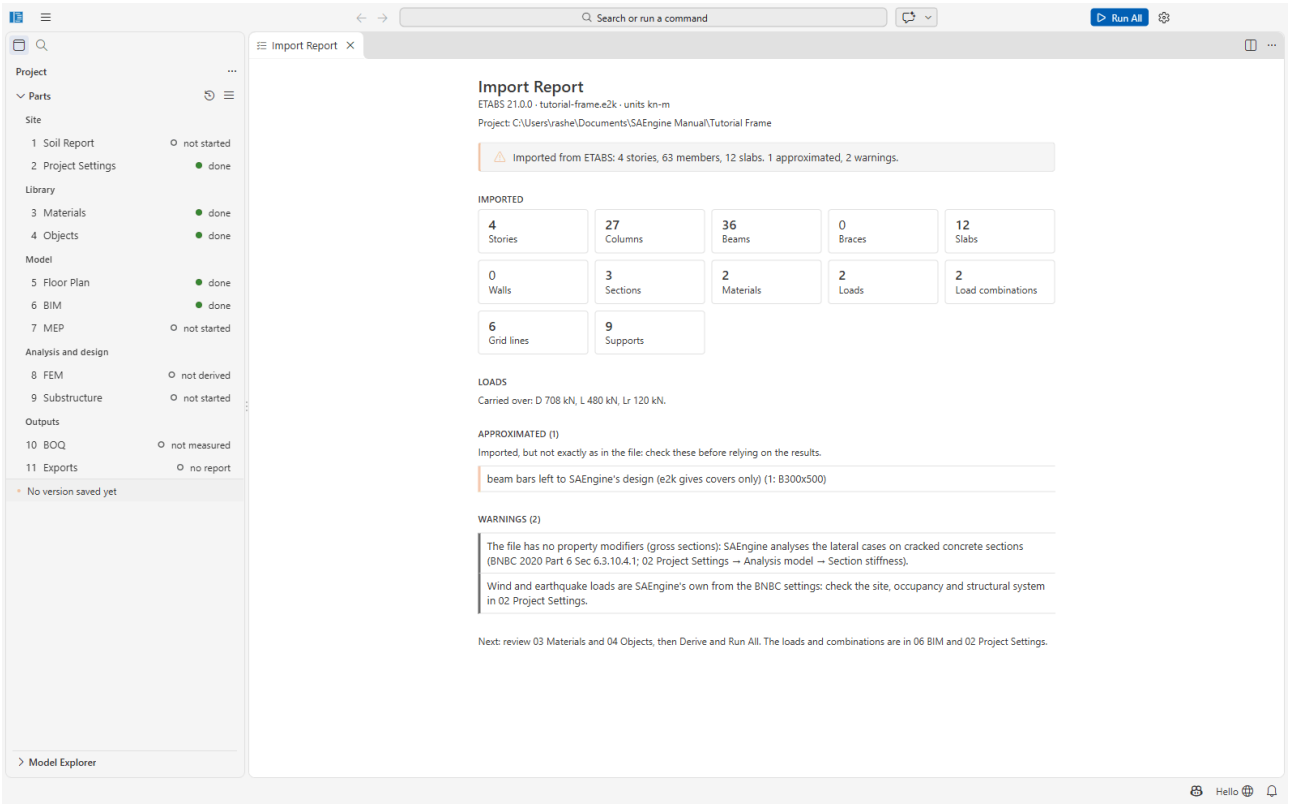


Figure 27: The import report

The Import Report has these sections:

Section	Meaning
Imported	what was read: stories, grid lines, columns, beams, slabs, supports, area loads, combinations, object types, materials
Loads	the loads carried over, and *Not carried over: ... (total ... kN). Add these loads in SAEngine, or check that its own loads cover them.* (point and frame loads)
Approximated	*Imported, but not exactly as in the file: check these before relying on the results.* (end releases, offsets, ribbed slabs as solid, beam bars left to SAEngine's design)
Skipped	*Not imported: add them in SAEngine if the design needs them.*
Warnings	e.g. *Wind and earthquake loads are SAEngine's own from the BNBC settings: check the site, occupancy and structural system in 02 Project Settings.*

It ends with: *Next: review 03 Materials and 04 Objects, then Derive and Run All. The loads and combinations are in 06 BIM and 02 Project Settings.* **Show Import Report** reopens the report of the last import made in the app. The importers also write the report into the project as `import-report.json`.

From the command line (the report is printed, and written to `import-report.json`):

```
saengine import "Office.e2k" "D:\Projects" --name "Office"
```

What the ETABS importer reads

- **Units** from `$ CONTROLS UNITS "KN" "M" "C"` (kN, kip, kgf, tonf; m, mm, in, ft). Without a units line: *kN and m assumed*.
- **Stories** (`STORY ... HEIGHT / ELEV`), **cartesian grids** (`GRID ... DIR "X" COORD`).
- **Materials** (concrete f′c, steel Fy/Fu, rebar, E, ν, unit weight).
- **Frame sections:** concrete rectangular and circular; I, tube/box, pipe, channel, angle, tee steel; a designation such as `W14X90` from SAEngine's section table. Concrete T/L and general sections are approximated. **Column bars** from `CONCRETESECTION` beam bars are left to SAEngine's design.
- **Shell properties:** slabs and walls with their thickness (ribbed/waffle and decks as solid; mat foundations skipped: foundations belong to 9 Substructure).
- **Points, lines, areas** and their **story assigns**. A point's third number puts it that far below its story.
- **Supports:** SAEngine fixes the column and wall bases of the lowest story itself. A pinned restraint there becomes a pinned support (the column base rotates freely); restraints elsewhere (a brace's foot, a support above the base) become supports too. A file **without restraints is an error** in the report. Point springs are not imported.
- **Walls:** a panel that is a rectangle between two stories is a wall; ETABS mesh pieces and the panels round doors and windows are merged into one wall with openings.
- **Releases:** steel beams and braces released in bending at both ends get a simple connection (fin plate or gusset; check it in 04 Objects), which pins them in the analysis. Other releases are reported, and the members stay continuous.
- **Property modifiers:** SAEngine has no per-member modifiers. Modifiers equal to BNBC 2020 6.3.10.4.1's cracked factors switch on SAEngine's cracked sections; other modifiers are reported as not applied. A file without modifiers gets the warning *The file has no property modifiers (gross sections): SAEngine analyses the lateral cases on cracked concrete sections*.
- **Steel sections named only by a designation** (`SHAPE "W14X90"`): SAEngine's section table, then FreeCAD's profile table (American W, HEA/HEB/HEM, IPE, UPE, UB, UC, CHS, RHS and Indian sections), then an estimate from the name (reported as *estimated from the designation — check*).
- **Load patterns** (Dead and Super Dead → D/SD, Live → L, Roof Live → Lr, wind and seismic → WX, WY, EX, EY by the X or Y in the name; fluid patterns such as *F* or *Water* → F), **uniform gravity area loads** (the slabs with the most area give the floor and roof loads, other groups become area loads of their own on their slabs), and **linear combinations** (envelopes and SRSS are skipped).
- Reported but not carried: point loads and frame loads (with their totals), end offsets, rigid-zone factors and insertion points, the mass source where it differs from SAEngine's seismic weight, auto seismic and wind definitions (SAEngine's BNBC loads apply). Ignored: diaphragms, piers, spandrels, design preferences.

After importing

1. Read the import report.
2. Check **02 Project Settings** (location, occupancy, structural system: the wind and earthquake loads are SAEngine's), **03 Materials** and **04 Objects**.
3. Run All. Then compare with the source program, as in Tutorial 4.

Not confirmed / limitations

- **There is no automatic comparison** of SAEngine's results with those of the source program. Compare by hand (Tutorial 4 shows how).
- The ETABS importer has been run on 58 real ETABS 22.5 exports (according to docs/import.md); real-file testing of the STAAD.Pro and SAP2000 importers could not be confirmed for this manual. Check the first import of each program's export against its own model (the report's counts and the base reactions under D).

Reference: commands, command line and files

Command Palette

Press **Ctrl+Shift+P** (or click the search box in the title bar) and type part of a command. SAEngine commands are in the category **SAEngine** floor-plan commands in **Floor Plan**.

Command	Where else
Show Start Window	
New Project…	File menu
Open Project…	File menu
Open Sample Project…	start page
New Building from Grid…	start page, File, ≡
Import from ETABS, STAAD or SAP2000…	start page, File, ≡
Show Import Report	
Project Overview, Project Settings, Exports	≡
Versions, Save Version…	≡, sidebar
Export Project…, Open Project Archive…	File, ≡
Open Folder in File Explorer	≡
Run All	title bar
Run Analysis, Run Selected Analysis Cases…, Show Analysis Log	FEM toolbar
Design Reinforcement	FEM toolbar, Project menu
Print Design Report	File, Exports page
Export Bill of Quantities as CSV…	BOQ, Exports page
Open DXF in CAD…	File
Add Borehole to the Soil Report, Import Boreholes from CSV/Excel…, Save Borehole Template…	Soil Report toolbar
Size Footings, Import Footings from the Building Model into the Substructure	Substructure toolbar
Suggest Sleeves from the MEP Clashes	MEP toolbar
Floor Plan: Toggle Command Line, Select All on Story, Select by Type…, Copy Selection to Stories…, Mirror Selection…, Delete Selection, Members Table	floor plan
Check Python Runtime	
Toggle Developer Mode	Help

The `saengine` command line

The installer can add `saengine` to the PATH (an opt-in task). Every step of the app can run without the window, for scripts and checking. Full description: docs/cli.md.

Command	Does
`saengine validate <project>`	checks every part file
`saengine status <project>`	the status of every part, as in the sidebar
`saengine run-all <project> [--steps ...]`	Run All (steps: derive analysis ssi design substructure mep boq report drawings)
`saengine derive, analyse, design, report <project>`	single steps
`saengine check <project>`	exits with 1 when a check fails
`saengine export <project> --format ifc|dxf|csv|pdf|drawings|civil`	the exports
`saengine import <file> <dest> [--name NAME]`	`.civil`, `.e2k`, `.std`, `.s2k`
`saengine new-building <spec.json> <dest>`	New Building from Grid from a JSON spec

Command	Does
`saengine versions status|list|save|diff|restore|issue`	versions
`saengine verify [--pdf]`	runs the verification benchmarks against closed-form and published references

`--json` writes one JSON document; `-q` hides the progress.

Files of a project

File	Written by
`<name>.saengine`	New Project
`01 Soil Report/soil-report.json`	Soil Report editor
`02 Project Settings/project-settings.json`	Project Settings editor (and the soil map's *Apply*)
`03 Materials/materials.json`	Materials editor
`04 Objects/objects.json`	Object Types editor, Design Reinforcement
`05 Floor Plan/floor-plan.json`, `Underlays/*.dxf`	Floor Plan editor
`06 BIM/bim.json`	Build from Floor Plan, Additions
`07 MEP/mep.json`, `IFC/`, `mep.results/`	MEP editor
`08 FEM/model.osmodel`, `model.results/`	Derive, Run Analysis, design checks
`09 Substructure/substructure.json`, `substructure.results/`	Substructure editor
`10 BOQ/boq.json`	Measure / Update
`11 Exports/report.pdf`, `Drawings/`, `IFC/`, `Floor Plan/`, `boq.csv`	Exports
`import-report.json`	the importers

All part files are JSON. *Reopen Editor With… → Text Editor* shows the text of any of them; edit it only if you know the format.

Verification

`saengine verify` runs SAEngine's verification pack: benchmark problems compared with an independent reference (closed-form solutions, published benchmarks, hand calculations) within a stated tolerance (1 % for closed-form frames, 0.5 % for design hand calculations, …). With `--pdf` it writes a verification manual. See docs/verification (`docs/verification/README.md`).

Glossary: SAEngine terms and the ETABS / AutoCAD terms you know

SAEngine	What it is	ETABS / SAP2000	AutoCAD / drafting
Project	A folder with a `.saengine` manifest and one folder per part	Model file (`.EDB`)	Drawing set folder
Part	One of the eleven numbered steps in the Project sidebar	Menus Define / Assign / Analyze / Design	—
Overview	The page with the status of every part and the next step	—	—
Run All	One click: derive, analyse, design, foundations, MEP checks, BOQ, report, drawings	Analyze → Run, then Design → Start Design/Check	—
Soil Report (Part 1)	Boreholes, strata, SPT, lab tests and the derived site class and bearing	Not in ETABS (site class typed in the seismic load)	—
Project Settings (Part 2)	Codes, occupancy, site, area loads, combinations, analysis options	Define → Load Patterns / Load Combinations, Auto Seismic/ Wind, Preferences	—
Code Values	The BNBC values (Z, V, I, R, Ω0, Cd, site class) worked out for the project	Auto seismic / wind load parameters	—
Area load	A named set of live, floor finish and partition loads for floors or the roof	Shell uniform load sets	—
Load case D, SD, L, Lr, WX, WY, EX, EY, F, T	Created by SAEngine; not user-defined	Load patterns + load cases	—
Load combination	Factors on the load cases (BNBC 2.7.3 by default)	Load combinations	—
Materials (Part 3)	Concrete, rebar, steel, masonry, mortar with their mix	Define → Material Properties	—
Object type (Part 4)	A section with its material, profile and bars	Frame section / Shell property / Concrete section	Block / member type
Floor Plan (Part 5)	The grids, storeys and members drawn storey by storey	Plan view drawing (Draw Columns, Beams, Floors)	Structural framing plan
Storey	A level with an elevation (top of floor)	Story	Level / floor
Typical floor (`similarTo`)	Storeys that repeat a master storey; edits apply to all	Similar stories / Master story	—
Window / crossing window	Drag left→right selects members inside; right→left selects members touched	Window / intersecting-line selection	Window / crossing selection
Fill	Places members on every grid intersection, segment or bay inside a window	Quick Draw Columns / Beams / Floors	—
Copy to Stories	Copies the selection to other storeys	Replicate → Story	Copy to layouts
Underlay	An architect's DXF shown greyed under the plan	DXF import as background	Xref
Command line (:)	Typed commands on the floor plan (`fill`, `copy`, `array`, `mirror`)	—	Command line
Building model / BIM (Part 6)	The 3D model generated from the floor plan, with openings and stairs	The 3D model	3D / IFC model
Additions	Openings, stairs, inclined members the building model adds to the plan	Openings, ramps, braces	—

SAEngine	What it is	ETABS / SAP2000	AutoCAD / drafting
MEP (Part 7)	Equipment loads, sleeves, shafts, plumbing, HVAC, electrical for the structure	Additional point/area loads, openings	MEP coordination drawing
Analysis model (`model.osmodel`, Part 8)	The finite-element model derived from the building model	The analysis model (meshed)	—
Derive	Create the analysis model from its sources	(automatic in ETABS when you run)	—
Out of date	A derived file whose sources changed	Model unlocked / analysis results deleted	—
FEM Model Viewport	The 3D view of the analysis model and its results	Model window with Display → Force/Stress Diagrams	—
Design Utilisation	Member colour by demand/capacity ratio	Concrete Frame Design → Display Design Info (P-M-M ratios)	—
Required Steel	The area of steel each member needs	Longitudinal reinforcing (As required)	—
Design Reinforcement / design proposal	Bars proposed for every object type and applied to Part 4	Concrete Frame Design summary (bars are not written back)	Reinforcement detailing
Section too small	No practical bars can carry the demand: enlarge the section	*O/S* (over-stressed) message	—
Steel short	Bars entered, but less than required	Rebar % < required	—
Seismic Design Category	BNBC category from Z, site class and occupancy	Seismic design category	—
Soil springs	Supports on springs from the soil report	Point springs / area springs	—
Substructure (Part 9)	Footings, combined and strip footings, pile caps, piles, rafts, tie beams	SAFE / CSi Detail / hand design	Foundation layout
Size Footings	Proposes footing sizes from the reactions	SAFE design strips / hand sizing	—
BOQ (Part 10)	Quantities of concrete, steel by diameter, formwork, earthwork, cement bags, sand, stone chips	Material takeoff	Quantity takeoff
Exports (Part 11)	Report, drawings, IFC, DXF, CSV	Reports / Export	Plot / Publish
Design report	PDF calculations	Design report / tables	—
Drawings / sheets S-101…	DXF sheets plus a PDF set and a bar bending schedule	—	Sheet set / layouts
Bar bending schedule (BBS)	`bbs.csv` with BS 8666 shape codes	—	BBS
Version	A saved snapshot of the project	(save as a new file)	Revision
Issued	A version marked as submitted, e.g. *Rev A*	—	Issue / revision cloud
.civil archive	The whole project in one file	`.EDB` / `.e2k` transfer	eTransmit
Developer mode	Shows the underlying code-editor tools	—	—

Abbreviations

BNBC	Bangladesh National Building Code 2020
SPT, N, N60, (N1)60	Standard penetration test blow count, corrected to 60 % energy, and to overburden

BNBC	Bangladesh National Building Code 2020
N̄, s̄u	Harmonic mean N and undrained shear strength over the top 30 m
qa	Allowable (net) bearing pressure
ks	Modulus of subgrade reaction
LPI	Liquefaction potential index
BTM	Bangladesh Transverse Mercator coordinates
R, Ω0, Cd	Response reduction factor, overstrength factor, deflection amplification factor
SD (load case)	Superimposed dead load; not to be confused with site class SD
RX, RY	Response spectrum cases
SDC	Seismic Design Category
IFC	Industry Foundation Classes (BIM exchange format)
DXF	AutoCAD drawing exchange format
cft	Cubic feet

Tutorials

Four step-by-step tutorials, each from a project you can make yourself. All data in them is synthetic.

Tutorial 1: The sample project — Run All, the design proposal, drawings

You will open the sample project, run the analysis and design of the whole building with one click, let SAEngine propose the reinforcement, size the missing footings, resolve the problems the drawings report, and produce the drawing set.

Time: about an hour, much of it waiting for Run All.

The *Sample Frame* is a small synthetic RC building: grids A–C (0, 6, 13 m) and 1–2 (0, 5 m), storeys at 3.2, 6.4 and 10.0 m, seven columns, five beams, a wall, two slabs, a stair, a steel brace, MEP equipment and tanks, three boreholes and one footing.

1. Open the sample

1. On the start page click **Open Sample Project...**.
2. Choose a folder for your copy (default `Documents\SAEngine Projects`). SAEngine copies the sample there and opens it. Your changes go to the copy; the original stays untouched.

The **Overview** shows every part. Most are *done* 8 FEM is *not derived* and the Overview's **Next** line says *Next: 8 FEM — derive the analysis model*.

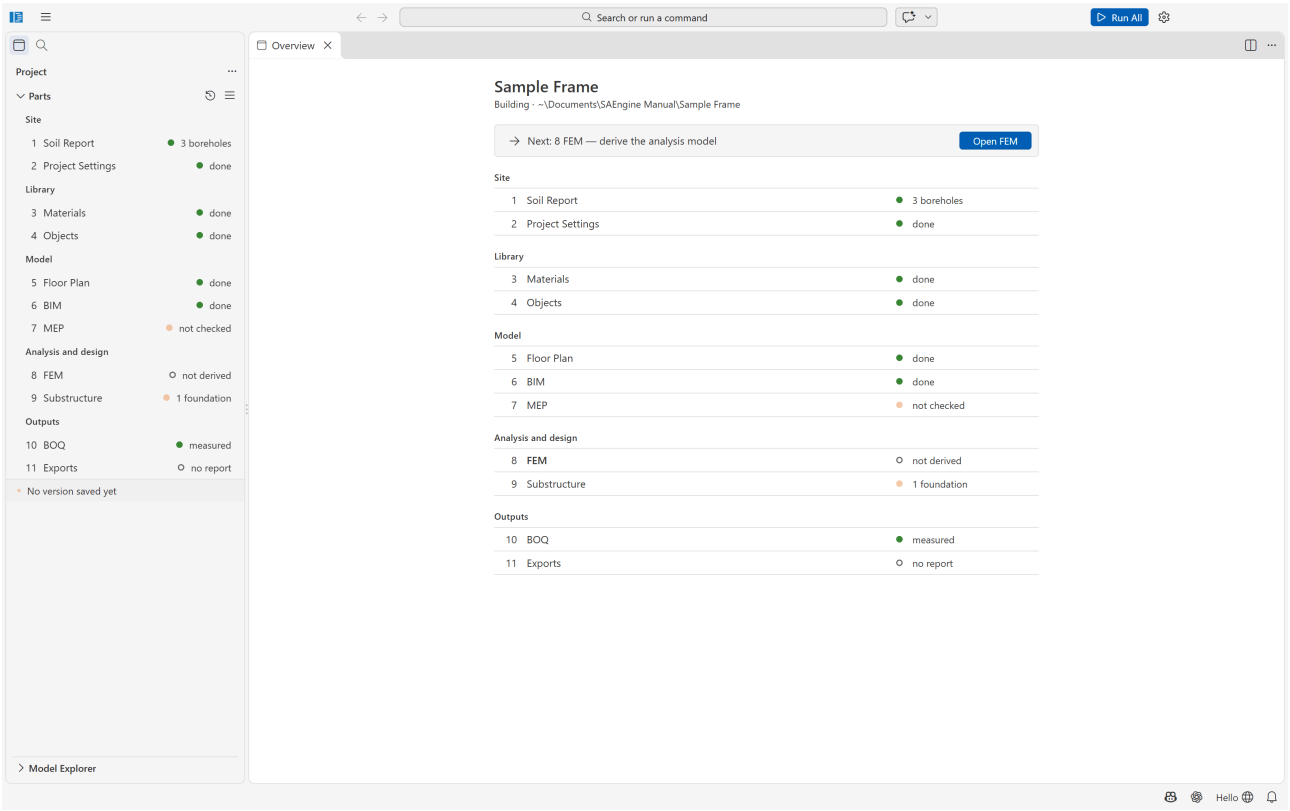


Figure 28: The Overview of the sample

Take a minute to click through the parts: the boreholes in 1 Soil Report, the grids and members in 5 Floor Plan (pick *Level 1 (+3200)* in the storey picker), the column type *C400x600* in 4 Objects with its bars.

2. Run All

Click **Run All** in the title bar. The left of the status bar shows the steps (*Running 2 of 9: analysis*, *Running 4 of 9: design checks*, ...). Click it if you want to cancel. On a typical PC the sample takes three to five minutes.

When it finishes, a notification sums up:

Done: 3 pass, 15 fail; steel short in 16 element(s) · foundations 0 pass, 1 fail · MEP: failing openings 0, open clashes 0 · plumbing: failing checks 1, ... · model derived · BOQ updated · report written

The sidebar now says *8 FEM — 15 fail* and *9 Substructure — 1 fail*. Fifteen members fail because their bars give too little steel (*steel short*): the sample's object types carry deliberately light bars.

Open **8 FEM**. The viewport shows the analysis model; pick a combination and a result in the two pickers, e.g. **1.2D + 1.6L + 0.5Lr** and **Moment My**.

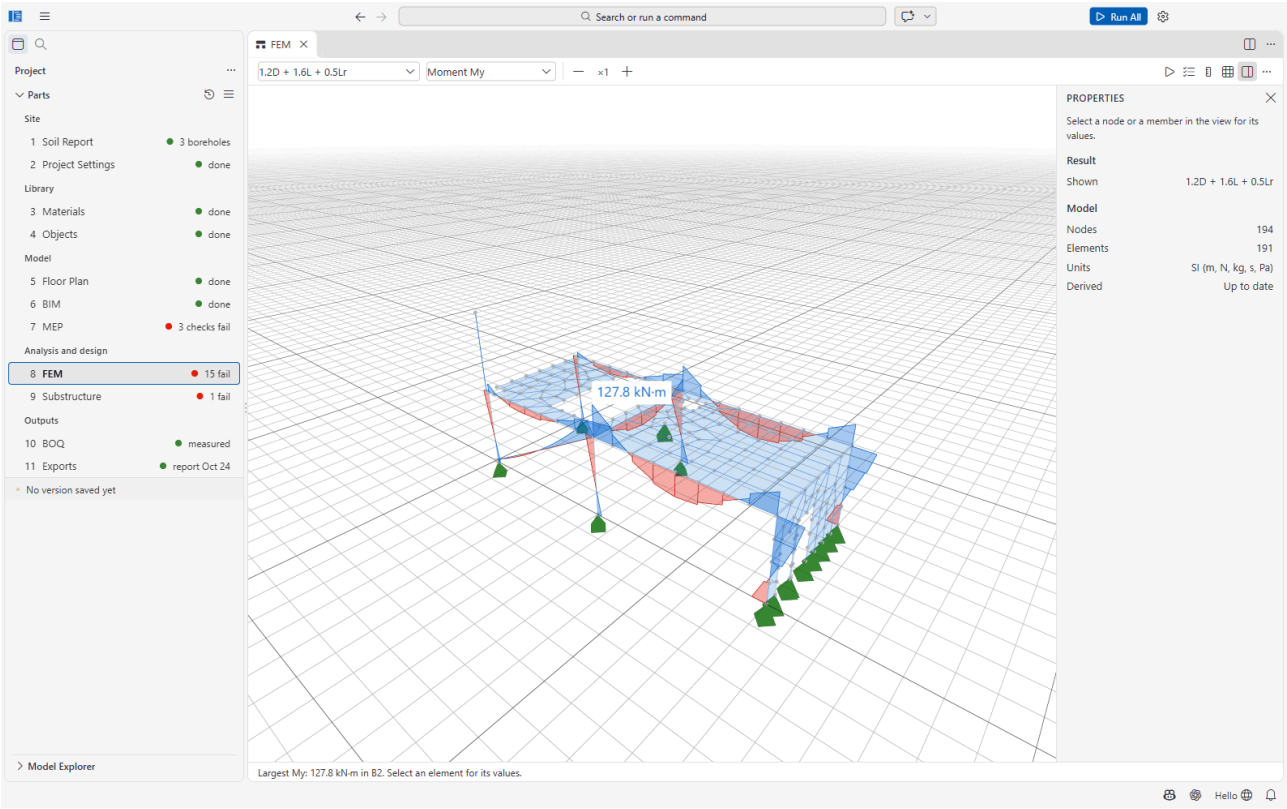


Figure 29: Moment My under 1.2D + 1.6L + 0.5Lr

Choose **Design Utilisation** in the second picker. After about a minute the members are coloured green (below 0.9), amber (0.9 to 1.0) or red (above 1.0), and the Properties panel lists the result: *3 members pass, 15 fail · largest utilisation 3.78 in C1*, *Steel short: 16 (B2, B4, C1, …)*, *Other checks fail: F1*. Click a red member to see its checks and its **Required Steel**.

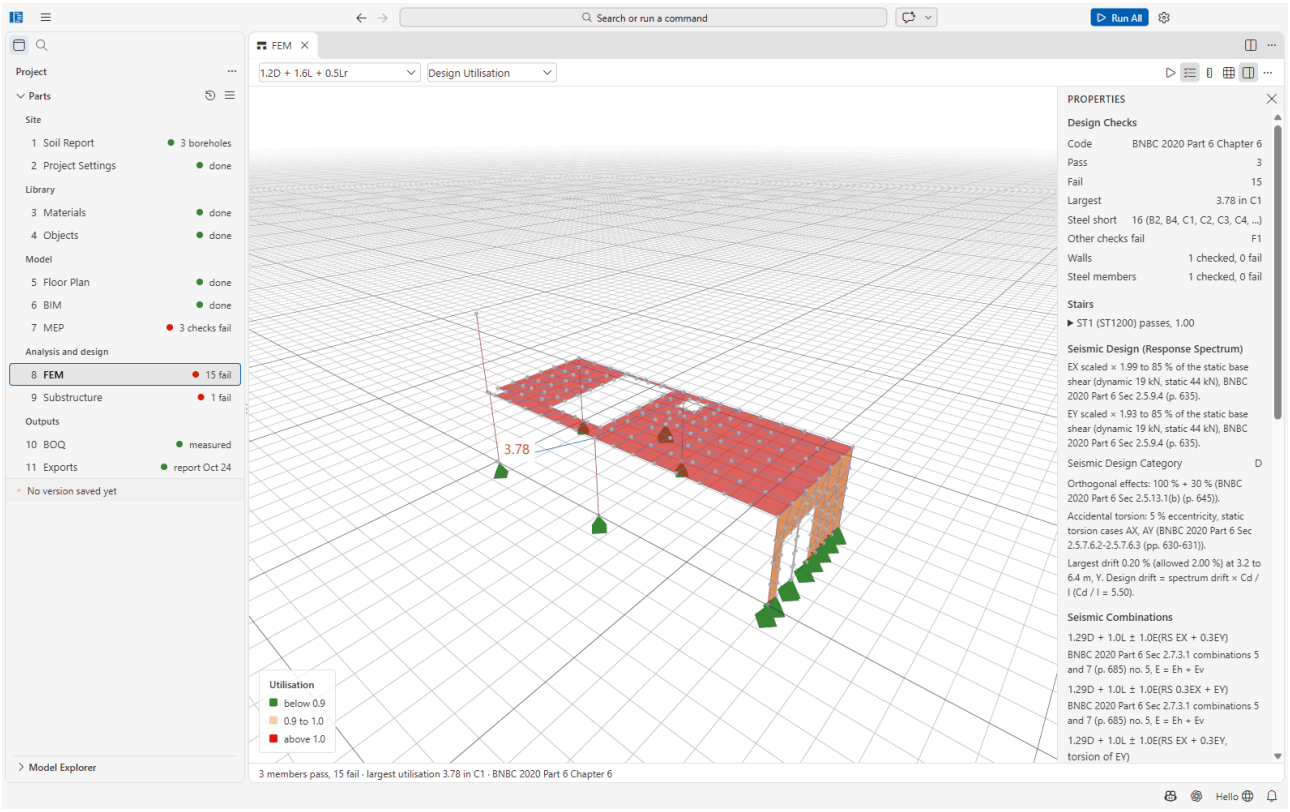


Figure 30: Design utilisation before the bars are designed

3. Let SAEngine propose the bars

1. Click **Design Reinforcement** in the FEM toolbar. SAEngine proposes bars for every object type from the steel its most demanding member requires (*Keep Shared Types*).
2. Read the proposal. For example *C400x600 — 10Ø20, ties Ø10 @ 150 → 12Ø16, ties Ø10 @ 75 (3 × 4 legs); governed by C1 · 8 members*. The brace type *BR-SHS* gets a lighter section of the same family (*SHS 150x150x8 → SHS 120x120x3*), and footing F1 gets new bars but also the note that bearing fails: *bars do not help; a larger foundation (or another type) is the remedy*.

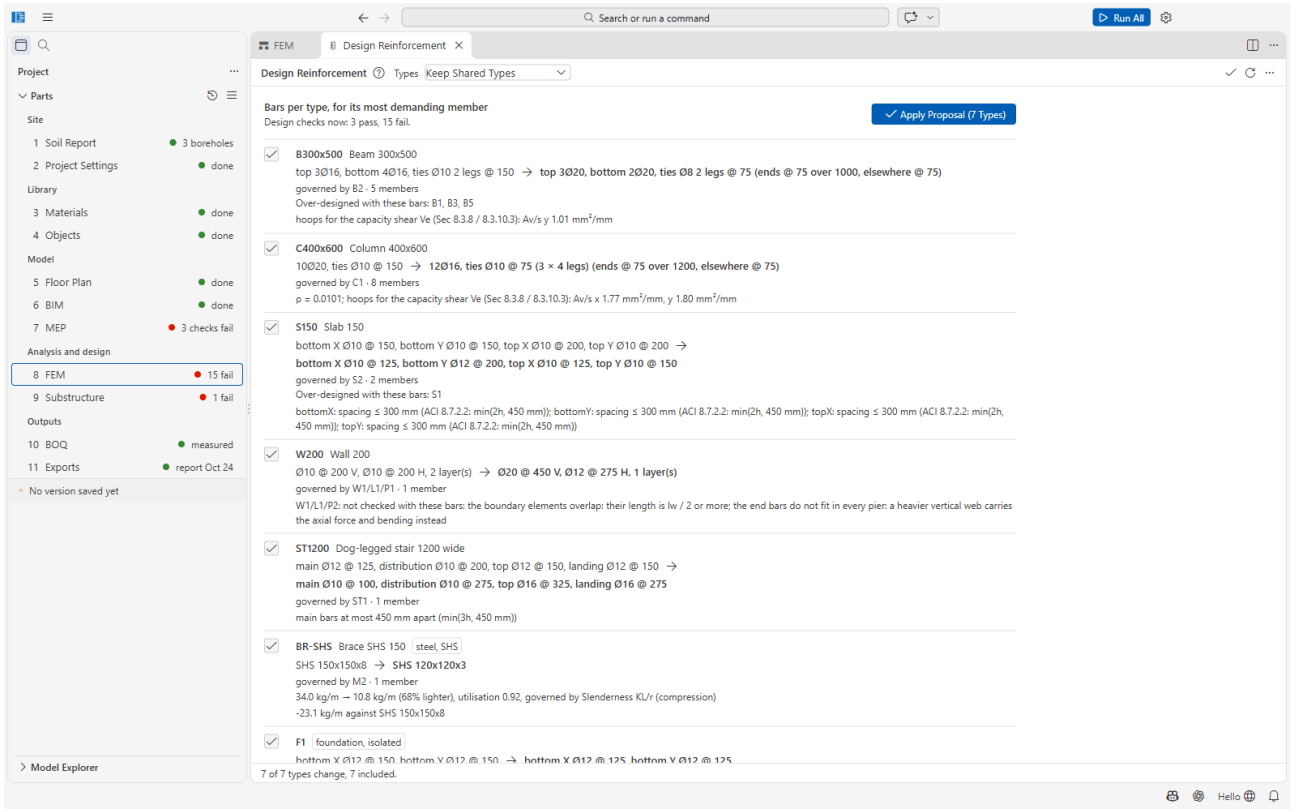


Figure 31: The design proposal

3. Click **Apply Proposal (7 Types)**. SAEngine writes the bars into 4 Objects and checks again: *Proposal applied — 3 pass, 15 fail → 18 pass, 0 fail.* (**Undo Apply** would revert it.)

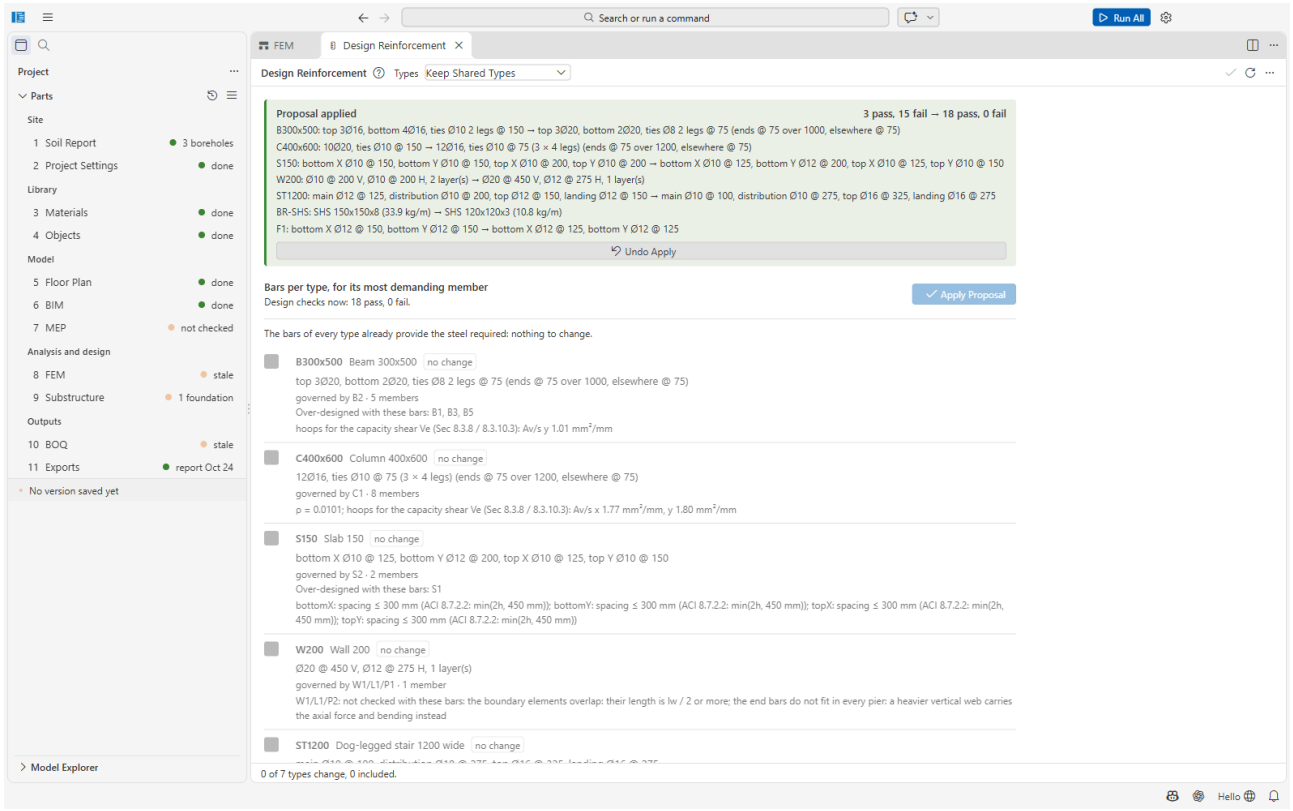


Figure 32: The proposal applied

The analysis model is now **out of date**: the brace section changed, so its stiffness did. You will run everything again after the foundations.

4. Size the footings

Footing F1 fails in bearing (*Bearing (service) 374 kPa against qa 150 kPa*) and five columns have no foundation.

1. Open **9 Substructure**. The plan shows F1 in red and the columns C2 to C6 without footings.
2. Click **Size Footings**. SAEngine proposes footings for the five columns, drawn dashed (*Proposed sizes for 5 foundations*). Click **Accept**.
3. Click F1 to select it, click **Size Footings** again, and **Accept** the new size.
4. Save (**Ctrl+S**): the checks run on the saved file. The **Results** view shows *6 foundations checked, 0 fail*.

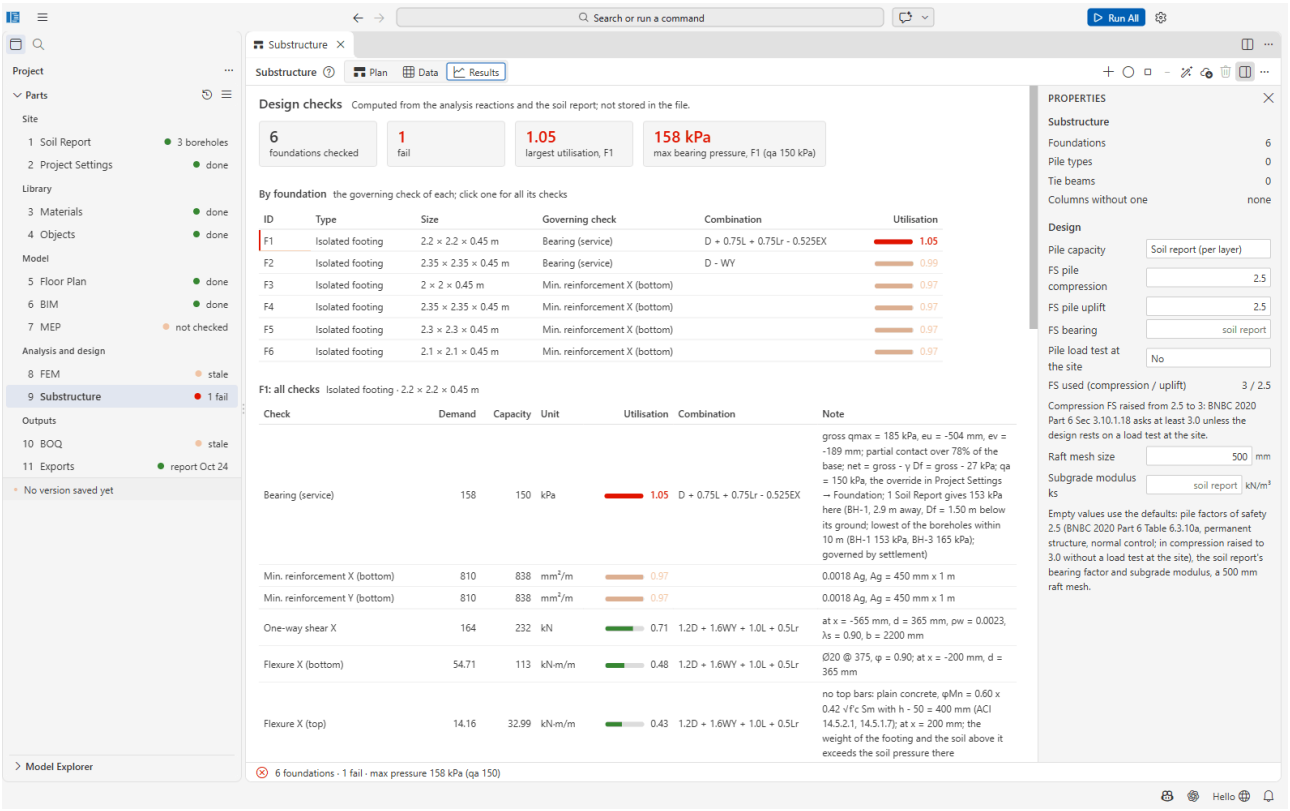


Figure 33: The foundation checks after sizing

5. Run All again, then the drawings

1. Click **Run All** again. It derives the model again (the objects changed), reruns the analysis and the checks: *Done: 18 pass, 0 fail · foundations 6 pass, 0 fail · ….*
2. Open **11 Exports**. Check the **Title block and sheets** (firm, engineer, checked by, …; **Edit in Project Settings** to change them) and leave the sheet size at **A1**.
3. Click **Generate Drawings**. After a few seconds the sheet list shows eleven sheets, S-101 *General notes* to S-111 *Bar bending schedule (2 of 2)*.

The drawings report **5 problems to resolve before issue**, for example:

- *C1: Column bars into foundation, C1: 328.53 mm needed, 251 mm provided, BNBC 2020 Part 6 Sec 8.3.11.2(a), (b); 8.2.6: not met.* The footings the sizing assistant chose for bearing and shear are too thin to develop the column bars.
- *S2: Top X bars Ø10 hooked into the edge support W1: 205.3 mm needed, 180 mm provided.* The 200 mm wall is too thin to anchor the slab's top bars.

Until they are resolved, the general notes sheet is marked **NOT FOR CONSTRUCTION**.

6. Resolve the drawing problems

Design is iterative: each fix changes the loads or the bars a little, so check again after each round. This is the sequence that resolves the sample's problems.

Round 1: thicker footings and wall.

1. In **9 Substructure**, click each of F1, F3, F5 and F6 and set **Thickness (mm)** to **450** in the Properties panel (press Enter). Save.
2. In **4 Objects**, select the wall type **W200** and set **Thickness** to **250**. Save.

3. **Run All**. Because a drawing set exists and is now out of date, Run All regenerates it at the end: *Done: 16 pass, 2 fail; steel short in 8 element(s) · foundations 2 pass, 4 fail · … · 11 drawing sheets drawn*.

The stiffer wall attracts more force (two members are now steel short) and the thicker footings need more bars: in 9 Substructure → **Results** the governing check of F1, F3, F5 and F6 is now *Min. reinforcement X (bottom)*.

Round 2: bars again.

4. **Design Reinforcement** → **Apply Proposal**: *16 pass, 2 fail → 18 pass, 0 fail*. The proposal also gives the footings heavier bars (*F3: bottom X Ø12 @ 200 → bottom X Ø20 @ 375*).
5. On the Exports page click **Regenerate Drawings**. One problem is left: *S2: Top X bars Ø16 hooked into the edge support W1: 328.5 mm needed, 230 mm provided.* The proposal chose Ø16 top bars, which a 250 mm wall cannot anchor. 9 Substructure shows F1 at 1.05 in bearing.

Round 3: engineering judgement.

6. In **4 Objects** select **S150** and change the top X bars from Ø16 @ 275 to **Ø10 @ 100** (about the same steel area, a much shorter anchorage). Save.
7. In **9 Substructure** select **F1** and set **Length** and **Width** to **2300**. Save.
8. **Regenerate Drawings**: *11 sheets, A1, current* — no problems left.
9. **Run All** once more so that the report and the bill of quantities match the final design.

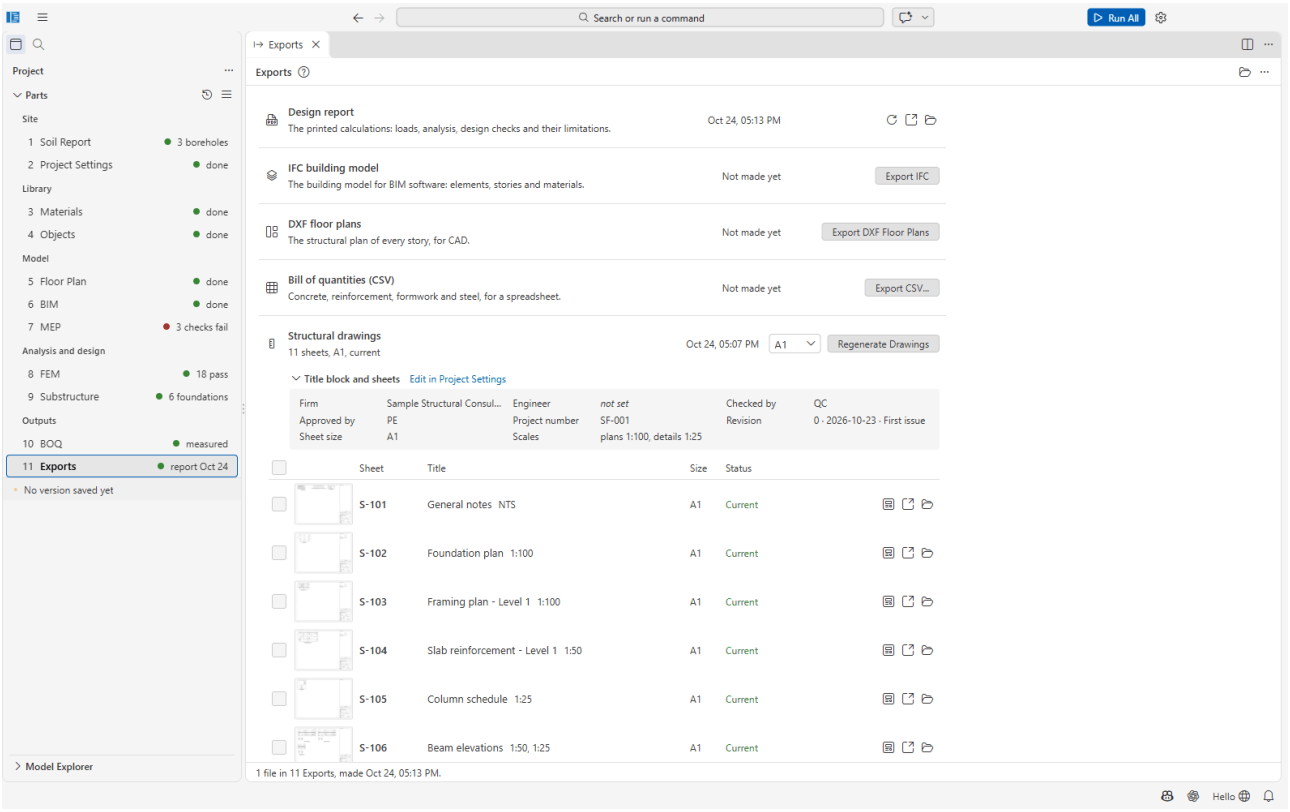


Figure 34: The drawing set on the Exports page

7. Look at the outputs

- **Preview** a sheet (e.g. S-106 *Beam elevations*); **Open in LibreCAD** opens its DXF; **Open PDF Set** the whole set; **Bar Bending Schedule (CSV)** the schedule.

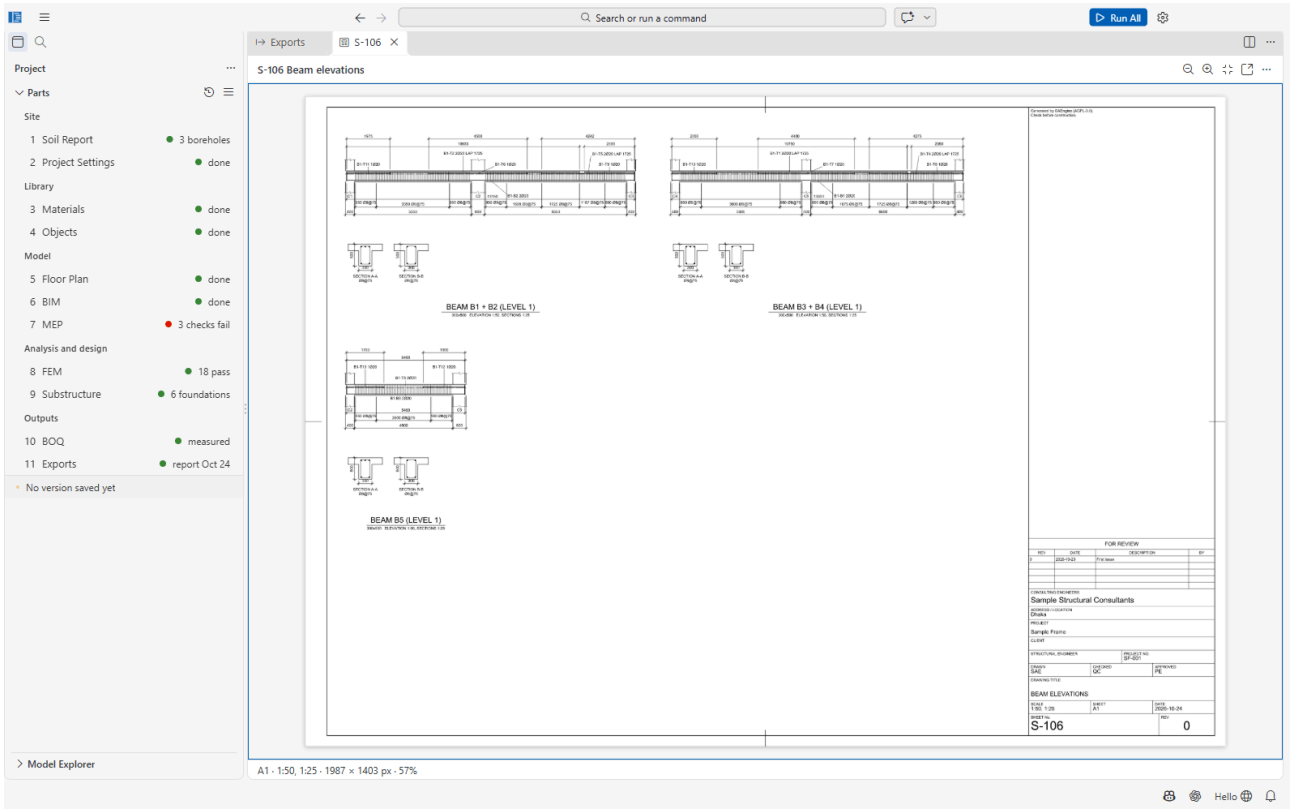


Figure 35: Sheet S-106 in the preview

- **Open Report** (or *Design report* on the Exports page) opens `11 Exports/report.pdf`, the printed calculations.
- **10 BOQ** shows the quantities measured after the last run.

Finally, save a version: **Save Version…** in the ≡ menu, message *Designed bars and footings*.

What you learned

- Run All runs the whole chain; failing checks are results, not errors.
- *Steel short* is fixed by **Design Reinforcement** a failing foundation by **Size Footings** or a bigger foundation; *section too small* by a bigger section (Tutorial 2).
- After a change to the object types, run the analysis again.
- The drawings check the detailing (anchorage, development lengths) and say what to fix before issue.

Tutorial 2: A new 6-storey RC building

You will generate a six-storey RC frame building from its grid, edit the floor plan with window selection, Fill and Copy to Stories, set the floor loads, run the analysis and design, fix the columns that come out *section too small*, and produce the drawings and the bill of quantities.

Time: about 1.5 hours, much of it waiting for Run All.

The building: a residential frame in Dhaka, three bays of 6 m in X and two bays of 5 m in Y, ground storey 3.6 m, five typical storeys of 3.2 m, concrete C27.6 and rebar B500DWR.

1. Generate the building

1. On the start page (or ≡ → **New Building from Grid…**) open the **New Building** wizard.
2. Fill in:

Field	Value
Name	*Six Storey Demo*
Folder	your projects folder
X (A, B, C …)	`3×6000`
Y (1, 2, 3 …)	`2×5000`
Number / ground, mm / typical, mm	`6` / `3600` / `3200`
System	RC frame
Concrete / Reinforcement	C27.6 / B500DWR
Footings	Isolated
Location	Dhaka (zone 2, 65.7 m/s)
Building use / Occupancy category	Residential / II

The preview draws the plan and the elevation and sums up the bays, storeys and columns.

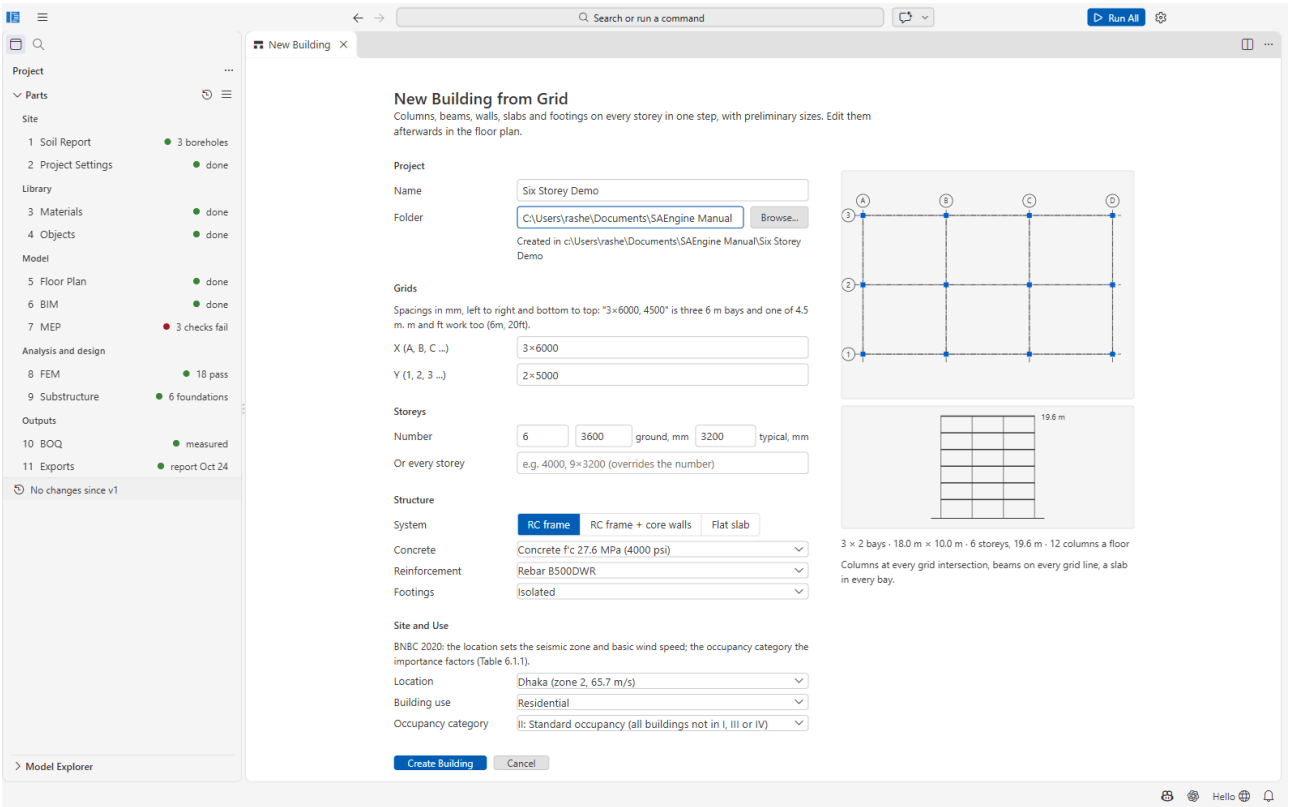


Figure 36: The New Building wizard

3. Click **Create Building**. The page *Six Storey Demo Is Ready* lists the **Generator Notes**: how each preliminary size was chosen. Read them, then click **Open Building**.

The first note is a warning worth reading:

> *WARNING - Seismic Design Category D: the column sizes are for GRAVITY ONLY. The earthquake moments (and, in a special moment frame, the strong-column / weak-beam and joint rules) usually govern the columns here, the corner and edge ones most. Expect Design mode to report some columns as 'section too small': enlarge those column types (Members Table or 04 Objects) after the first Run All.*

The other notes give the Seismic Design Category (*D, zone 2, site class SD, occupancy category II; system RC special moment frame*) and the rule behind each size.

The project opens at the floor plan (if the plan is still blank, click **5 Floor Plan** in the sidebar again). Look at what was generated:

- **4 Objects:** column types C300x300, C350x350 and C400x400 (by storey group and position: interior, edge, corner), beam types B250x400 and B250x350, slab types S200 and S175 — all **without bars**.
- **5 Floor Plan:** the grids A–D and 1–3, the storeys Base and Level 1 to Level 6, a column at every intersection, beams on every grid line, a slab in every bay, and **typical floors**: the storey picker shows *Level 2 (+6800) (typical)*, *Level 3 (+10000) = Level 2*, *Level 4 (+13200) (typical)*, *Level 5 = Level 4*, *Level 6 = Level 4*. A new family starts where the column types change.
- **9 Substructure:** twelve isolated footings sized for an assumed qa = 100 kPa.
- **2 Project Settings:** residential, Dhaka, structural system *RC special moment frame*.

2. Edit the floor plan

The architect needs a stair and lift void in bay **B–C / 2–3** on every floor, and a stair room on the roof over it.

a) Remove the slab of the void with a crossing window.

1. In the storey picker choose **Level 1 (+3600)**. Click **Zoom Extents** hide the Properties panel if you want more room.
2. With the **Select** tool, drag **from right to left** inside bay B–C / 2–3, without touching the grid lines: a dashed *crossing* window. The status line says *Crossing window: 1 member(s) touched. · Selected 1: 1 slab*.

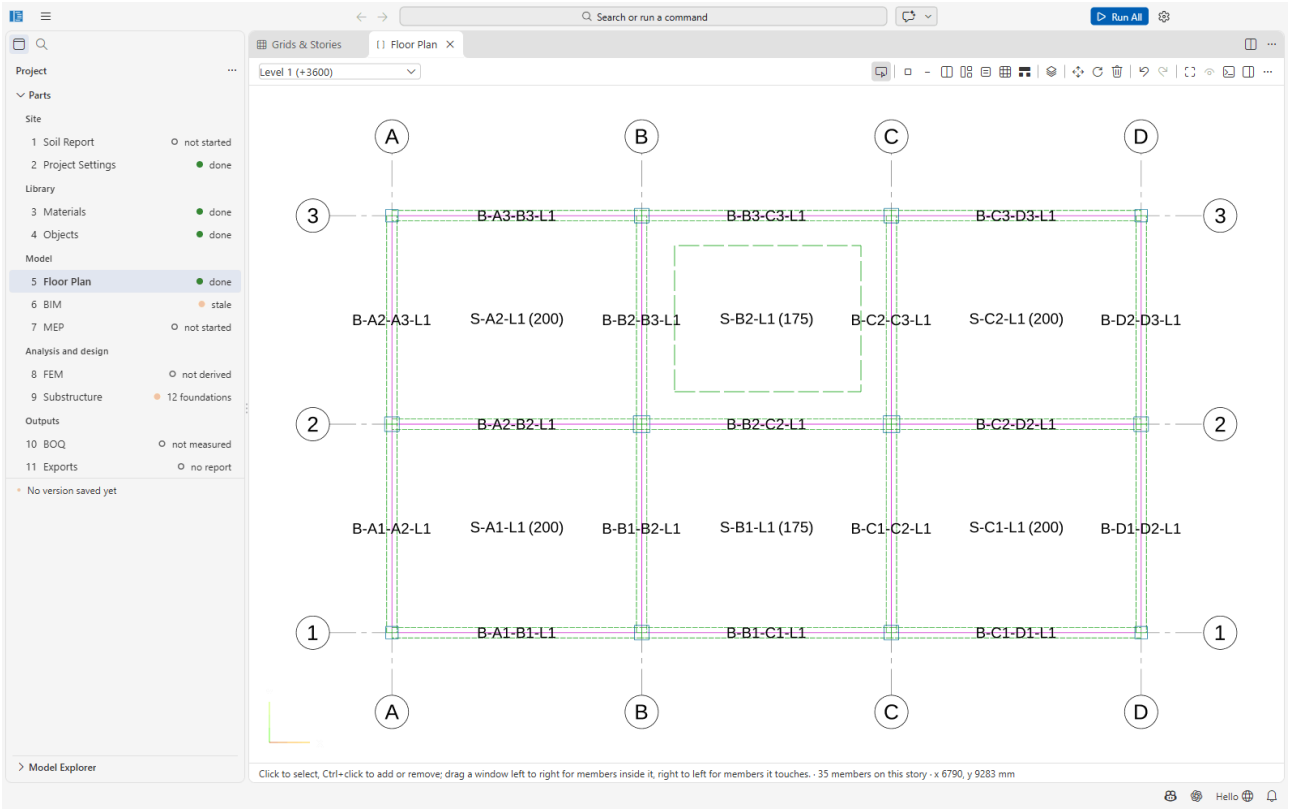


Figure 37: A crossing window (right to left) inside the bay

3. Press **Delete**: *Deleted 1 member(s)*.
4. Choose **Level 2 (+6800) (typical)** and do the same. The status line says *Typical floor Level 2: edits repeat on all 2 stories*: Level 3 loses its slab too.

5. Choose **Level 4 (+13200) (typical)** and do the same (Levels 5 and 6 follow).

b) Add a storey for the stair room.

6. In **…** choose **Edit Grids and Stories**. Under the storeys click **Add Story** and enter ID *SR*, Name *Stair room*, Elevation *22400* (2.8 m above Level 6). Close the form and save.

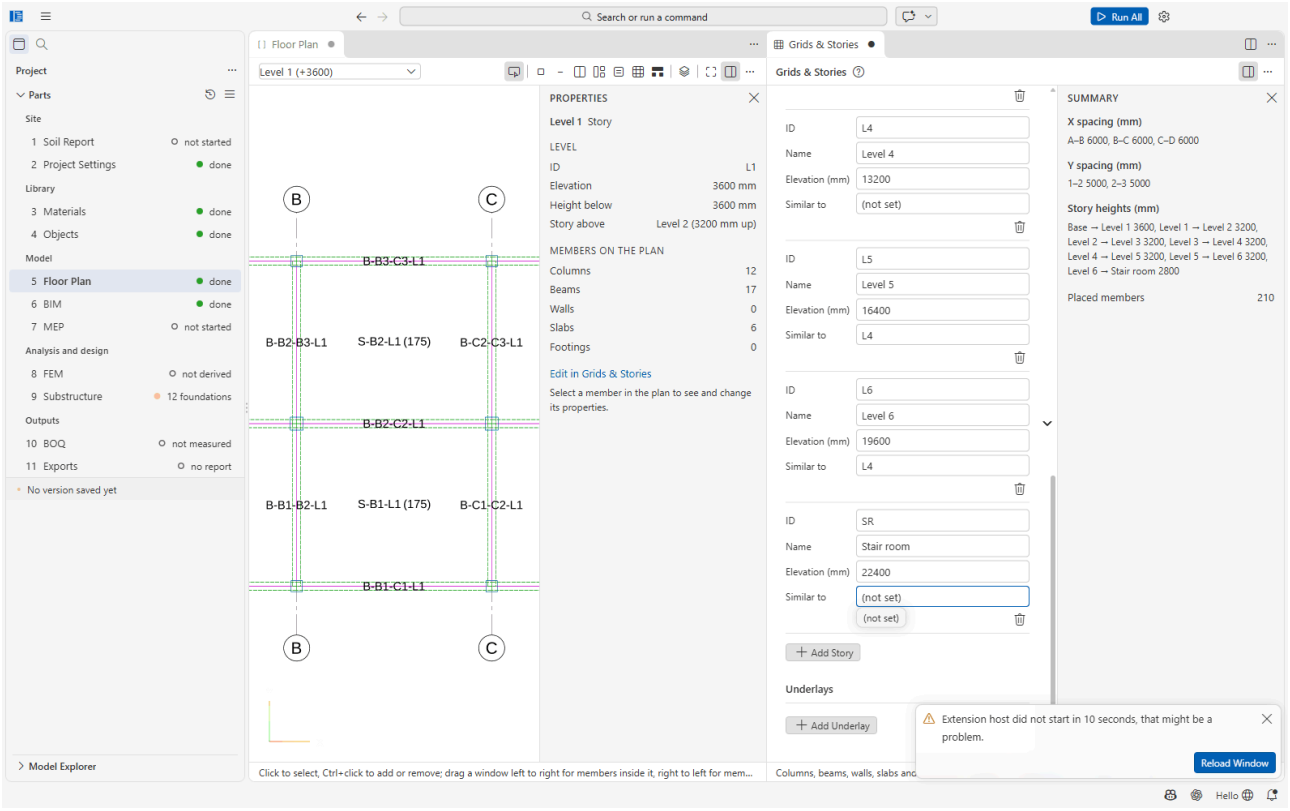


Figure 38: Grids & Stories with the new storey

c) Copy the frame of the bay up with a window and Copy to Stories.

7. Choose **Level 6 (+19600) = Level 4**. Drag **from left to right** a window a little larger than bay B–C / 2–3: a solid *window* selects what lies fully inside: *Window: 8 member(s) fully inside. · Selected 8: 4 columns, 4 beams*.

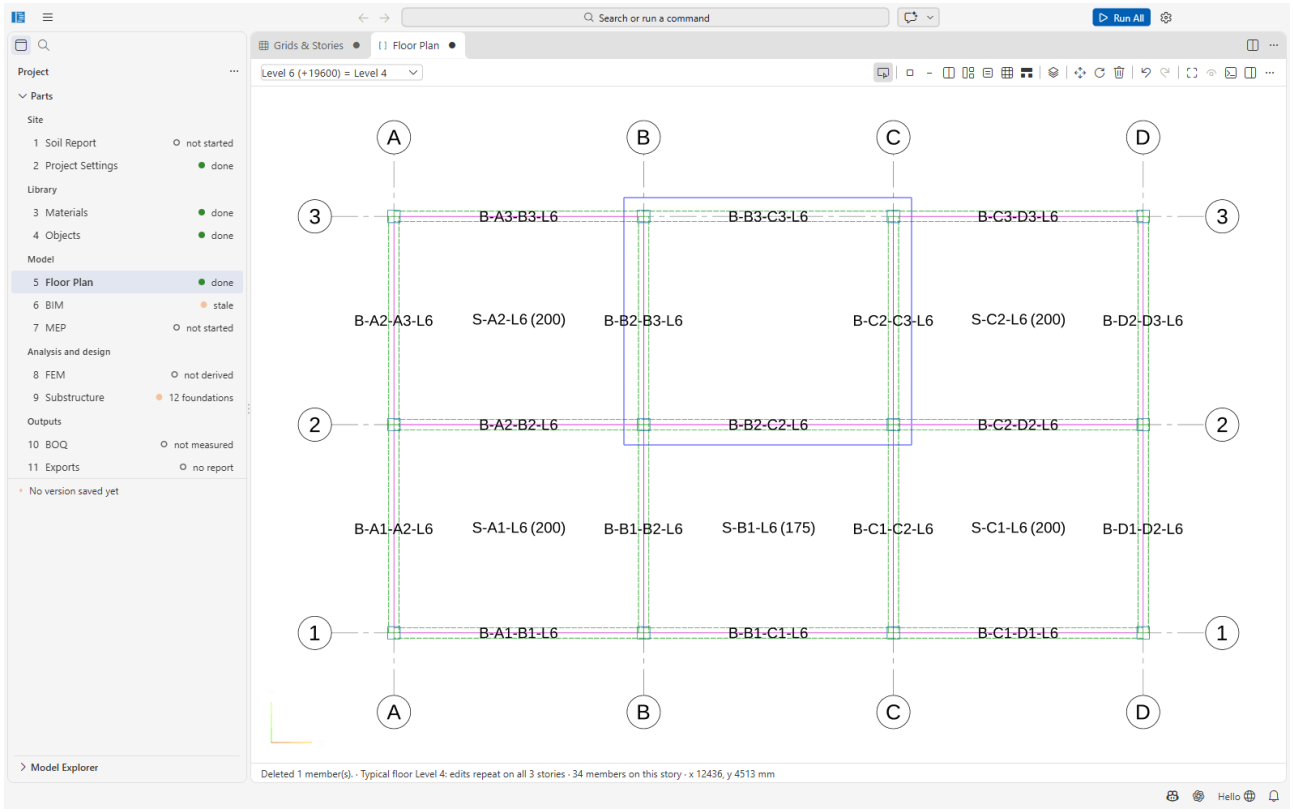


Figure 39: A window (left to right) selecting the four columns and four beams round the bay

8. In **...** choose **Copy Selection to Stories...**, tick **Stair room+22400** and click **OK**: *Copied 8 member(s) to Stair room (and their typical floors); skipped 0 already there.* The copied columns rise from Level 6.

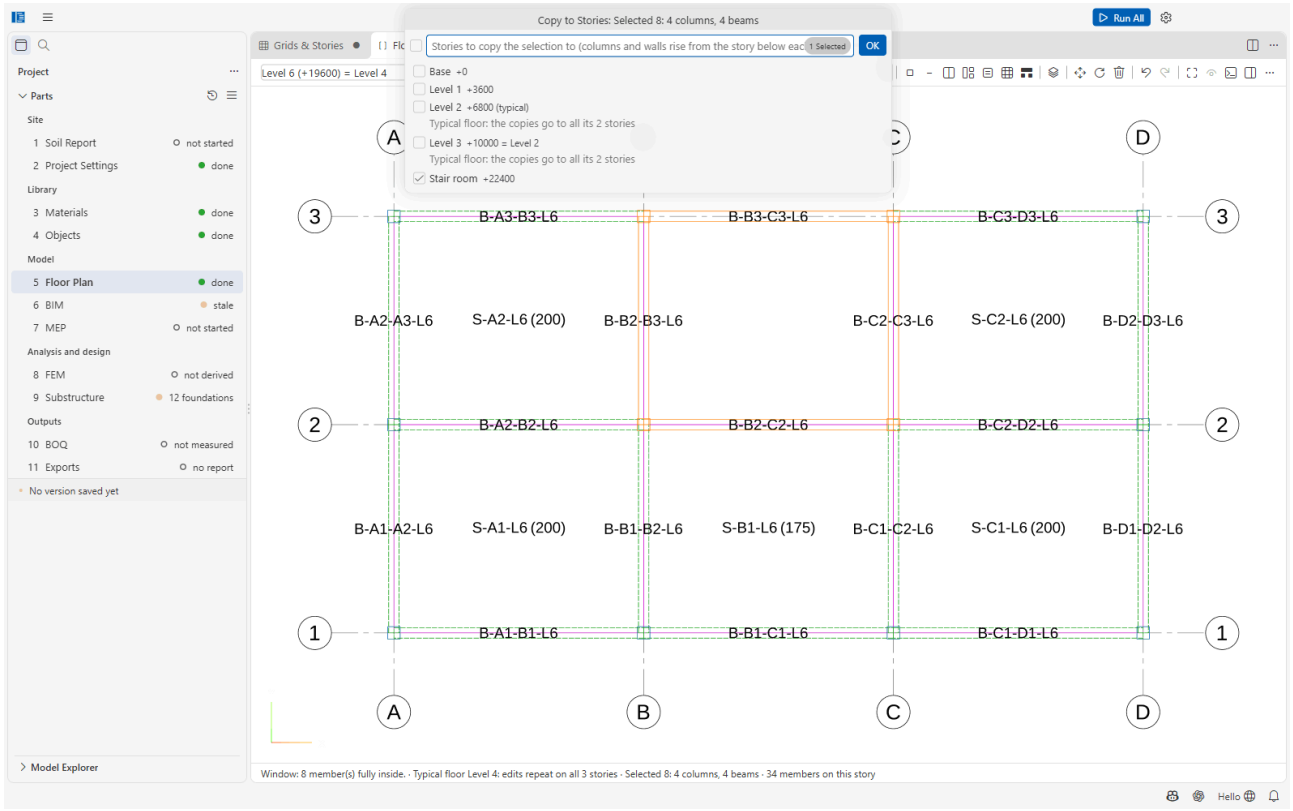


Figure 40: Copy to Stories

d) Fill the roof slab of the stair room.

9. Choose **Stair room (+22400)**. Click **Fill** in the pickers next to it choose **Slabs in Bays** and the type **Slab 200 (S200)**.

10. Drag a window over the bay: *Placed 1*. Press Esc and save (Ctrl+S).

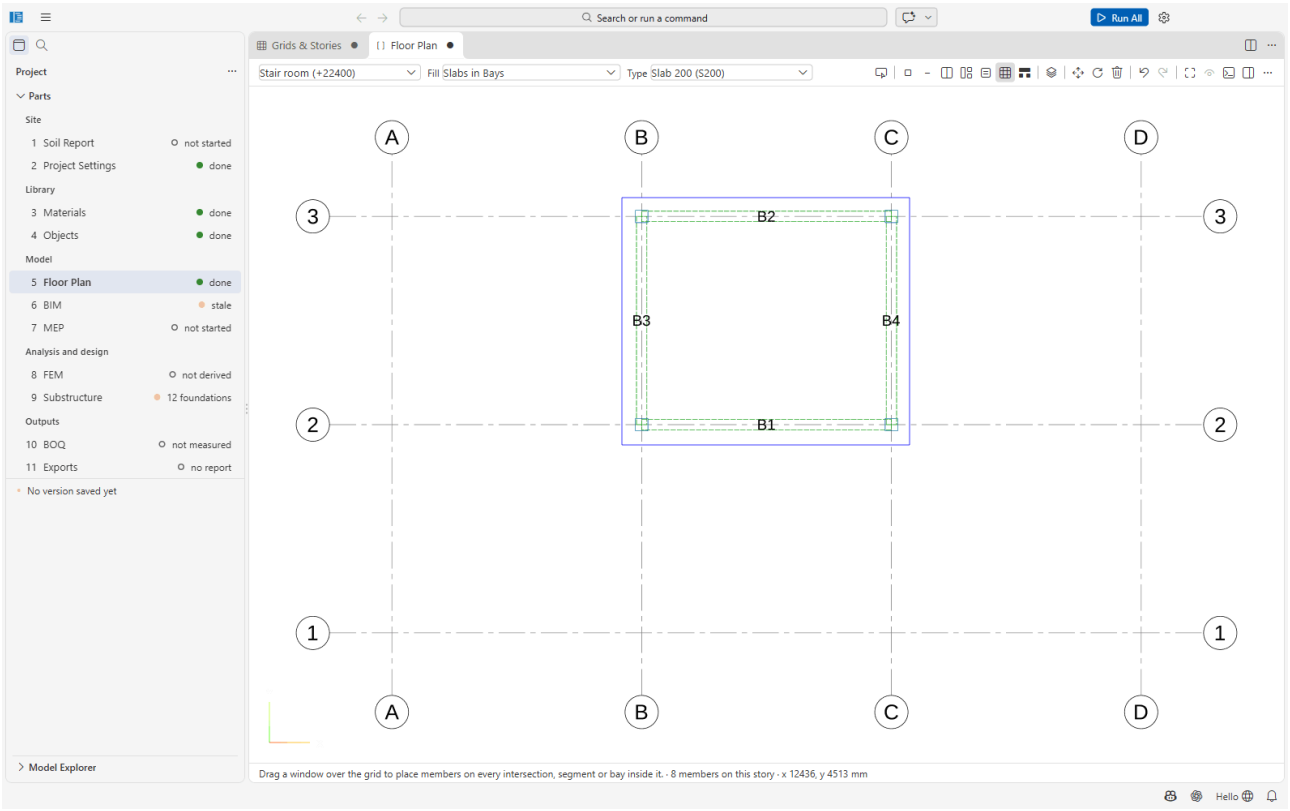


Figure 41: Fill: slabs in every whole bay inside the window

The same tools work at any scale: **Fill** → **Columns at Intersections** over the whole grid places every column of a storey in one drag, and the command line (press **:**) takes `fill beams`, `copy 6000,0 3` or `tostories all`.

3. Floor loads

Open **2 Project Settings** → **Area Loads**. The generator set the residential floor load:

Area load	Use	Live	Floor finish	Partitions
FL Floors	Residential	2.0	1.2	1.0
RF Roof	Roof	1.0	1.5	—

Change them to your project's values (for example a heavier floor finish), and check that **Floors** is *FL* and **Roof** is *RF*. A live load below the BNBC minimum for the use is flagged. Save (Ctrl+S).

Self-weight is computed from the sections; there are no load entries on the plan.

4. Build and Run All

1. Open **6 BIM** and click **Rebuild from Floor Plan** (it is highlighted because you edited the plan). Check the building in the **3D View**.
2. Click **Run All**. A six-storey building takes noticeably longer than the sample (several minutes to a quarter of an hour, depending on the PC).

On the PC used for this manual it took about 20 minutes, most of it in the design checks.

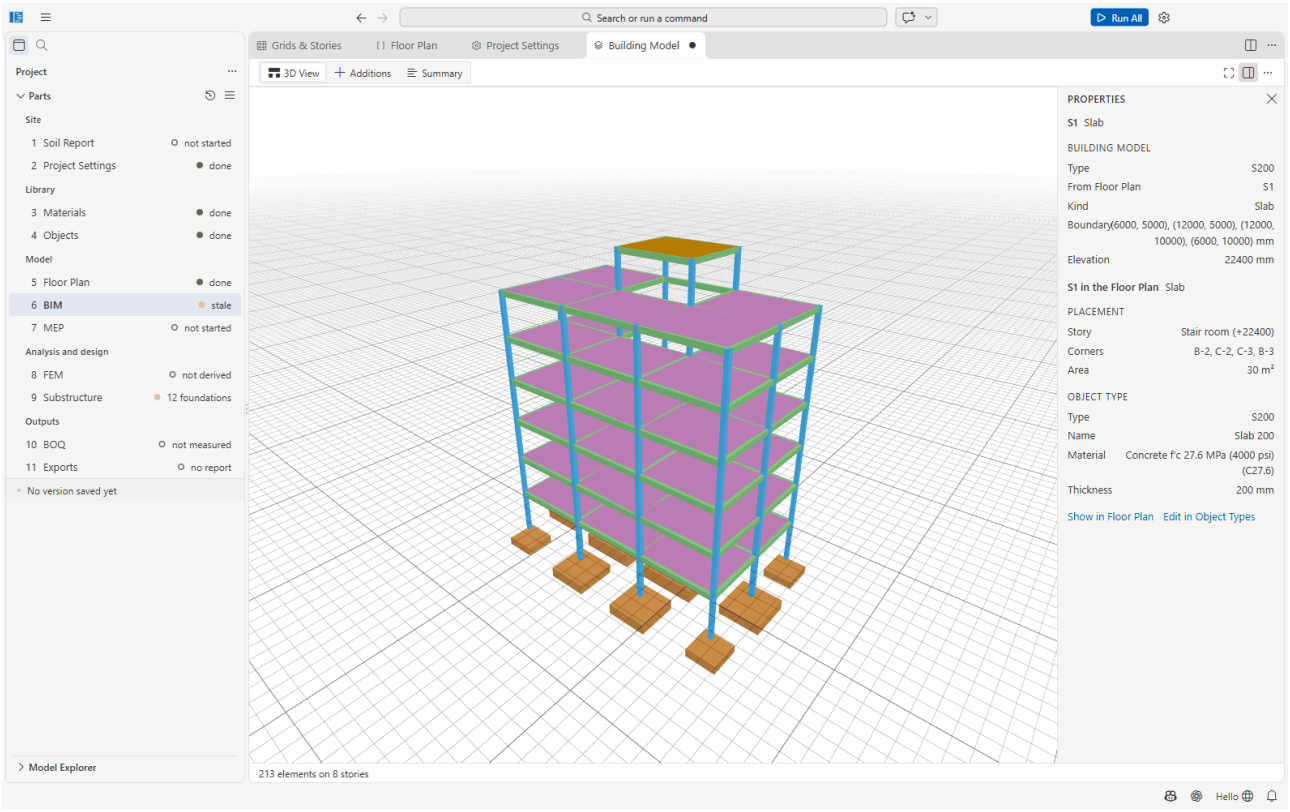


Figure 42: The building model with the void and the stair room

The result:

Done: 0 pass, 0 fail, 36 section(s) too small · foundations 0 pass, 0 fail · model derived · BOQ updated · report written

No member passes or fails yet: the generated types have **no bars**, so the checks report the steel each member requires. But 36 columns are **too small**: no practical bars can carry their demand. Open **8 FEM** and choose **Design Utilisation**: the Properties panel lists them under **Too small** (*C-A1-L1, C-A3-L1, C-D1-L1, …*: corner and edge columns of the lower storeys first, as the generator warned) and every type under **No bars yet**. The response-spectrum summary shows the earthquake governs: *EX scaled × 3.22 to 85 % of the static base shear*, and the largest design drift is 1.67 % against 2.00 % allowed.

5. Fix the columns that are "section too small"

1. Click **Design Reinforcement**. The proposal can only give bars to the beam type B250x400 (*Apply Proposal (1 Type)*); the other types are marked **section too small** with a suggestion:

Type	Members	Message
C300x300	52	*Section too small: no layout with ρ up to 0.08 (ACI 10.6.1.1) carries the demand; try 400 x 400 mm (+100 mm) with about ρ = 0.025; the analysis must be run again*
C350x350	18	*… try 500 x 500 mm (+150 mm) …*
C400x400	6	*… try 550 x 550 mm (+150 mm) …*
B250x350	50	*Section too small: no practical bars found for the demands; try 250 x 400 mm (depth +50 mm); the analysis must be run again, the stiffness changes*

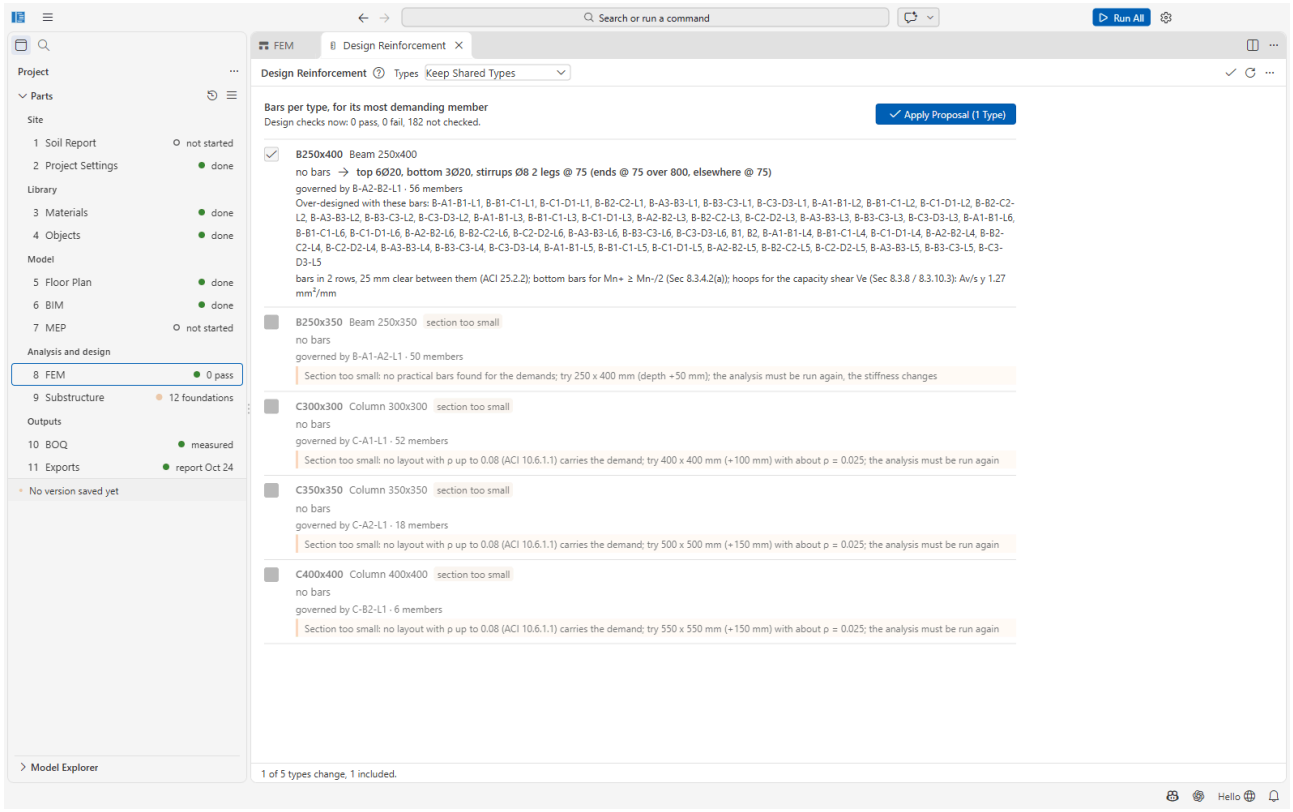


Figure 43: The proposal with the types that are too small

Do not apply yet: the bars of B250x400 would be designed for forces that change when the columns grow.

2. Click **Open Object Types** (or open **4 Objects**). For each type, select it and change **Width** and **Depth** as suggested, and its **Name** so that it still describes it: C300x300 → 400 × 400, C350x350 → 500 × 500, C400x400 → 550 × 550, B250x350 → 250 × 400. The IDs stay the same, so every member on the floor plan keeps its type. Save.

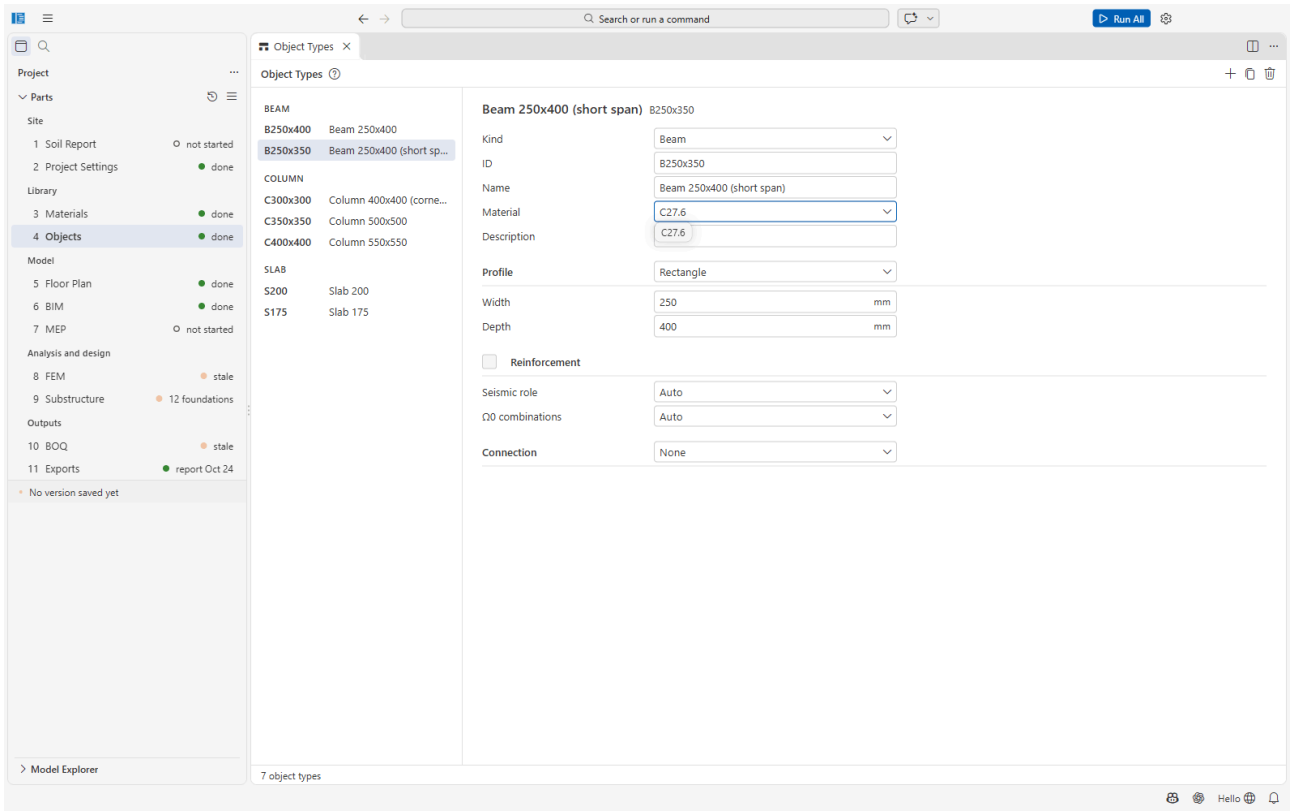


Figure 44: A column type made bigger in 4 Objects

To give only some members a bigger section, create a new type instead (**Duplicate**) and assign it to those members on the floor plan (select them, then **Type** in the Properties panel, or the Members Table).

3. **Run All** again: the model is derived again (the stiffness changed) and every member is checked again.

Done: 0 pass, 0 fail · foundations 0 pass, 0 fail · model derived · BOQ updated · report written — no section is too small any more; every type still waits for its bars.

4. **Design Reinforcement** again. Now all five frame types get bars (*Apply Proposal (5 Types)*), for example *C300x300: no bars → 12Ø20, ties Ø10 @ 100 (4 legs)* and *B250x400: no bars → top 3Ø25, bottom 4Ø16, stirrups Ø8 3 legs @ 125*. Click **Apply Proposal**: *0 pass, 0 fail, 182 not checked → 132 pass, 50 fail*. The proposal then says *The bars of every type already provide the steel required: nothing to change.*

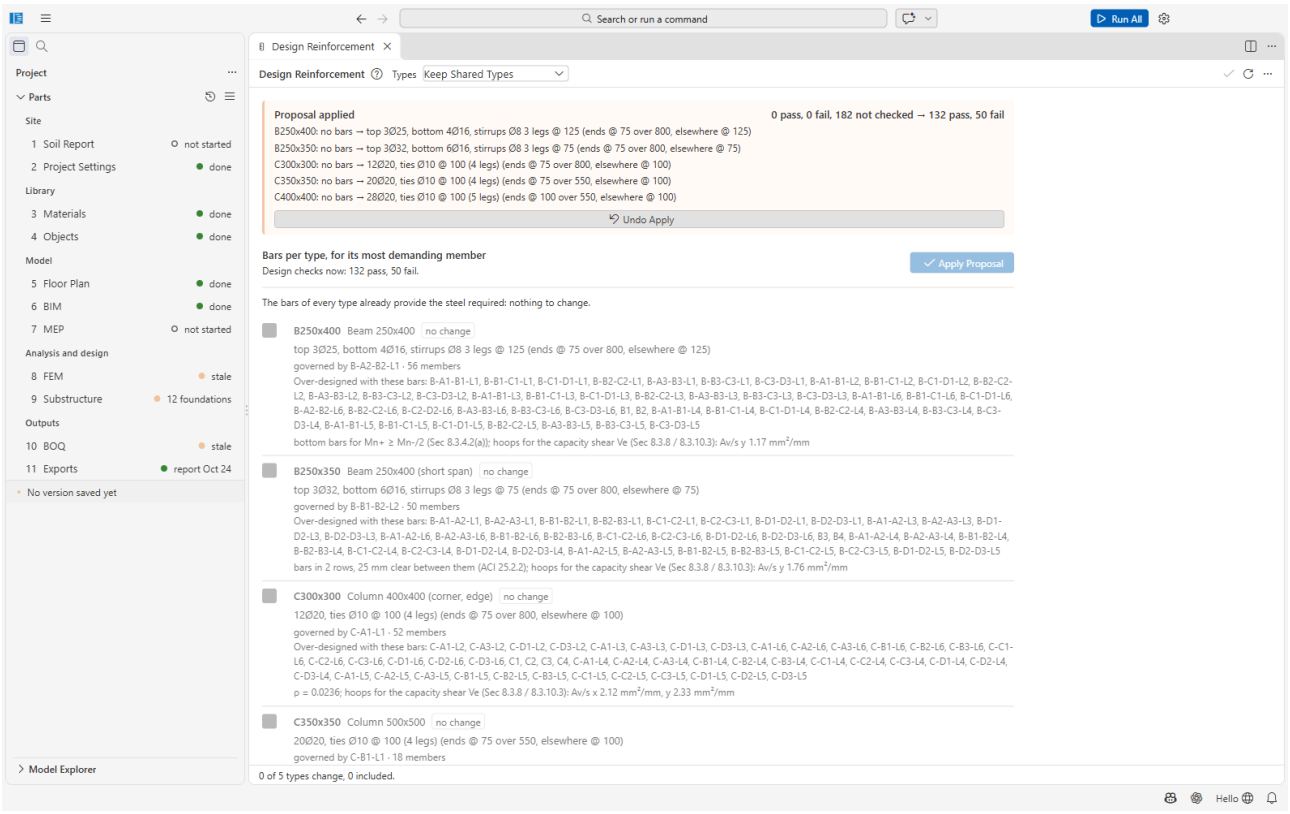


Figure 45: The proposal applied

5. **Run All** once more (the bars are new, and the model is out of date): *Done: 132 pass, 50 fail; steel short in 6 element(s)* · …

Why 50 still fail, and what to do. Open the report (`11 Exports/report.pdf`, chapter *Member design*): every failing element is a beam of type **B250x350**, governed by *Detailing: Largest steel ratio, top (8.3.4.2(a))*: 3.0 % against the 2.5 % that BNBC 2020 Sec 8.3.4.2(a) allows in the beams of a special moment frame. The proposal had to put 3Ø32 on top of a 250 × 400 beam to carry the earthquake moments. More bars cannot fix a steel-ratio limit: the beam needs a bigger section, exactly like the columns before.

6. In **4 Objects** make **B250x350** wider and deeper (for example 300 × 450), save, **Run All**, **Design Reinforcement** → **Apply Proposal**, and **Run All** again. Repeat until 8 FEM reports no failing member. The six *steel short* elements are marginal (for example *required 1476.6 mm², provided 1472.6 mm²*) and are fixed by applying the proposal after the last analysis.

The manual's walk-through stops at this point; the loop is always the same: **enlarge what is too small or over the steel limit → Run All → Design Reinforcement → Run All**.

> On the PC used for this manual each Run All of this building took 20 to 35 minutes, most of it in the design checks; the checks run again whenever you choose *Design Utilisation* after a change. Plan the iterations, and save a version (≡ → **Save Version…**) after each one.

6. Drawings and bill of quantities

Bill of quantities. Run All measured the bill: open **10 BOQ**. It lists earthwork, concrete by grade and element (columns, beams, slabs, footings), reinforcement by bar diameter, formwork, and the cement (kg and 50 kg bags), sand and stone chips (cft) of the mixes. Click a row for its breakdown by element and storey; **Export CSV…** writes `11 Exports/boq.csv` for your rates.

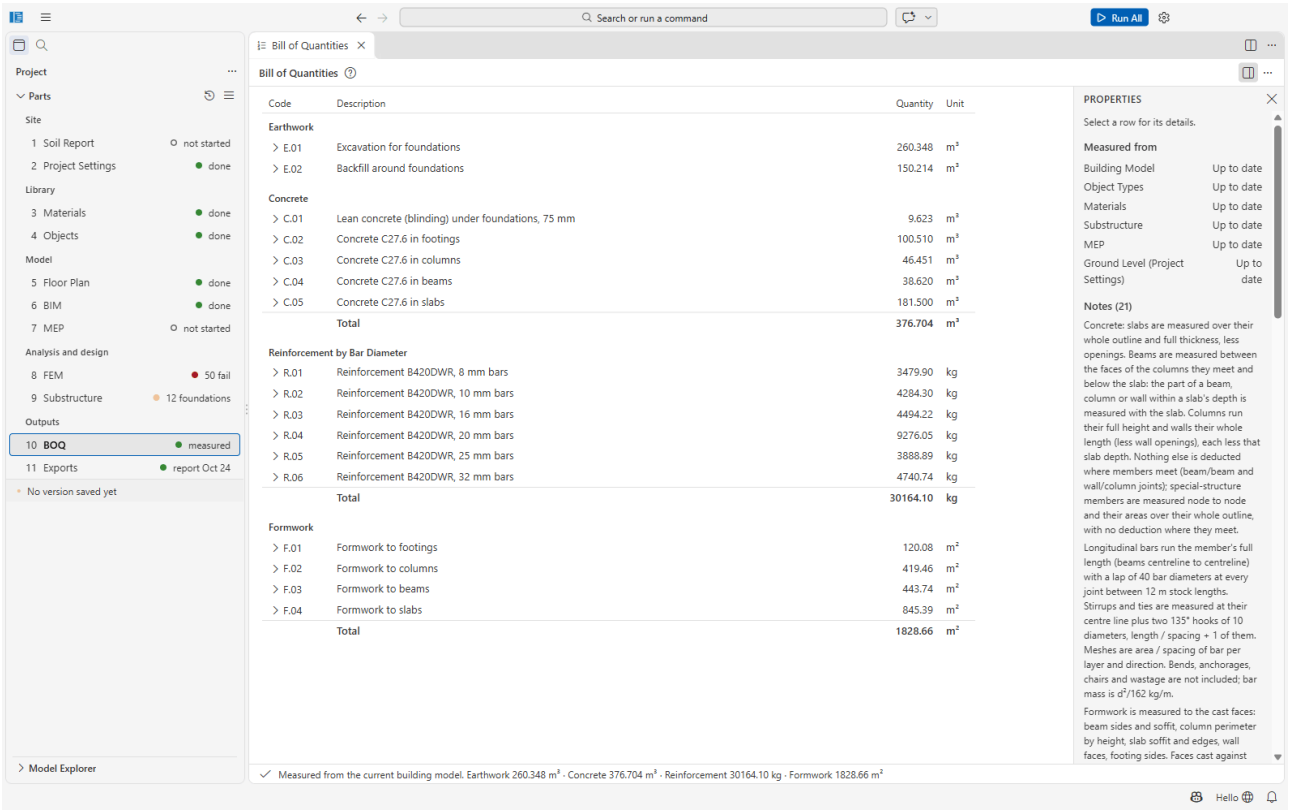


Figure 46: The bill of quantities of the six-storey building

Drawings. Open **11 Exports**, check the **Title block and sheets**, and click **Generate Drawings** (it runs the design checks first, so it takes as long as they do). While members still fail, the general notes sheet says **NOT FOR CONSTRUCTION until these are resolved: …** and the problems are listed under the sheet list: finish step 5 first, then **Regenerate Drawings**.

For this building, at the state where the walk-through stopped, the set has **30 sheets** on A1 (*30 sheets, A1, current; 108 problems to resolve before issue*): general notes, the foundation plan, framing plans (*Framing plans - Level 1, Level 2, … Level 6* on one sheet, and the stair room's own), slab reinforcement plans two levels per sheet, the column schedule, beam elevations on four sheets, foundation details and the bar bending schedule. Fill in **Firm**, **Engineer**, **Checked by** and the other title-block fields (*not set* here) before issuing.

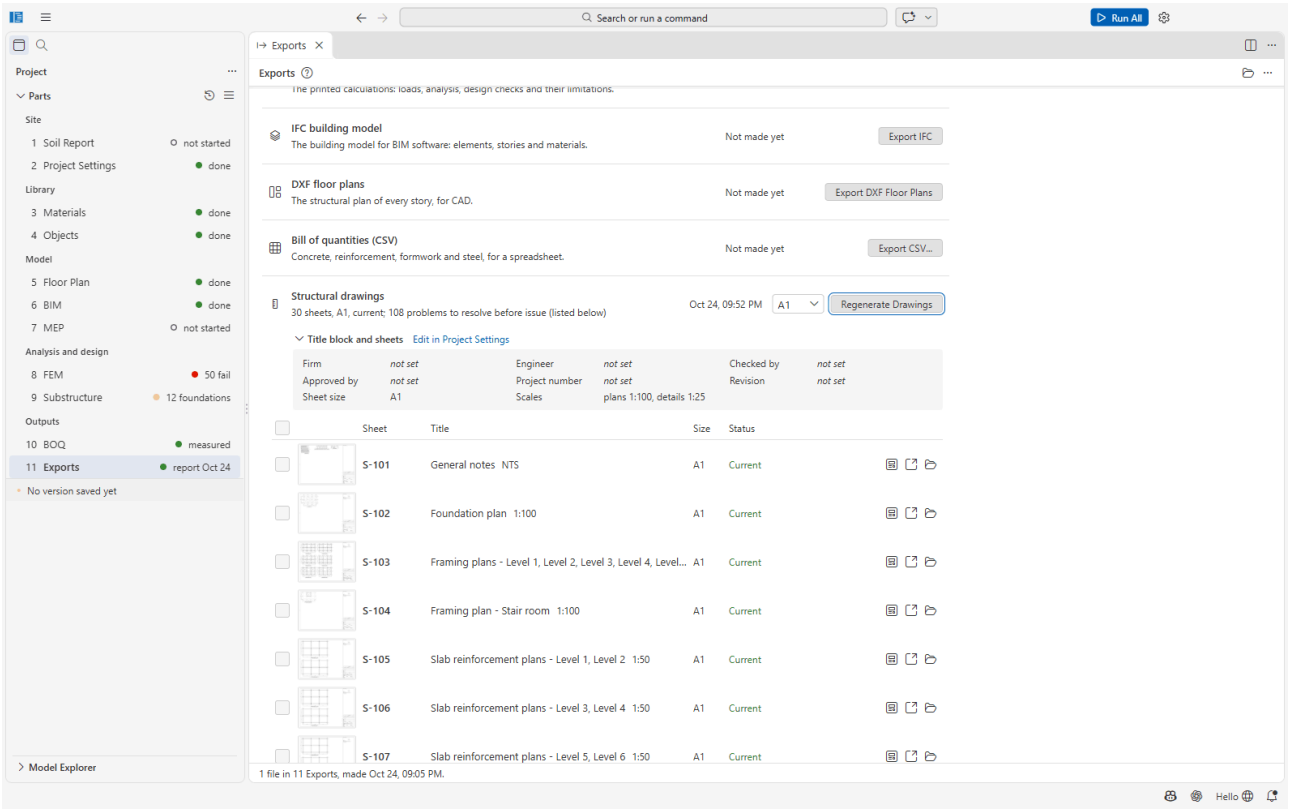


Figure 47: The drawing set of the six-storey building

What you learned

- New Building from Grid gives a complete model with preliminary sizes in one step.
- Window (left to right) and crossing (right to left) selection, Fill, and Copy to Stories.
- Floor loads are area loads in Project Settings.
- *Section too small* is fixed in 4 Objects (a bigger section), followed by Run All and Design Reinforcement.

Tutorial 3: Soil report, site map and footing design

You will enter a borehole log sheet by hand, import a second borehole from a CSV file, place the site on the map, read the bearing capacity, and design the footings of the building from Tutorial 2 in 9 Substructure.

Time: about 40 minutes.

You need the project *Six Storey Demo* from Tutorial 2 (analysed with Run All), and the file tutorial-boreholes.csv. All borehole data in this tutorial is synthetic.

1. Fill in the report

1. Open **1 Soil Report**. It shows *No boreholes yet*.
2. Click **Add Borehole** (it creates BH-1). Select **Report** in the list on the left and fill in the Properties panel: Title *Sub-soil investigation (tutorial, synthetic)*, Consultant *Example Geotechnics (fictitious)*, Ground level *0*.
3. Under **Design** leave the defaults (textbook practice, Idriss & Boulanger 2014, hammer energy 60 %, factor of safety 3, settlement 25 mm), or choose **Bangladesh consultant practice** if that is how your consultant works.

2. Enter the log sheet of BH-1

Select **BH-1** and switch to the **Data** view.

Header strip: ID *BH-1*, Water table *2* (m below ground), Termination depth *20*, X *-2*, Y *-2* (m, next to grid A-1), Date drilled *2026-10-10*, Method *Wash boring, 100 mm*. Leave Ground level empty: the report's 0 applies.

Strata (click a cell and type; Tab moves right, Enter down). The new borehole starts with one clay layer 0–10 m: overwrite it and add rows.

Top	Bottom	Soil type	USCS	Description
0	1.5	Fill		Silty clay with brick chips (fill)
1.5	6	Clay	CL	Stiff brown silty clay
6	10.5	Silt	ML	Medium dense grey sandy silt
10.5	20	Sand	SP-SM	Dense grey fine sand with mica

The folded column groups (*Unit weight*, *Strength*, *Index*, *Consolidation*) take γ, cu, c′, φ′, E, fines, PI and the rest when the report gives them; this tutorial leaves them empty, so SAEngine works from the SPT values.

Samples and SPT (one row per test, *From* the depth; *To* fills in as From + 0.45 m):

From (m)	1.5	3	4.5	6	7.5	9	10.5	12	13.5	15	16.5	18	19.5
N	6	9	11	12	14	16	22	26	30	33	36	40	44

If your log sheet gives the three 150 mm increments, type them in *Seat*, *N1* and *N2* instead: N is worked out as N1 + N2. You can also copy a block of cells from Excel and paste it with **Ctrl+V** at the first cell (a block that is longer than the sheet asks *Paste 13 rows × 8 columns from row 1? It adds 12 row(s).* — click **Paste**).

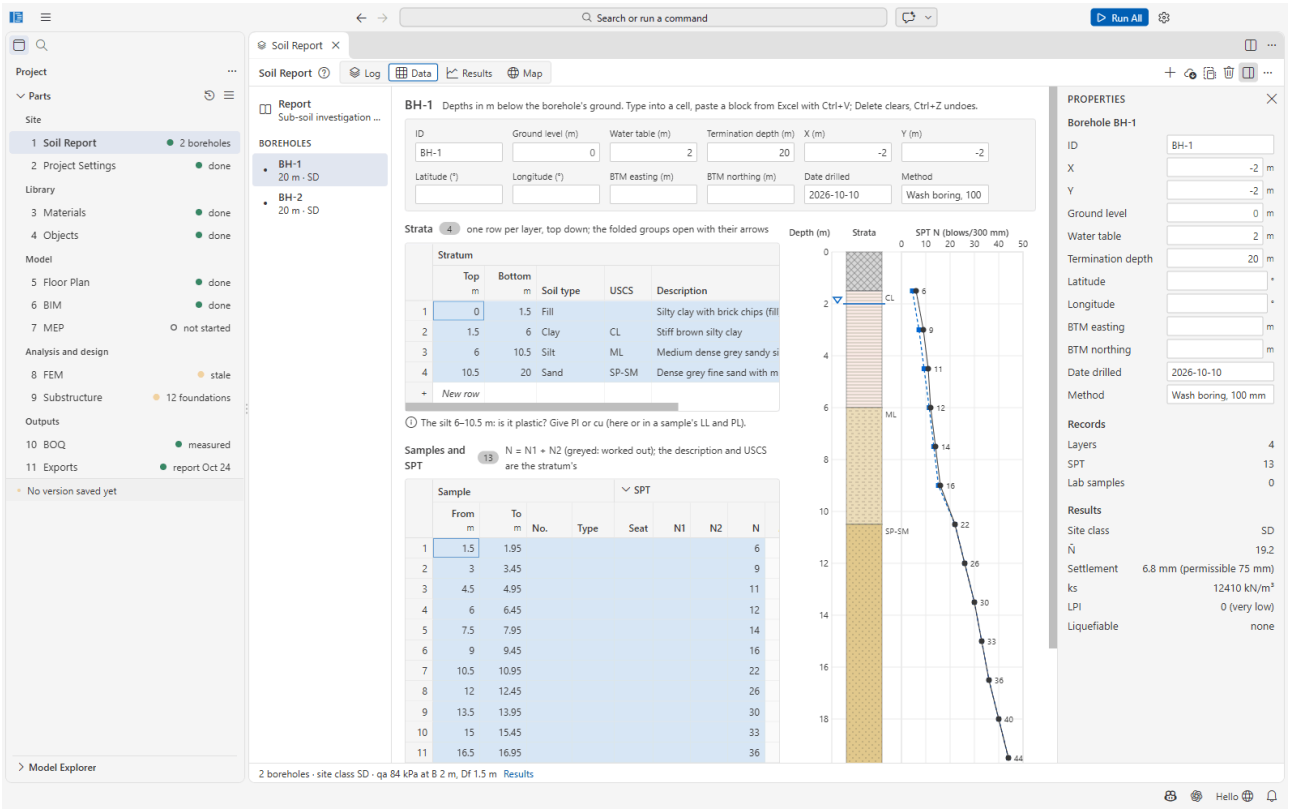


Figure 48: The log sheet of BH-1

SAEngine fills in *To* (From + 0.45 m), the stratum's description and USCS, and the corrected **N60** and **(N1)60** of every row; the compact log beside the sheet draws the strata and the SPT values as you type. The status line now reads *1 borehole · site class SD · qa 84 kPa at B 2 m, Df 1.5 m*. Fix anything listed under *Problems* (for example *BH-1: layer 1 ends at or above its top.* if a row is shifted).

3. Import BH-2 from a file

1. Click **Import CSV/Excel...** and pick `tutorial-boreholes.csv`. SAEngine reports *Imported tutorial-boreholes.csv: 1 new borehole(s).*
2. Look at the file in Excel: one row per record, `Record = layer` or `spt`, the borehole's position and water table on its first row. **Save Template...** writes an empty template in the same layout for your own data.
3. Click **Show All Boreholes Side by Side** in the **Log** view to compare BH-1 and BH-2.

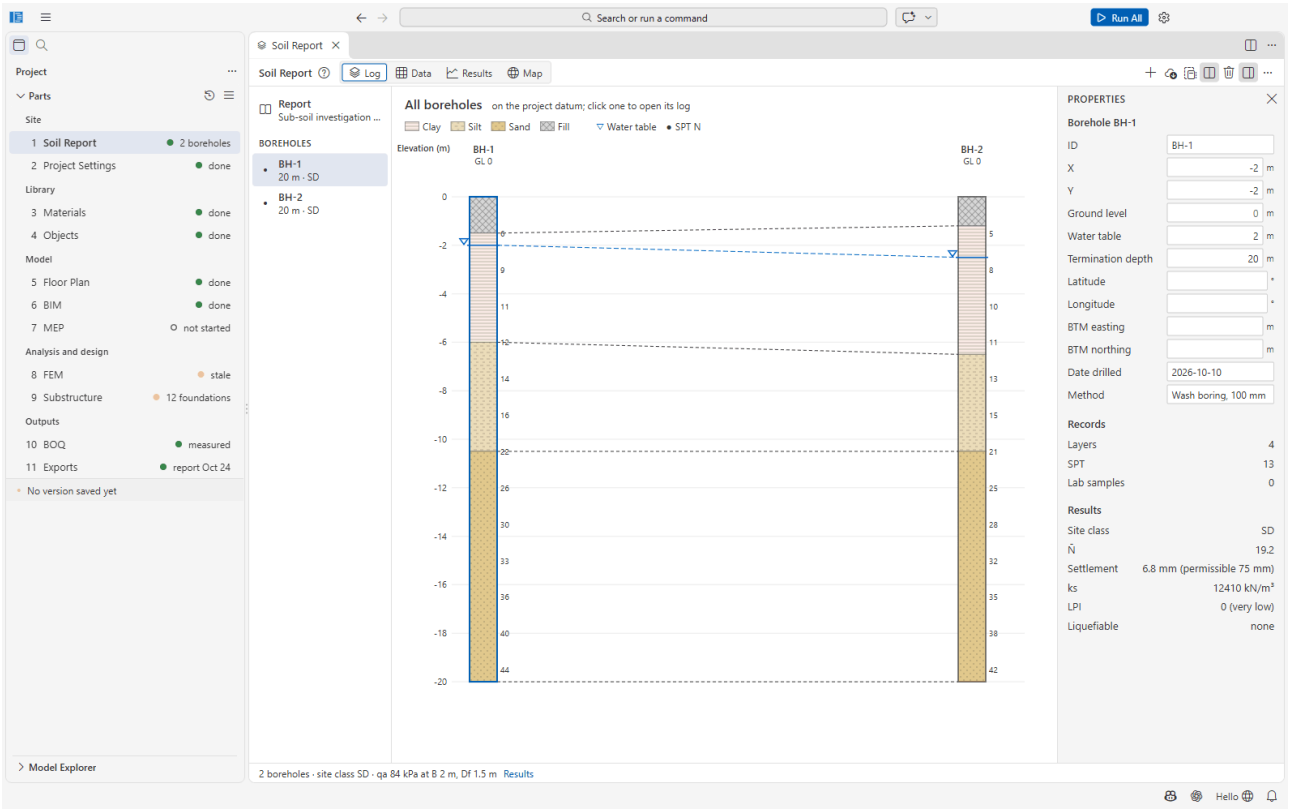


Figure 49: Both logs side by side

4. Place the site on the map

1. Switch to the **Map** view.
2. Under **Site**, type the latitude and longitude of the project point (grid A-1) in the box (placeholder *23.8103, 90.4125*); for the tutorial `23.7800, 90.4000`, and press Enter. Leave **Grid bearing** at 0 (the X axis points east).
3. SAEngine shows the point in degrees, minutes and seconds (*23°46'48.00" N, 90°24'00.00" E*) and in BTM (*E 541 036.80 m, N 629 632.10 m*), and places BH-1 and BH-2 from their X and Y (*BH-1: 23.779982° N, 90.399980° E*). Drag a marker to a surveyed position if you have one; that borehole then keeps its own latitude and longitude.
4. When the nearest BNBC town's seismic zone or wind speed differ from those in 2 Project Settings, a banner names them with **Apply to Project Settings**. The tutorial building is already set to Dhaka, so no banner appears.
5. **Export GeoPackage…** (or **Export GeoJSON…**) writes the site, the boreholes and the building outline for QGIS, into `11 Exports` by default.

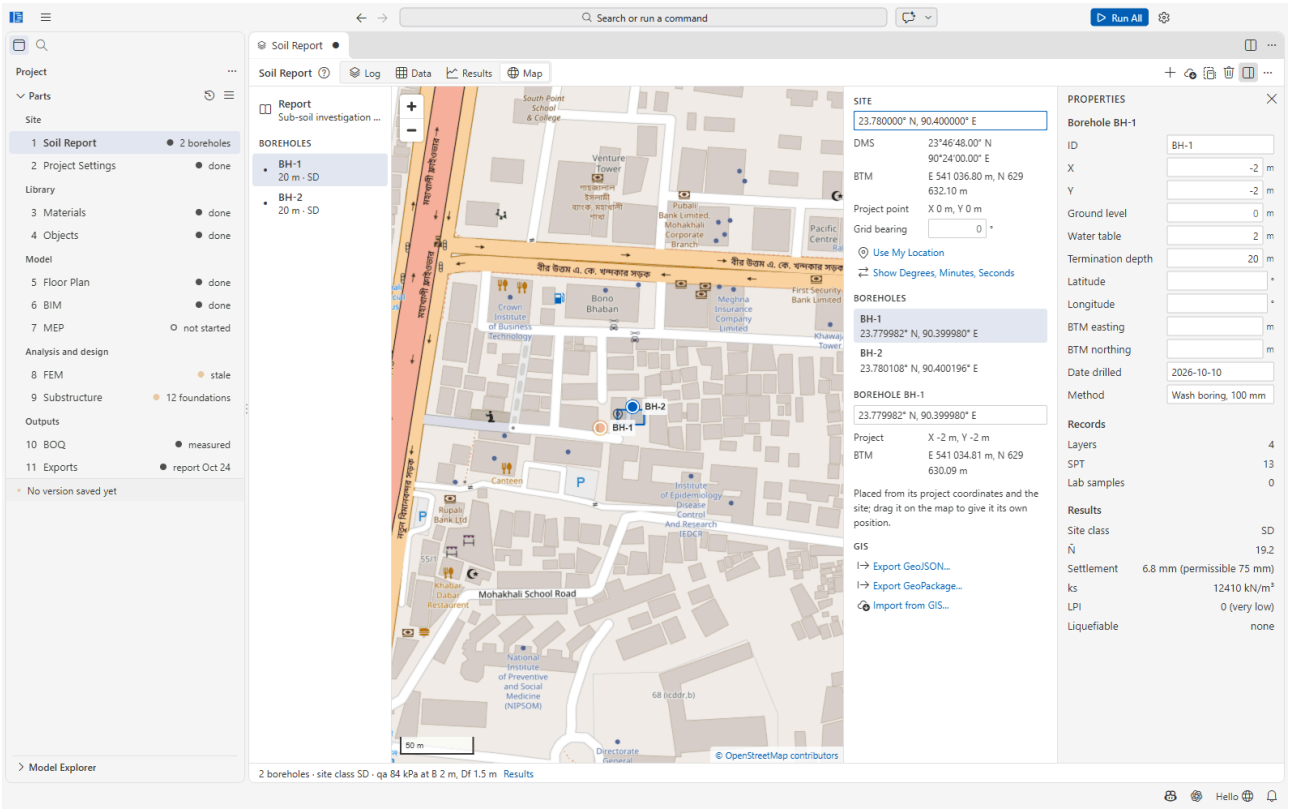


Figure 50: The site map with the two boreholes

5. Read the results

Select **Report** in the list and switch to **Results**:

- **Site class:** *SD — N̄ 17.4 · s̄u 33 kPa · over 20 m · governed by BH-2* (BH-1: N̄ 19.2, BH-2: N̄ 17.4). Project Settings → Site → *Site class source: From the soil report* uses it for the earthquake loads; it is the same class the generator assumed.
- **Allowable bearing pressure** (net, FS 3, settlement ≤ 25 mm, the lowest of the two boreholes):

Df \ B (m)	1	1.5	2	3	4
1	—	—	—	—	—
1.5	90	91	84	78	74
2	93	89	91	82	78
3	97	93	90	91	84

At 1 m the fill has no strength data, so no value is given. At the generator's founding level of 1.5 m, qa is 74 to 91 kPa: **less than the 100 kPa the generator assumed**.

- **Settlement** of the reference footing: 6.8 mm (BH-1) and 6.9 mm (BH-2) against the 75 mm BNBC Table 6.3.8 permits on plastic clay. **Liquefaction**: LPI 0 (very low).

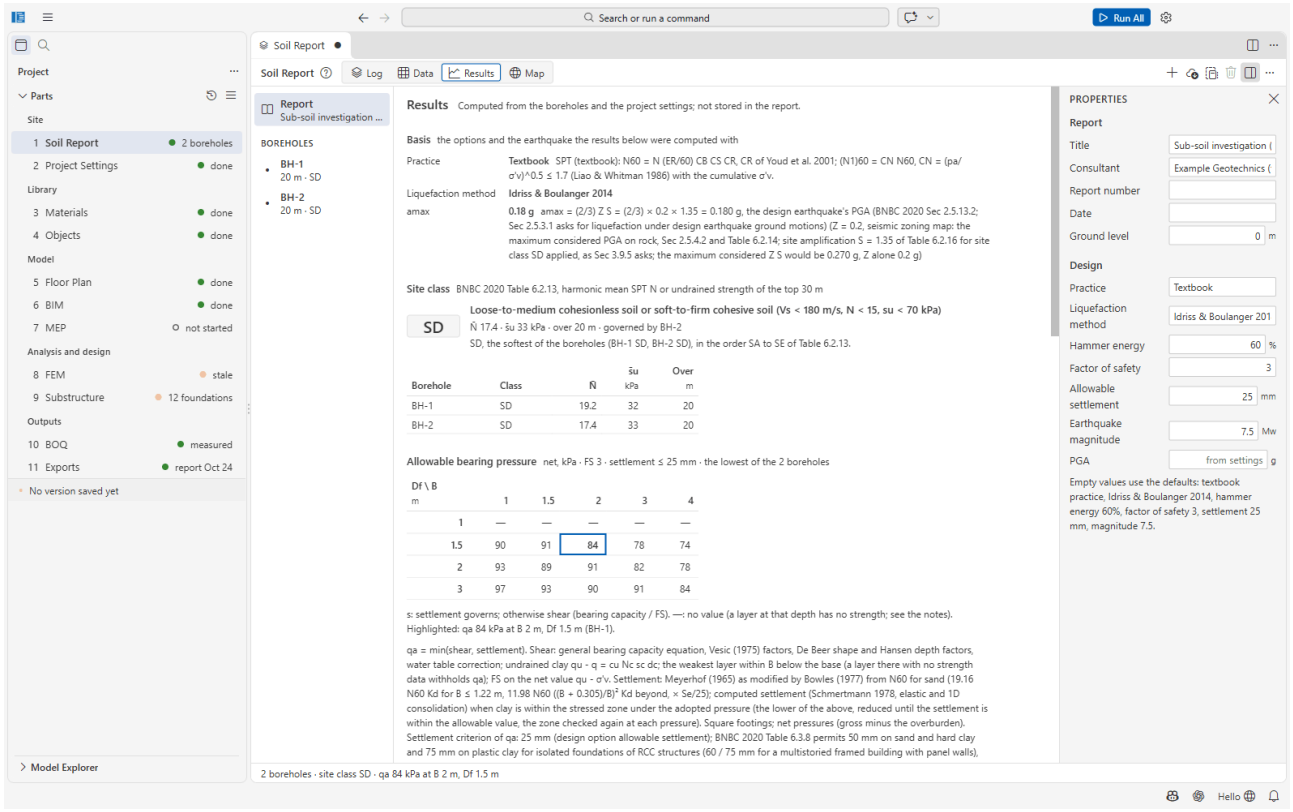


Figure 51: Results of the soil report

6. Design the footings

1. Open **9 Substructure** and its **Results** view. All twelve footings say *not checked: no reinforcement*: the generator sized them (for qa = 100 kPa) but gave them no bars.
2. On the **Plan**, select all twelve footings: click the first, **Shift+click** the others. The Properties panel says *12 foundations*.
3. Click **Size Footings**. The assistant sizes each from its column reactions and the soil report's qa at its position (the lowest borehole within 10 m) and proposes bars: *Proposed sizes for 12 foundations (dashed)*. Click **Accept**. The footings grow from 2.3–4.3 m to 3.55–5.4 m, for example F-A1 *3.65 × 3.65 × 0.45 m* with bottom bars *20@325* both ways.

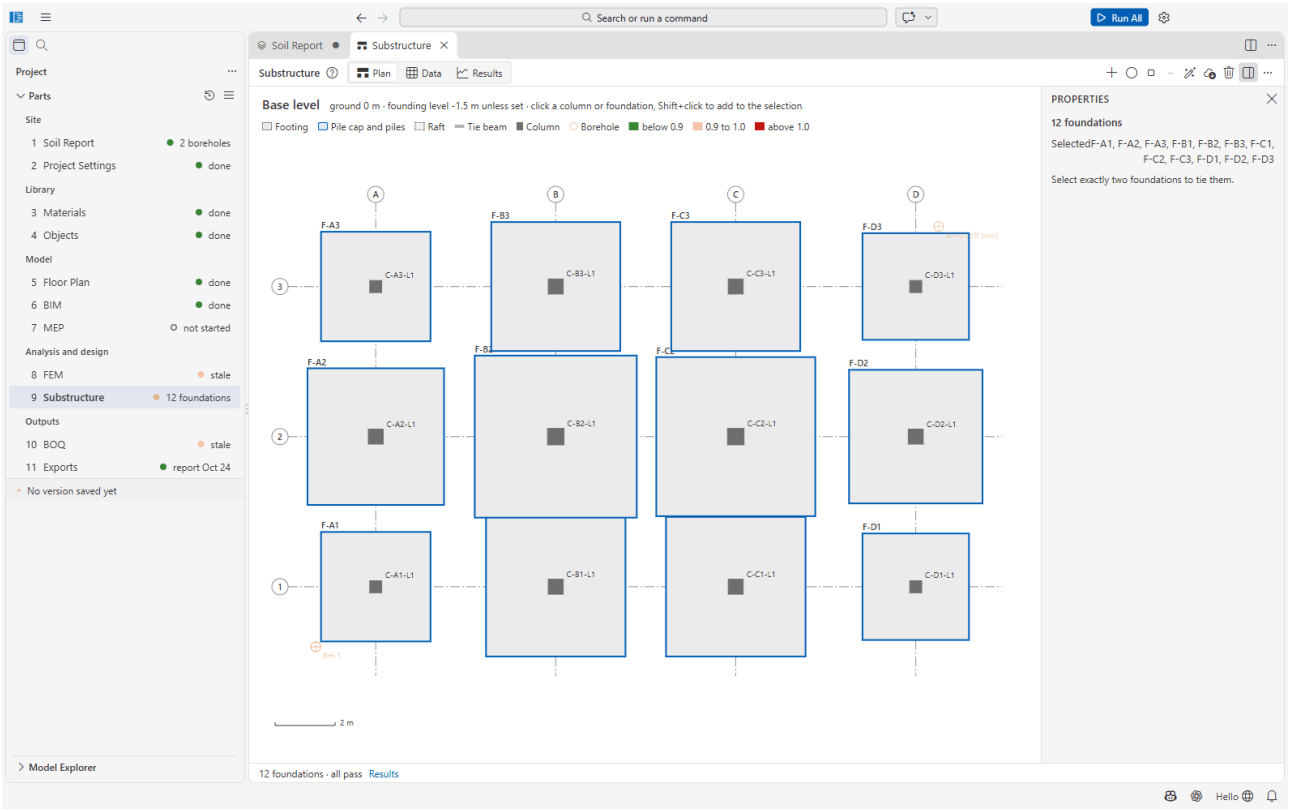


Figure 52: The footings after sizing with the soil report

4. Save (**Ctrl+S**): the checks run on the saved file. If the Results still show the old state, choose **…** → **Check the Foundations Again**.
5. Results: *12 foundations checked, 0 fail · 0.999 largest utilisation, F-B1 · 78 kPa max bearing pressure, F-D1 (qa 78 kPa)*. Bearing governs the inner footings (*Bearing (service)* under *D − WY* or *D + 0.75L + 0.75Lr ± 0.75WY*), one-way shear the corner ones. Click a footing for all its checks: the note of *Bearing (service)* explains the pressures and where qa came from.

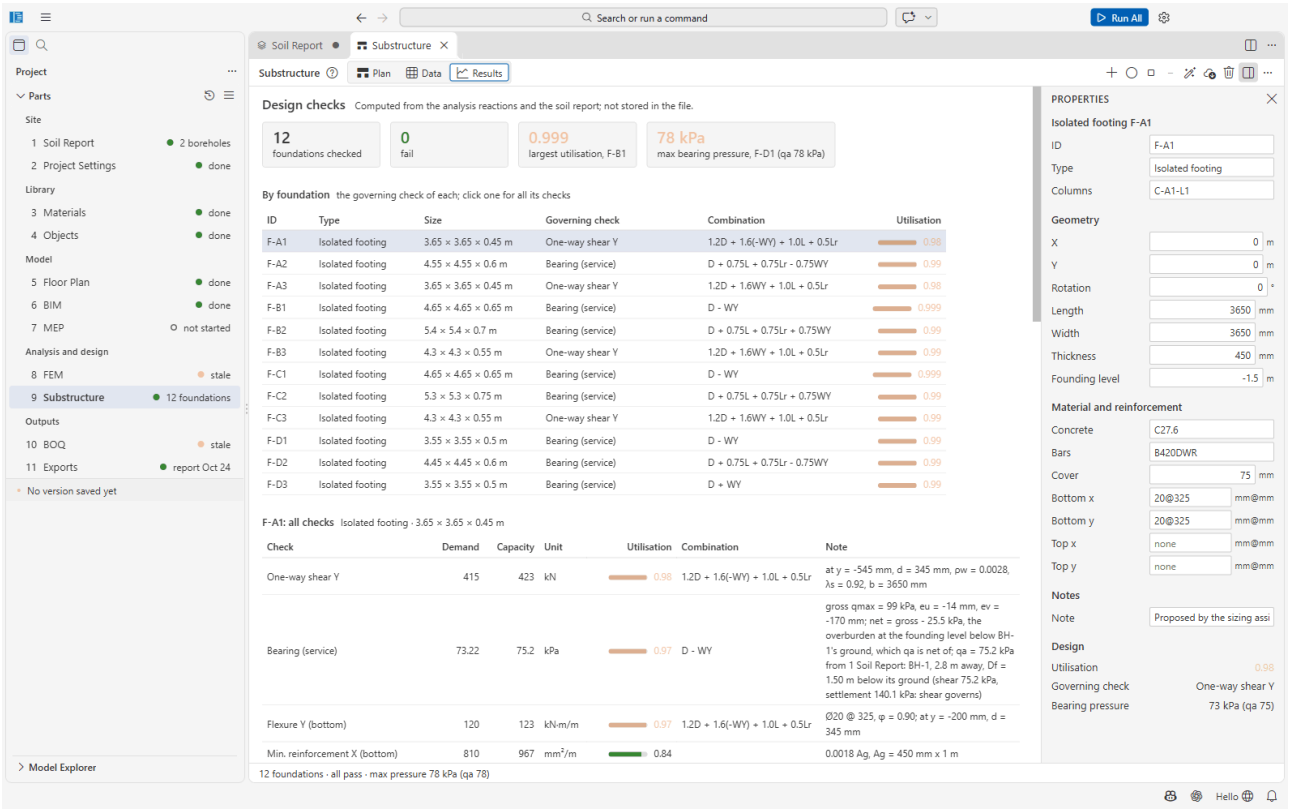


Figure 53: All footings pass

6. Look at the plan critically: with 6 m × 5 m bays, the footings 4.65 m and 5.4 m wide meet (F-B1 with F-B2, and F-C1 with F-C2: 2.33 m + 2.70 m > 5 m). All checks pass, but footings that touch are not isolated footings any more. Where they overlap, select the two columns and use **Add Combined Footing under the 2 Selected Columns**, or a raft (**Add Raft under the Selected Columns**), and check again. Tie beams: select two foundations and **Add Tie Beam between the Two Selected Foundations**.

7. Finish

Run All once more: the bill of quantities measures the new footings (excavation, blinding, concrete, bars) and the drawings' *Foundation plan* and *Foundation details* show them.

What you learned

- Entering a log sheet in the Data view, and importing boreholes from CSV or Excel.
- Placing boreholes and the site on the map, and taking the BNBC location from it.
- How the soil report feeds the site class (Project Settings) and the bearing capacity (9 Substructure).
- Sizing and checking footings from the analysis reactions.

Tutorial 4: Importing an ETABS model and comparing results

You will import a small ETABS model from its `.e2k` text file, read the import report, run the analysis and design in SAEngine, and compare the results with hand calculations and with the ETABS results of the same model.

Time: about 30 minutes, most of it waiting for Run All.

> SAEngine has **no automatic comparison** with ETABS results: you compare by hand, with the table at the end of this tutorial. Check your first real import as carefully as this tutorial does.

The model

The file tutorial-frame.e2k is a synthetic model written by hand for this manual (it is not an export of a real project). It has the structure ETABS writes:

- units kN, m (`$ CONTROLS  UNITS "KN" "M" "C"`);
- three storeys of 3.2 m (`Story1` to `Story3` above `Base`);
- grids A, B, C at 0, 6, 12 m and 1, 2, 3 at 0, 5, 10 m: two bays each way, 12 m × 10 m;
- columns C400x400, beams B300x500, slabs S150, concrete f′c 28 MPa, rebar B420; column bars 12 Ø16 in `$ CONCRETE SECTIONS`
- fixed supports at the nine base points;
- load patterns Dead (self-weight), SDL, Live, RLive with uniform area loads: SDL 2.2 kN/m² on the floors and 1.5 kN/m² on the roof, Live 2.0 kN/m² on the floors, roof live 1.0 kN/m²;
- two combinations, U1 = 1.4(Dead + SDL) and U2 = 1.2(Dead + SDL) + 1.6 Live + 0.5 RLive.

If you have ETABS, you can open this file there (File → Import → ETABS .e2k Text File) to get the ETABS results for the comparison. With your own model, export it from ETABS with **File → Export → ETABS .e2k Text File**.

1. Import the file

1. On the start page (or the ≡ menu) choose **Import from ETABS, STAAD or SAP2000...**.
2. Pick `tutorial-frame.e2k`.
3. **Choose Where to Put the Imported Project**: for example `Documents\SAEngine Projects`.
4. **Name the Imported Project**: *Tutorial Frame*.

SAEngine writes the project, opens it, and shows the **Import Report** beside it:

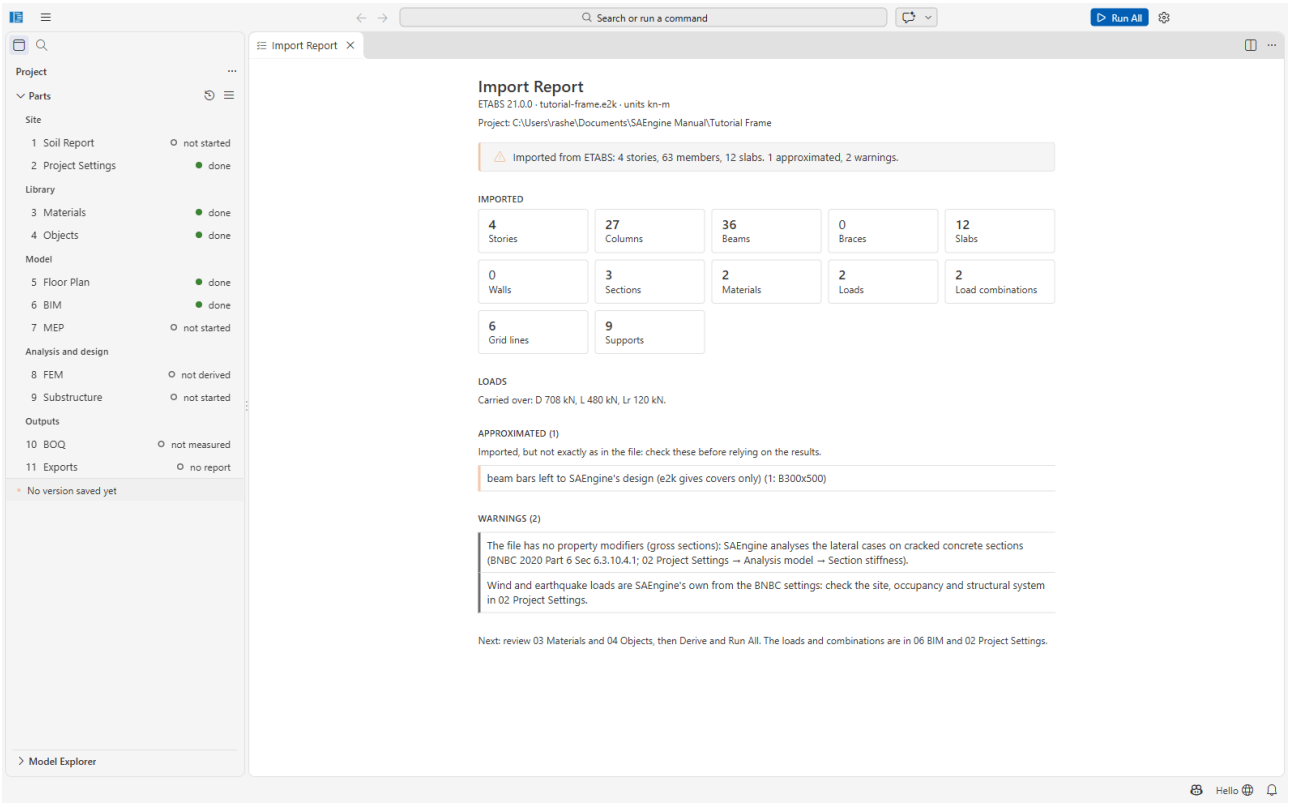


Figure 54: The import report of the tutorial frame

Read it section by section:

Section	What it says here	What to do
Imported	4 stories, 27 columns, 36 beams, 12 slabs, 3 sections, 2 materials, 2 loads, 2 load combinations, 6 grid lines, 9 supports	Check the counts against ETABS (9 columns × 3 storeys, 12 beams × 3, 4 slabs × 3).
Loads	*Carried over: D 708 kN, L 480 kN, Lr 120 kN.*	These are the area loads; nothing was lost.
Approximated	*beam bars left to SAEngine's design (e2k gives covers only)*	Expected: SAEngine designs the beam bars.
Warnings	*The file has no property modifiers (gross sections): SAEngine analyses the lateral cases on cracked concrete sections (BNBC 2020 Part 6 Sec 6.3.10.4.1; …)* and *Wind and earthquake loads are SAEngine's own from the BNBC settings: check the site, occupancy and structural system in 02 Project Settings.*	Note the first for the period comparison; the second is step 2.

The same import from the command line:

```
saengine import tutorial-frame.e2k "D:\Projects" --name "Tutorial Frame"
```

2. Check what was imported

1. **02 Project Settings.** The importer sets Dhaka, residential, occupancy II, site class SD, *RC intermediate moment frame*. Set them to your project's site and system: wind and earthquake are computed by SAEngine from these settings, not taken from the file. The **Area Loads** show *Floors (imported)* (live 2.0, finish 2.2) and *Roof (imported)* (live 1.0, finish 1.5): the file's SDL became the floor finish. The **Load Combinations** are the file's U1 and U2 only. Click **Reset Combinations to BNBC 2020…** if you want the full BNBC set including wind (the earthquake combinations of the response-spectrum design are added automatically).
2. **03 Materials:** C28 (f′c 28 MPa) and B420 (fy 420 MPa) with the file's E and unit weight.
3. **04 Objects:** C400x400 with its 12 Ø16 bars, B300x500 without bars, S150.
4. **05 Floor Plan:** the grids A–C, 1–3 and the members on each storey. **6 BIM** has been built.

3. Run All

Click **Run All**. On a typical PC this takes several minutes. When it finishes:

Done: 0 pass, 27 fail; steel short in 27 element(s) · model derived · BOQ updated · report written

All members fail because the beams have no bars and the columns' 12 Ø16 are not enough for the BNBC seismic design: *steel short*, not *section too small*. Open **8 FEM**, choose **Design Reinforcement**, and **Apply Proposal** as in Tutorial 1.

4. Compare

Open the report (**Open Report**, or `11 Exports/report.pdf`), chapter *Analysis results*.

Gravity loads by hand. Plan area 12 × 10 = 120 m², three floors.

Case	Hand calculation	SAEngine (report 4.2, ΣFz)
D, self-weight	slabs 0.15 × 120 × 3 = 54.0 m³; columns 0.4 × 0.4 × 3.2 × 9 × 3 = 13.8 m³; beams 0.3 × 0.5 × (3 × 12 + 3 × 10) × 3 = 29.7 m³; total 97.5 m³ × 23.56 kN/m³ = **2297 kN**	**2298.0 kN**
SD	2.2 × 120 × 2 + 1.5 × 120 = **708 kN**	**708.0 kN**
L	2.0 × 120 × 2 = **480 kN**	**480.0 kN**
Lr	1.0 × 120 = **120 kN**	**120.0 kN**

The self-weight matches the gross volumes: SAEngine counts the slab, beam and column volumes where they overlap. Check how your ETABS model treats these overlaps before comparing.

Against ETABS. Fill in the ETABS column from ETABS's own tables (Display → Show Tables → *Base Reactions*, *Modal Participating Mass Ratios*, *Story Drifts*):

Quantity	SAEngine 0.0.42	ETABS	Difference
ΣFz Dead	2298.0 kN		
ΣFz SDL	708.0 kN		
ΣFz Live	480.0 kN		
ΣFz Roof live	120.0 kN		
Seismic weight	3071.7 kN		
Period, mode 1 (X)	0.720 s		
Period, mode 2 (Y)	0.686 s		
Mode 1 mass participation	83.7 %		
Static base shear EX = EY	276.5 kN		
Largest design storey drift X	0.94 % (allowed 2.00 %)		

The SAEngine values above are those of the tutorial file in SAEngine 0.0.42; yours should be the same.

Why results can differ (check these before suspecting an error):

- **Gravity reactions** should agree within rounding. If not, look at the import report's *Loads* and *Skipped* sections (point and frame loads are not carried over) and at the unit weight of the materials.
- **Periods** depend on the stiffness: SAEngine uses cracked sections for the lateral cases (BNBC 6.3.10.4.1: columns 0.70 Ig, beams 0.35 Ig, slabs 0.25 Ig) and ignores end offsets and rigid zones. Compare with ETABS using the same property modifiers. They also depend on the mass source: compare SAEngine's seismic weight (shown when the model is derived: *Derived … nodes and … elements; base shear … kN, seismic weight … kN*) with the ETABS mass first.
- **Seismic and wind forces** are SAEngine's own from Project Settings, not ETABS's auto loads: compare them only when the site, occupancy, system and the code parameters are the same.
- **Member forces** at the ends of members differ where ETABS has end releases, insertion points or end offsets: SAEngine ignores them (*Approximated* in the import report).
- **Designs** differ because SAEngine designs to BNBC 2020 Part 6 Chapter 6 with its own seismic detailing rules (Sec 8.3), and proposes bars for each object type.

What you learned

- How an `.e2k` file maps to SAEngine's parts, and what the import report tells you.
- That wind and earthquake loads always come from SAEngine's BNBC settings.
- How to check an imported model with hand calculations and against the source program.